USER GUIDE

Power Vision 4

98100027.03

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Power Vision 4 User Guide.

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Printed in USA.

Part Number: 98100027 Version 3 (08/2025)
Power Vision 4 App

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CHAPTER 1

WELCOME TO DYNOJET POWER VISION 4 APP

The Software Engineers at Dynojet understand your need to attain the maximum performance from the vehicles you evaluate and tune. The Power Vision 4 (PV4) is the most flexible and powerful flash tuner in the world. The Power Vision 4 uses reflash technology to change the parameters inside the stock Engine Control Module (ECM) that affects the way the engine will run. Use one of Dynojet's pre-configured calibrations, or create a custom tune using Power Core software.

The Power Vision 4 User Guide provides information and step-by-step guidance for common tasks. This guide is designed to be a reference tool in your everyday work and includes the following chapters and information:

Diagnostic Connector Locations

This section will guide you through locating the diagnostic connector on your vehicle and connecting the Power Vision 4 to your vehicle.

Power Vision 4 App

This section will guide you through using the Power Vision 4 App.

Power Core Software Download

This section will guide you through downloading the Power Core Software Suite.

Troubleshooting

This section will guide you through a few of the more common troubleshooting topics.

DISCLAIMER

Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

NOTICE

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Power Vision 4 App

Power Vision 4 User Guide Part Number 98100027.03

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CONVENTIONS

Power Vision 4 App documentation uses consistent conventions to help you identify items. The following table summarizes these conventions.

example of convention	description
Bold	Highlights items you can select on in the app interface, including buttons and menus.
>	The arrow indicates a menu item and menu command that you need to click in sequence. For example, "select **File >Open**" means "select the **File** menu, then select the **Open** choice on the **File** menu."
Blue	Words highlighted in blue indicate a link.

SUPPORT

Dynojet Research Inc. is committed to providing the best customer service to all our clientele. Need technical assistance? Try our remote support assistance or watch one of our informative videos. Stay informed about product releases and new map downloads by signing up for the Dynojet Newsletter.

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CHAPTER 2

DIAGNOSTIC CONNECTOR LOCATIONS

This section will guide you through locating the diagnostic connector on your vehicle and connecting the Power Vision 4 to your vehicle.

This section is divided into the following categories:

- Locating the Diagnostic Connector—Harley-Davidson, page 2-2
- Locating the Diagnostic Connector—Polaris, page 2-8
- Locating the Diagnostic Connector—Indian, page 2-13
- Locating the Diagnostic Connector—Can-Am, page 2-15

LOCATING THE DIAGNOSTIC CONNECTOR—HARLEY-DAVIDSON

Use the following information to locate the diagnostic connector on your vehicle and connect the Power Vision 4. After the vehicle has been flashed, the Power Vision 4 does not need to remain connected to have full benefits of the device, however, the Power Vision 4, paired with your phone and the Dynojet PV4 app, provides a window into your vehicle. Gain access to customizable gauge screens, along with data logging screens and troubleshooting of Diagnostic Trouble Codes.

2012-2017 DYNA MODELS

The diagnostic connector is located under the left hand side electrical cover. This connector is light gray in color with a black rubber plug installed.

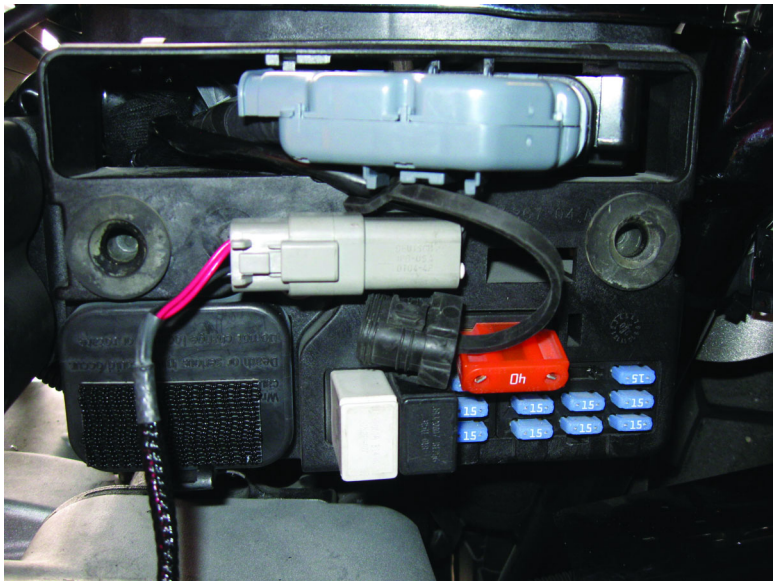


Figure 2-1: Diagnostic Connector—2012-2017 Dyna Models

2014-2025 Sportster Models Including XR Models

The diagnostic connector is located under the left hand side electrical cover. This connector is light gray in color with a black rubber plug installed.

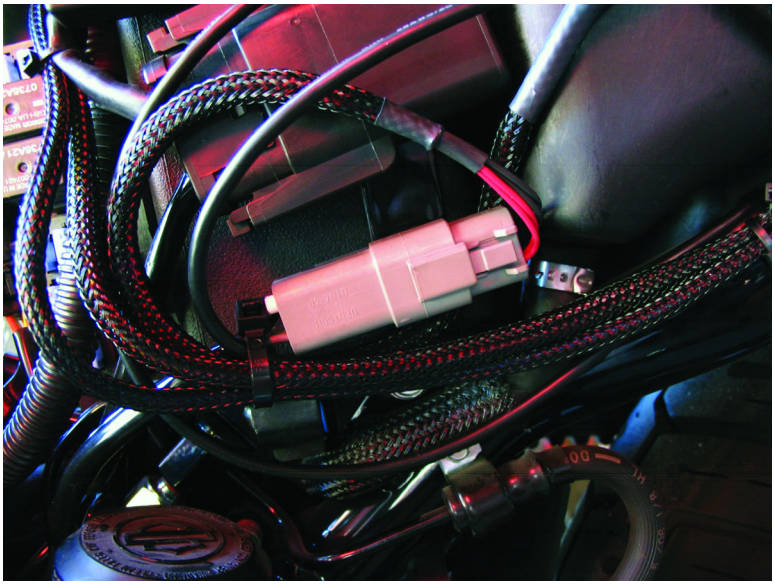


Figure 2-2: Diagnostic Connector—2014-2021 Sportster Models

2011-2020 Softail Models

The diagnostic connector is located under the seat attached to the frame by the rear fender. This connector is light gray in color with a black rubber plug installed.

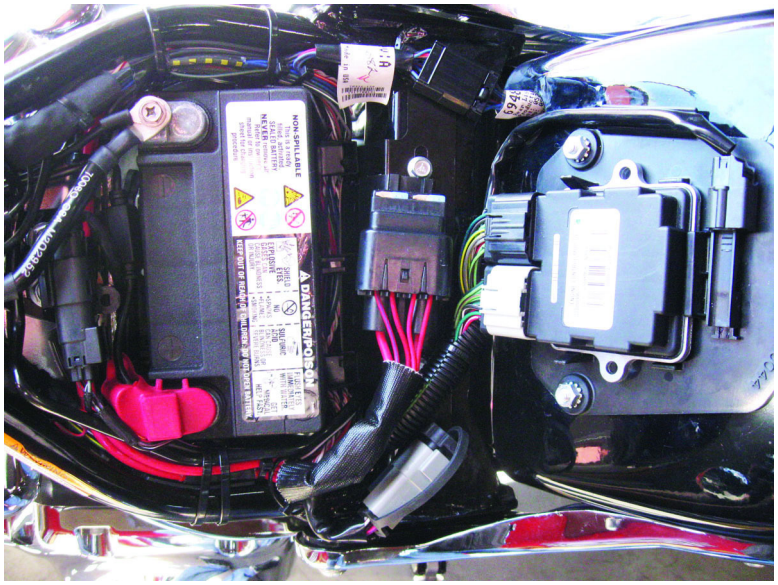


Figure 2-3: Diagnostic Connector—2011-2020 Softail Models

2021+ Softail Models

The diagnostic connector is located behind the left hand side cover. This connector is red in color. Some Softail models require removing the left saddle bag.

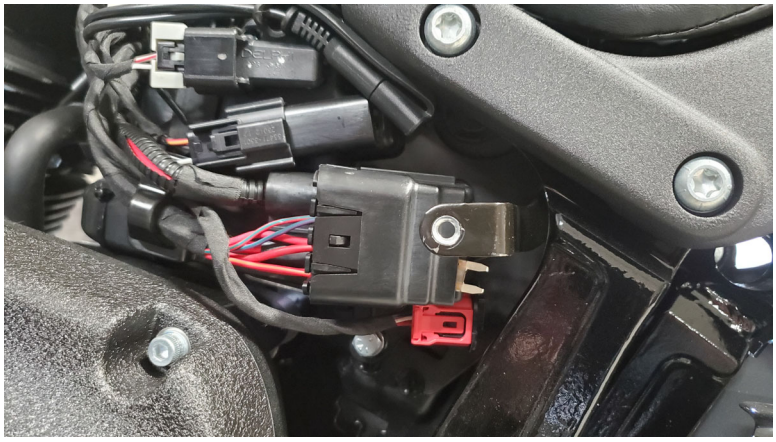


Figure 2-4: Diagnostic Connector—2021+ Softail Models

2015-2021 Street 500 and 750 Models

The diagnostic connector is located under the right hand side electrical cover. This connector is light gray in color with a black rubber plug installed.

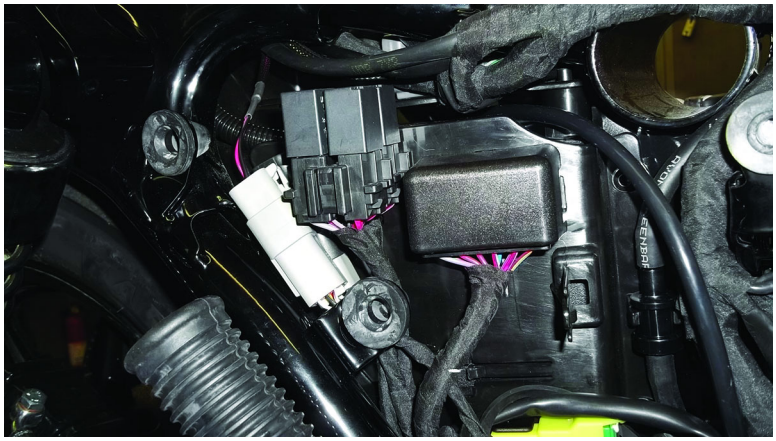


Figure 2-5: Diagnostic Connector—2015-2021 Street 500 and 750 Models

2002-2007 Touring Models

The diagnostic connector is located under the right hand side electrical cover. This connector is light gray in color with a black rubber plug installed. Requires removing the right saddle bag.

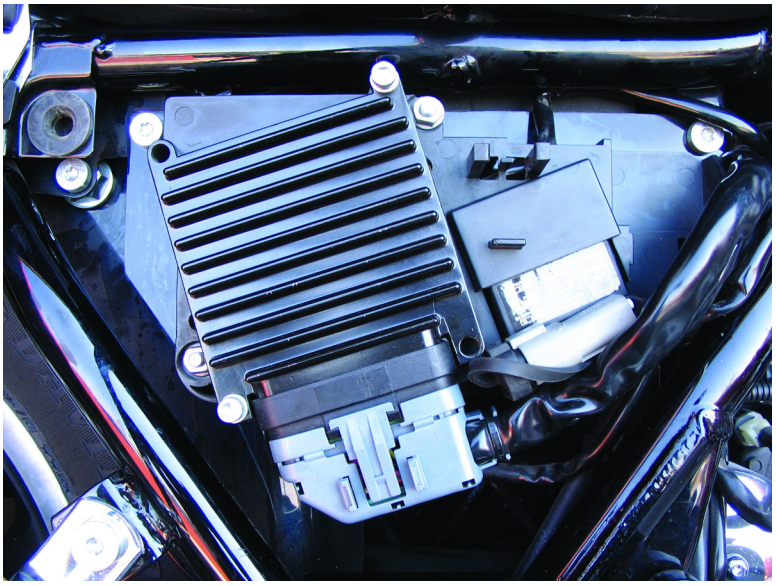


Figure 2-6: Diagnostic Connector—2002-2007 Touring Models

2008-2020 Touring Models

The diagnostic connector is located under the left hand side electrical cover. This connector is light gray in color with a black rubber plug installed. Requires removing the left saddle bag.

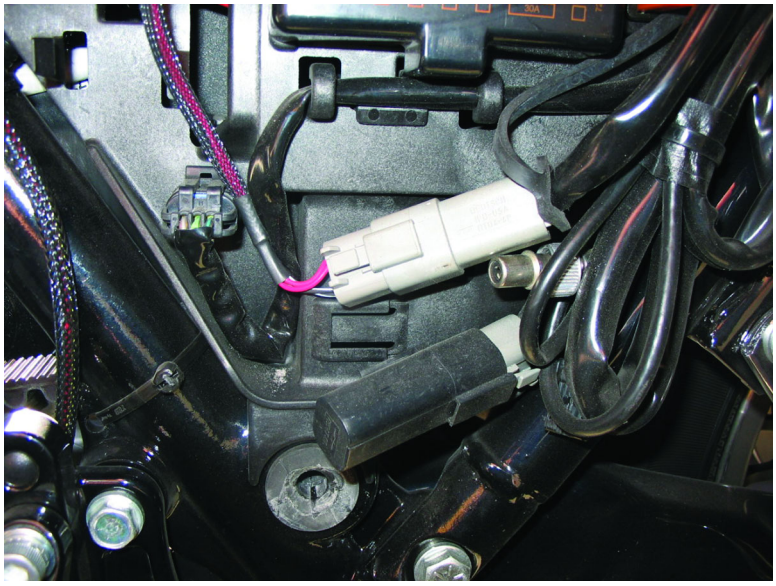


Figure 2-7: Diagnostic Connector—2008-2020 Touring Models

2021-2025 Touring Models

The diagnostic connector is located behind the left hand side cover. This connector is red, 6-pin. Requires removing the left saddle bag

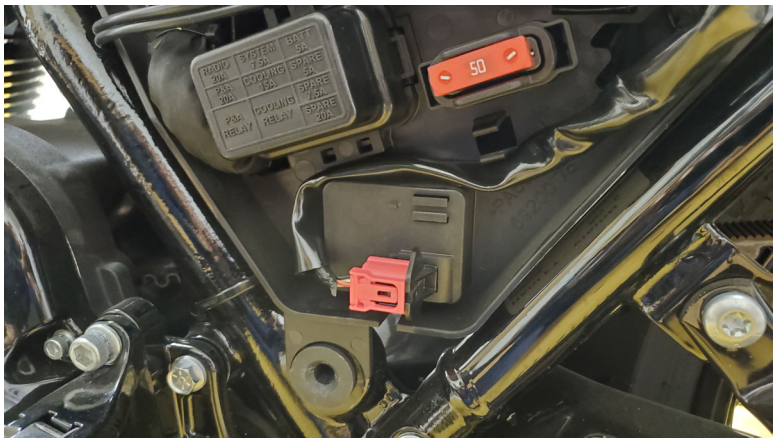


Figure 2-8: Diagnostic Connector—2021-2022 Touring Models

2002-2017 V-Rod Models

The diagnostic connector is located under the right front frame cover. This connector is light gray in color with a black rubber plug installed. Requires tools to access.

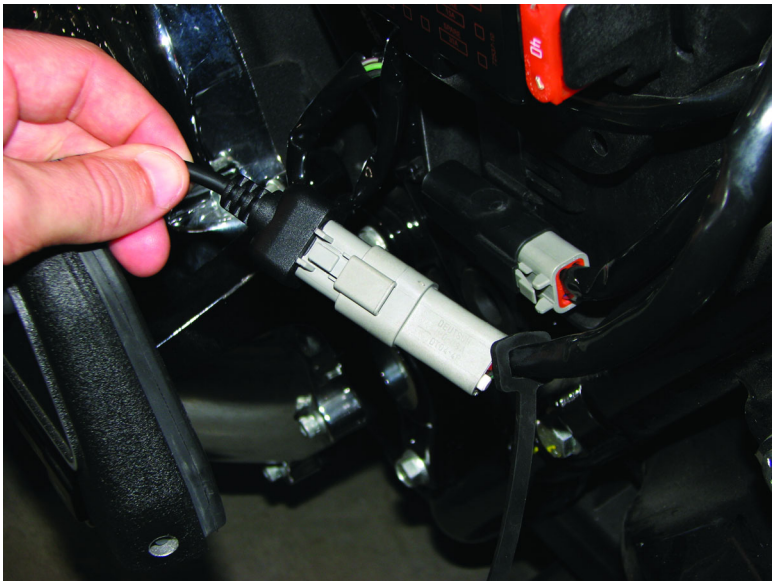


Figure 2-9: Diagnostic Connector—2002-2017 V-Rod Models

2021-2025 Sportster S

The diagnostic connector is located underneath the seat. This connector is red, 6-pin.

Figure 2-10: Diagnostic Connector—2021-2022 Sportster S

2021-2025 Pan America

The diagnostic connector is located underneath the seat. This connector is red, 6-pin.

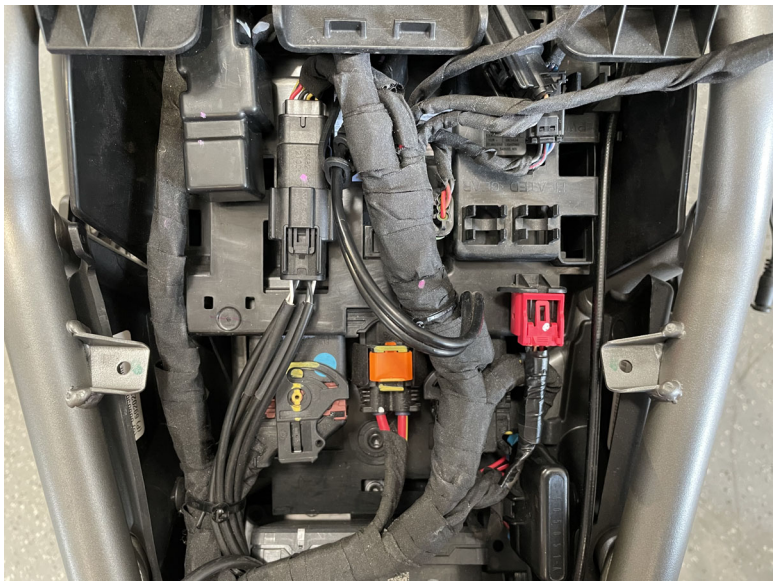


Figure 2-11: Diagnostic Connector—2021-2022 Pan America

LOCATING THE DIAGNOSTIC CONNECTOR—POLARIS

Use the following information to locate the diagnostic connector on your vehicle and connect the Power Vision 4. After the vehicle has been flashed, the Power Vision 4 does not need to remain connected to have full benefits of the device, however, the Power Vision 4, paired with your phone and the Dynojet PV4 app, provides a window into your vehicle. Gain access to customizable gauge screens, along with data logging screens and troubleshooting of Diagnostic Trouble Codes.

RANGER MODELS

The diagnostic connector is located under the seat.

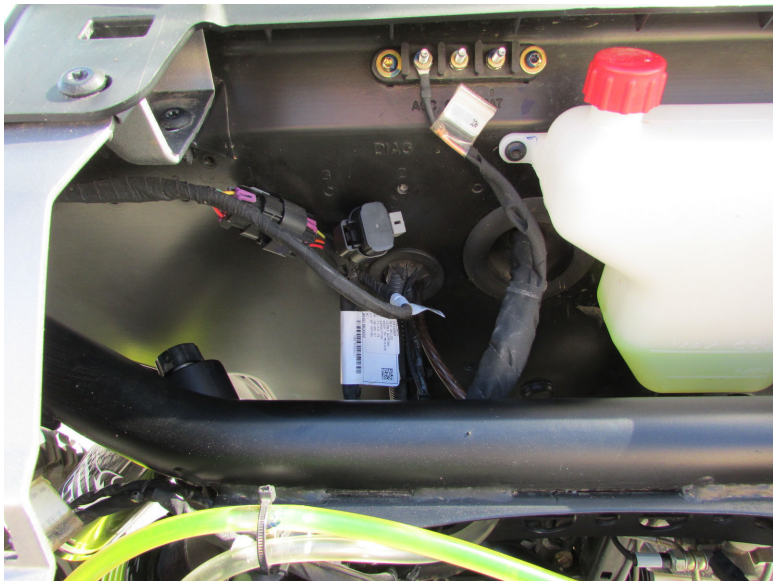


Figure 2-12: Diagnostic Connector—Ranger Models

RZR MODELS

The diagnostic connector is located under the front cover near the coolant reserve bottle.

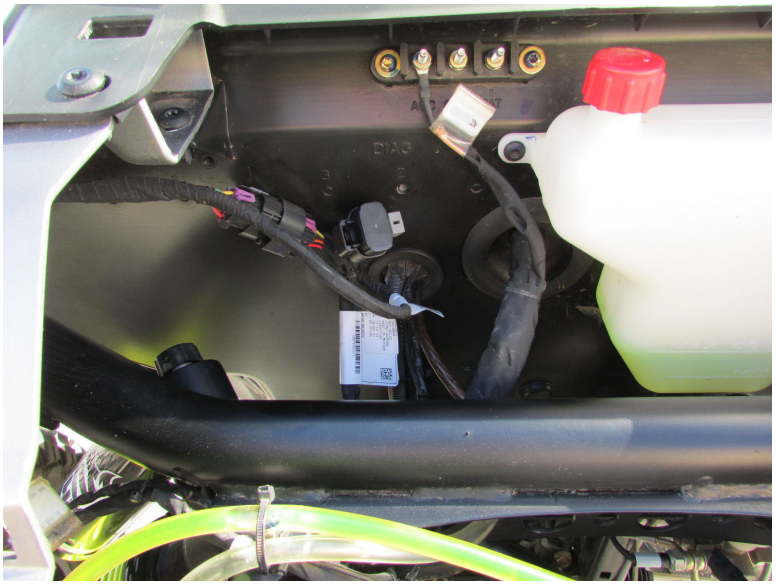


Figure 2-13: Diagnostic Connector—RZR Models

RZR PRO XP MODELS

The diagnostic connector is located under the steering wheel.

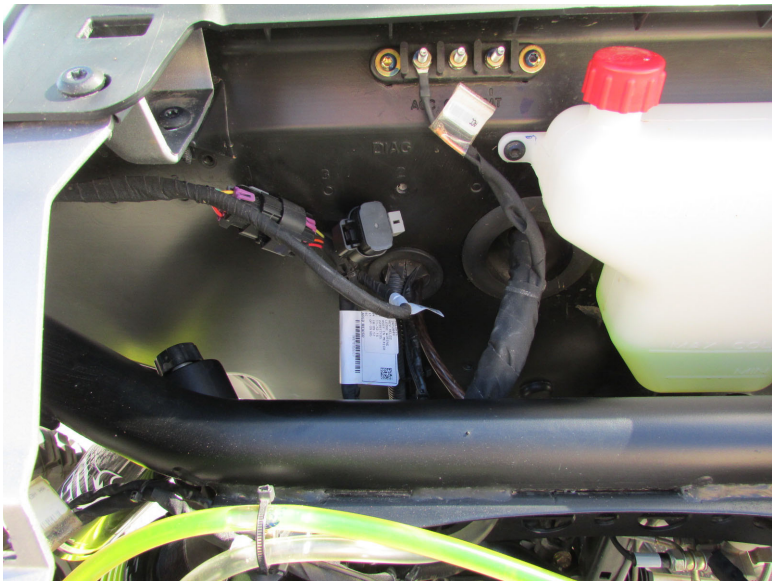


Figure 2-14: Diagnostic Connector—RZR Pro XP Models

General Models

The diagnostic connector is located under the front hood cover.

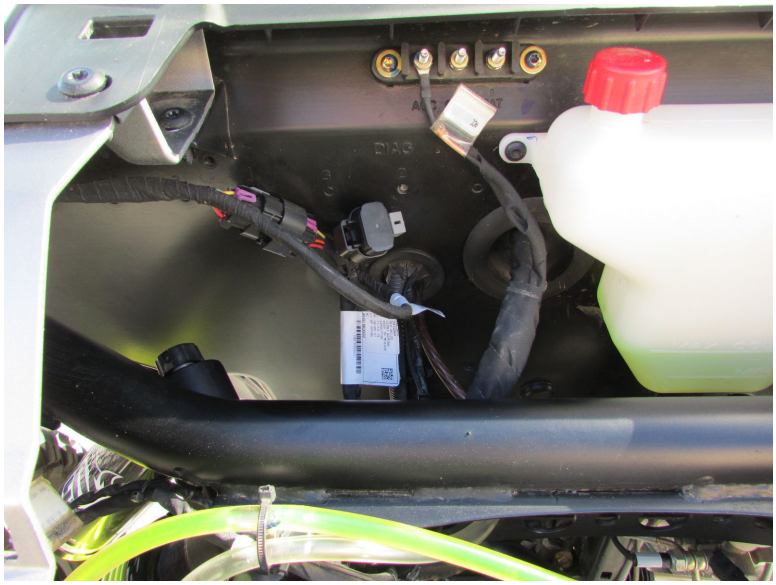


Figure 2-15: Diagnostic Connector—General Models

Ranger 800 Models

The diagnostic connector is located under the front hood.

Figure 2-16: Diagnostic Connector—Ranger 800 Models

2010-2014 RZR 800/XP900 MODELS

The diagnostic connector is located under the dash near the speedometer gauge cluster.

Figure 2-17: Diagnostic Connector—2010-2014 RZR 800/XP 900 Models

SPORTSMAN MODELS

The diagnostic connector is located under the storage compartment.

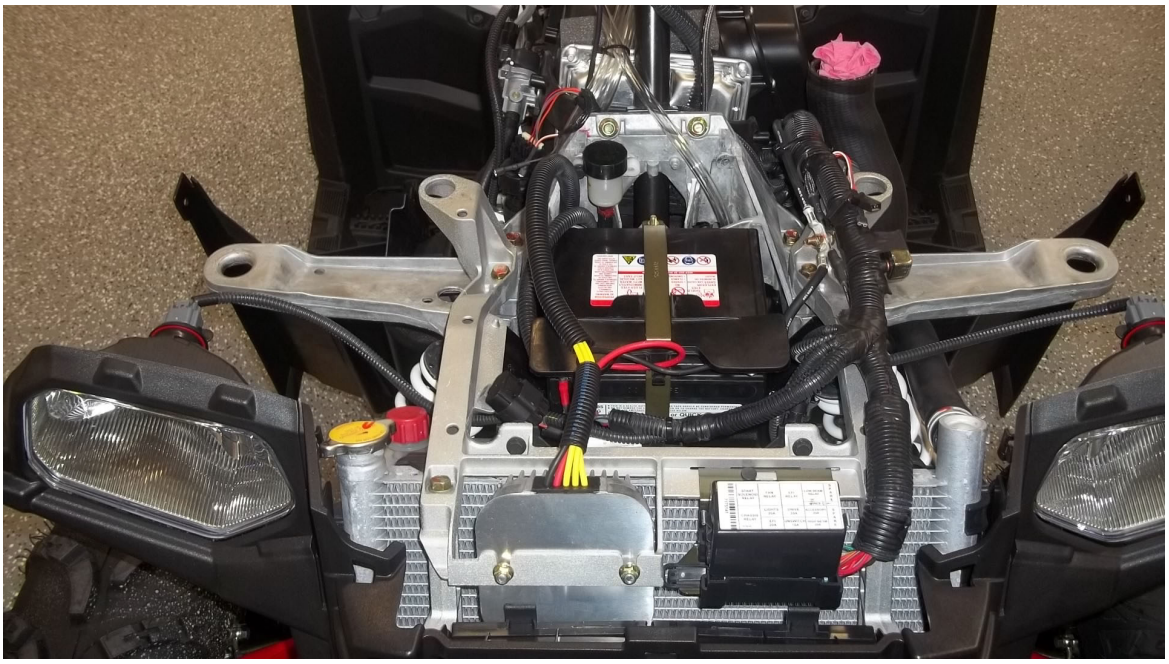


Figure 2-18: Diagnostic Connector—Sportsman Models

ACE MODELS

The diagnostic connector is located under the front dash.

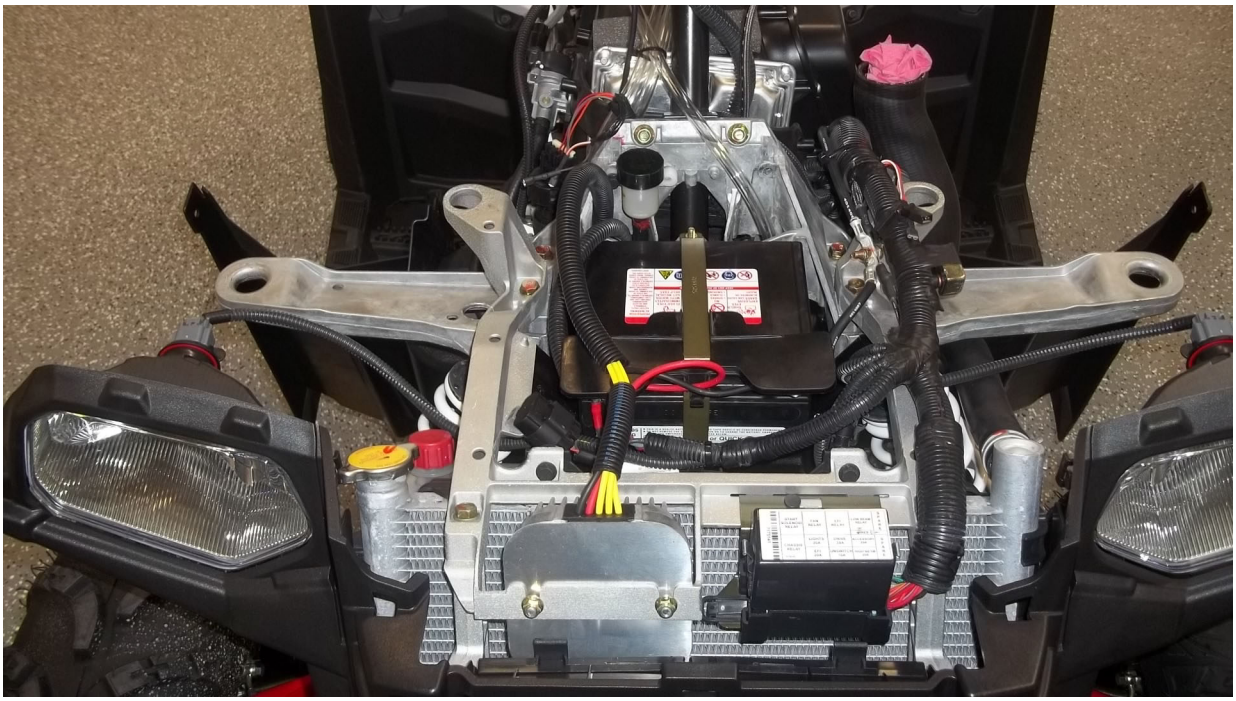


Figure 2-19: Diagnostic Connector—Ace Models

LOCATING THE DIAGNOSTIC CONNECTOR—INDIAN

Use the following information to locate the diagnostic connector on your vehicle and connect the Power Vision 4. After the vehicle has been flashed, the Power Vision 4 does not need to remain connected to have full benefits of the device, however, the Power Vision 4, paired with your phone and the Dynojet PV4 app, provides a window into your vehicle. Gain access to customizable gauge screens, along with data logging screens and troubleshooting of Diagnostic Trouble Codes.

SCOUT MODELS

The diagnostic connector is located under the seat.

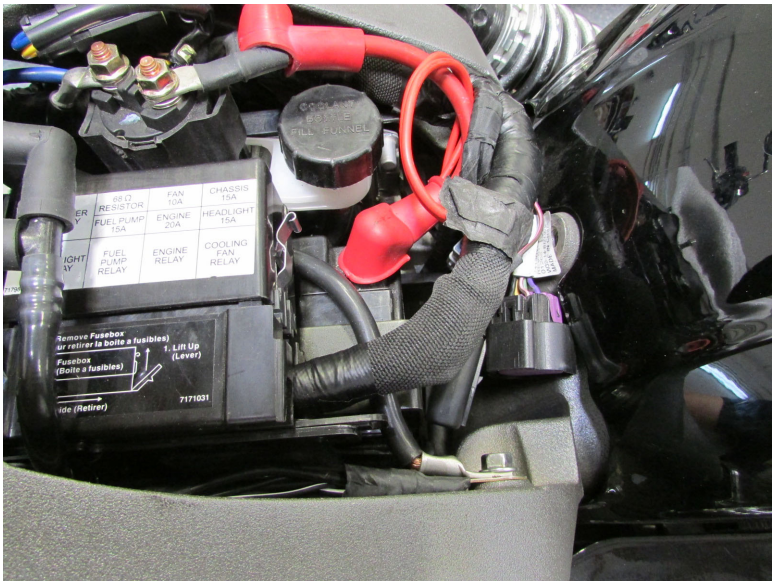


Figure 2-20: Diagnostic Connector—Scout Models

TS111 Models

The diagnostic connector is located under the left hand side cover.

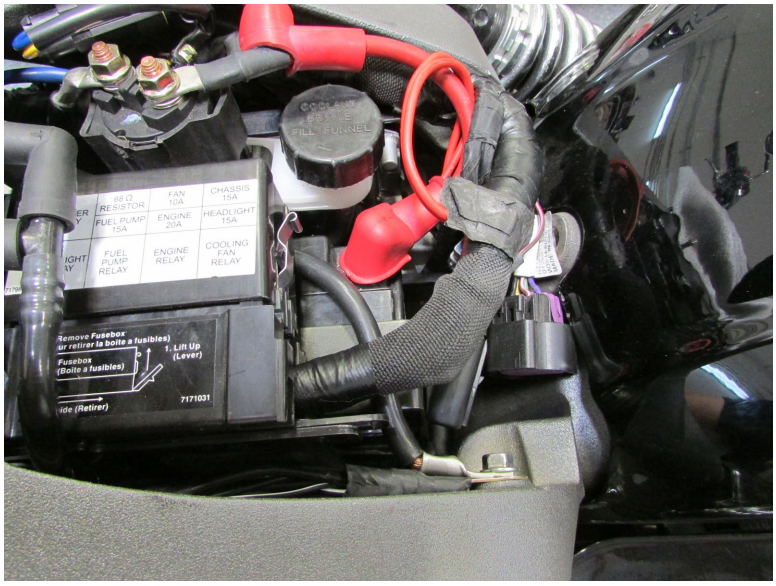


Figure 2-21: Diagnostic Connector—TS111 Models

FTR Models

The diagnostic connector is located behind the body panel on the right side and in front of the fuse block.

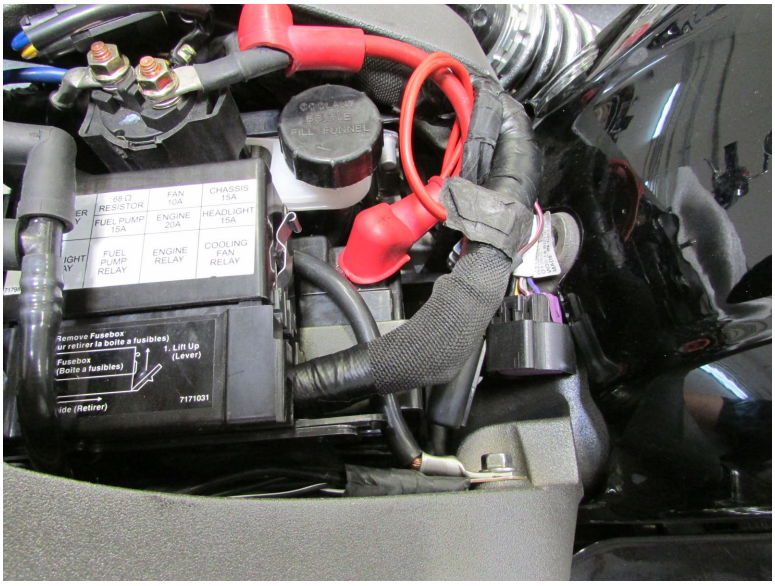


Figure 2-22: Diagnostic Connector—FTR Models

LOCATING THE DIAGNOSTIC CONNECTOR—CAN-AM

Use the following information to locate the diagnostic connector on your vehicle and connect the Power Vision 4. After the vehicle has been flashed, the Power Vision 4 does not need to remain connected to have full benefits of the device, however, the Power Vision 4, paired with your phone and the Dynojet PV4 app, provides a window into your vehicle. Gain access to customizable gauge screens, along with data logging screens and troubleshooting of Diagnostic Trouble Codes.

MAVERICK X3 MODELS

The diagnostic connector is located under the glove box lid.

Note: Remove the diagnostics port holder and insert the Power Vision cable to hold the cable securely in place.

Figure 2-23: Diagnostic Connector—Maverick X3 Models

TRAIL MODELS

The diagnostic connector is located under the dash to the left of the steering column.

Figure 2-24: Diagnostic Connector—Trail Models

MAVERICK MODELS

The diagnostic connector is located near the fuse box.

Note: Remove the diagnostics port holder and insert the Power Vision cable to hold the cable securely in place.

Figure 2-25: Diagnostic Connector—Maverick Models

SPYDER MODELS

The diagnostic connector is located under the front hood/luggage compartment area behind the small radiator cap access panel.

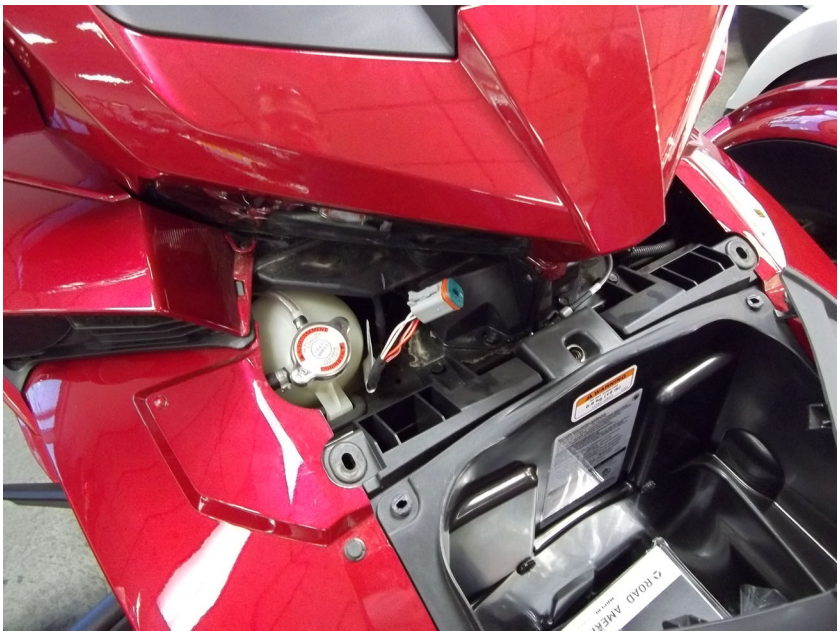


Figure 2-26: Diagnostic Connector—Spyder Models

COMMANDER MODELS

The diagnostic connector is located under the dash on the driver's side near the battery and fuse box.

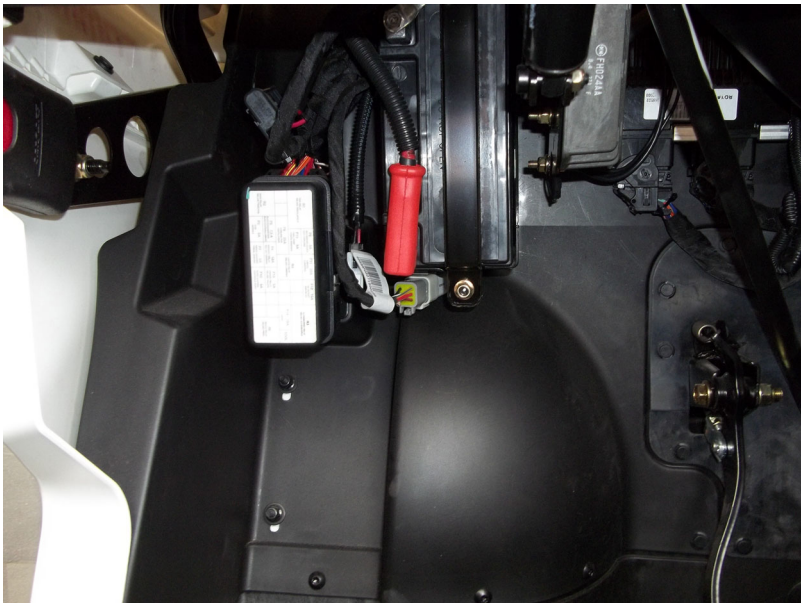


Figure 2-27: Diagnostic Connector—Commander Models

DEFENDER MODELS

The diagnostic connector is located under the front hood.

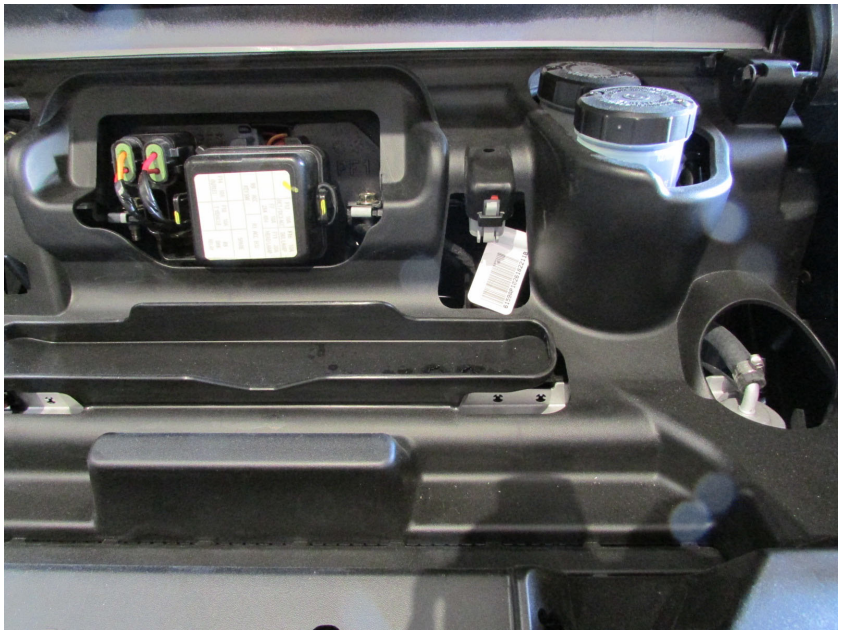


Figure 2-28: Diagnostic Connector—Defender Models

RYKER MODELS

The diagnostic connector is located under the front cover.

Figure 2-29: Diagnostic Connector—Ryker Models

OUTLANDER MODELS

The diagnostic connector is located under the seat.

Figure 2-30: Diagnostic Connector—Outlander Models

CHAPTER 3

POWER VISION 4 APP

The Power Vision 4 (PV4) is the most flexible and powerful flash tuner in the world.

This section will guide you through downloading the PV4 app, flashing your vehicle, and using the PV4 app.

This section is divided into the following categories:

- Power Vision 4 Overview, page 3-2
- Getting Started, page 3-6
- Flashing Your Vehicle, page 3-19
- Editing Tunes with Power Core, page 3-22
- License Manager, page 3-25
- Viewing Live Gauges, page 3-28
- Logging Data, page 3-42
- Viewing the Diagnostic Codes, page 3-43
- Using Autotune, page 3-44
- Viewing the Device Information, page 3-57
- Disconnecting from the Power Vision 4, page 3-59
- Restore to Stock, page 3-60
- Report a Problem, page 3-61

POWER VISION 4 OVERVIEW

The Power Vision 4 (PV4) is the most flexible and powerful flash tuner in the world. The Power Vision 4 uses reflash technology to change the parameters inside the stock Engine Control Module (ECM) that affects the way the engine will run.

After the vehicle has been flashed, the Power Vision 4 does not need to remain connected to have full benefits of the device, however, the Power Vision 4, paired with your phone and the Dynojet PV4 app, provides a window into your vehicle. Gain access to customizable gauge screens, along with data logging screens and troubleshooting of Diagnostic Trouble Codes.

POWER VISION 4 OVERVIEW—LED INDICATOR LIGHTS

The Power Vision 4 has three indicator lights. Press **Menu** and press **LED Legend**.

- Red LED—Communication. Solid red is communication on bus; rapid flashing is recovery mode.
- Green LED—Power. Short blink is power; long blink is connected to ECU.
- Blue LED—Bluetooth status. On is connected to Bluetooth; off is not connected.

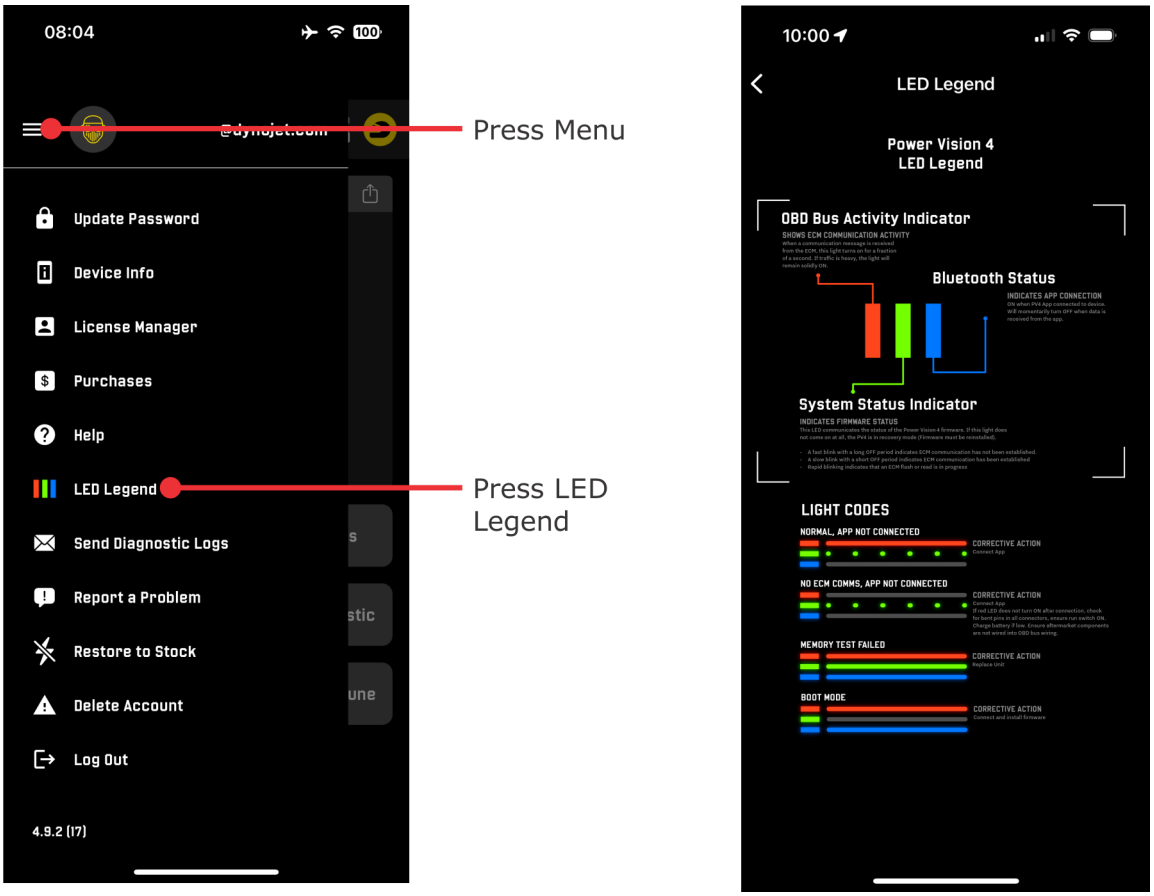


Figure 3-1: Power Vision 4 Overview—LED Indicator Lights

POWER VISION 4 OVERVIEW—HARLEY-DAVIDSON

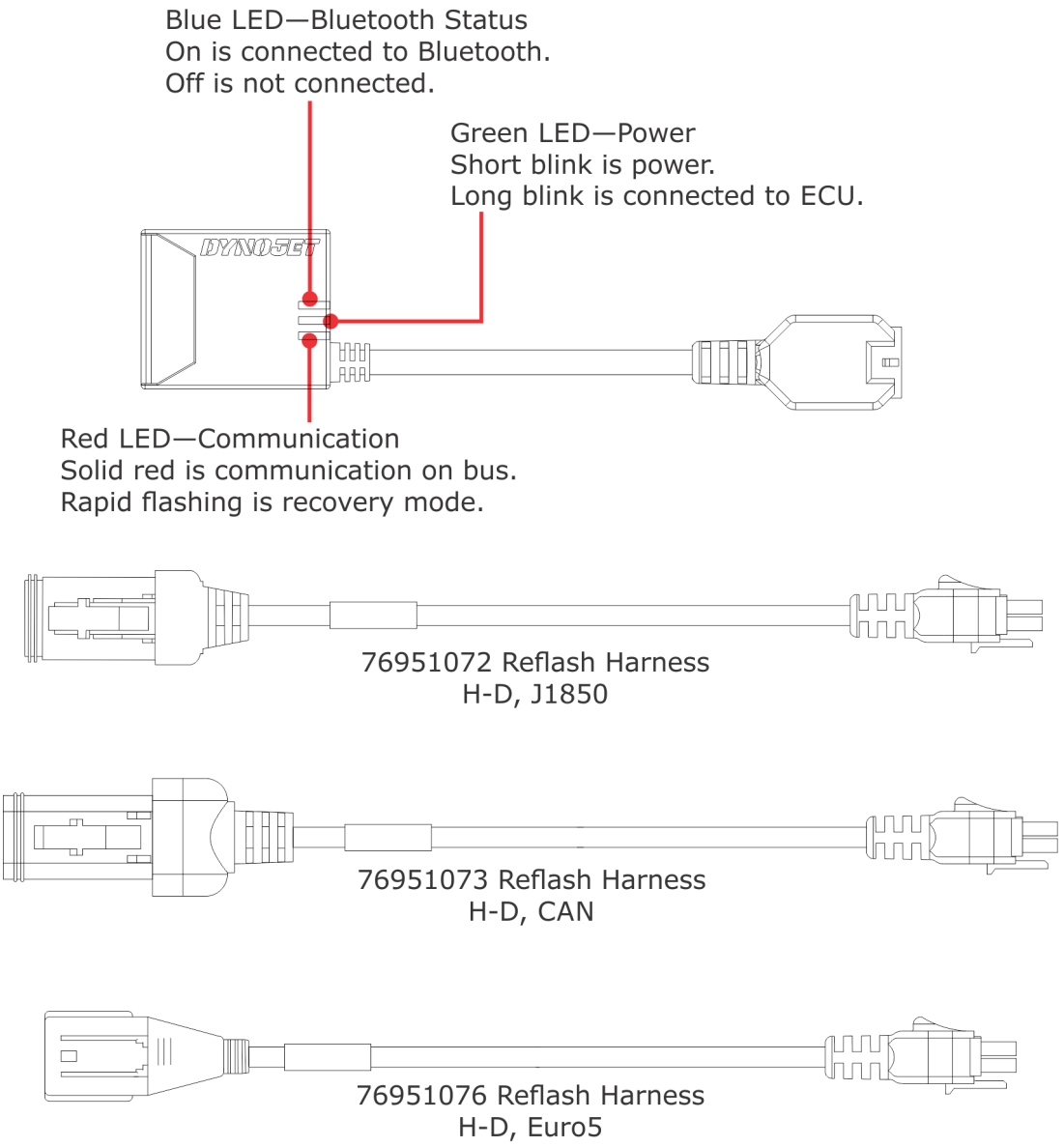


Figure 3-2: Power Vision 4 Overview—Harley-Davidson

Power Vision 4 Overview—Polaris and Indian

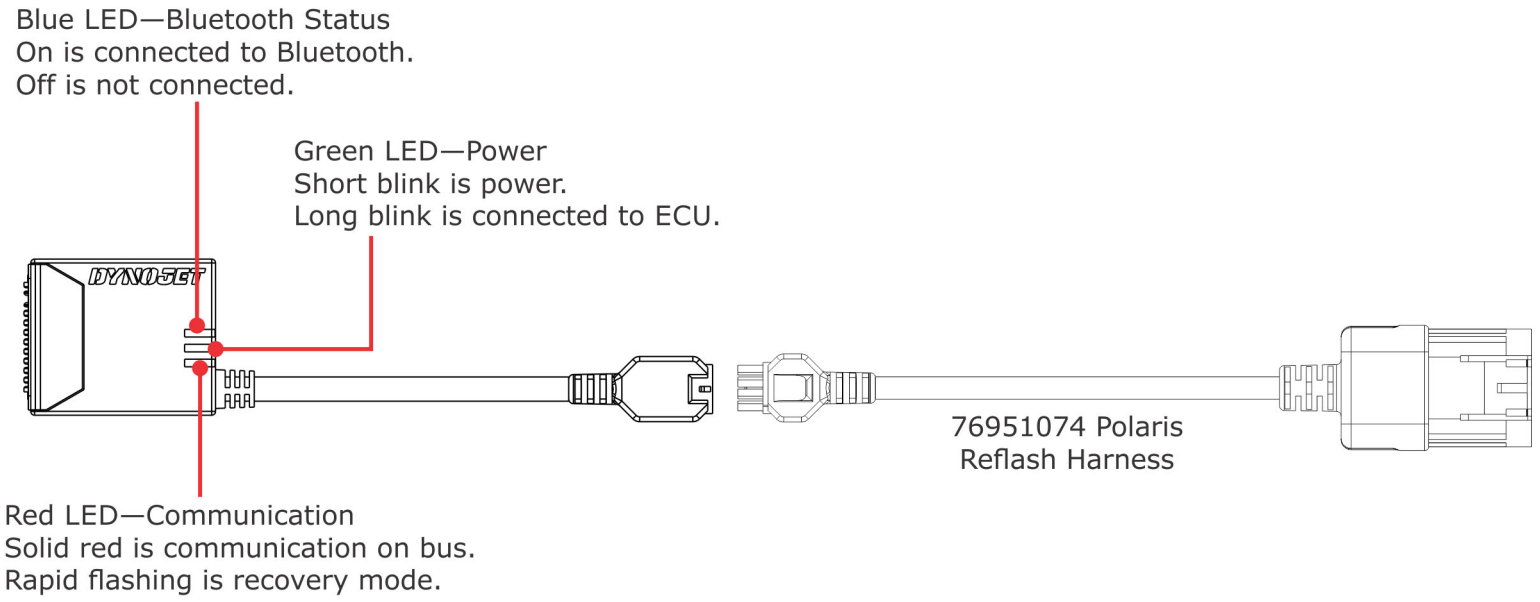


Figure 3-3: Power Vision 4 Overview—Polaris and Indian

POWER VISION 4 OVERVIEW—CAN-AM

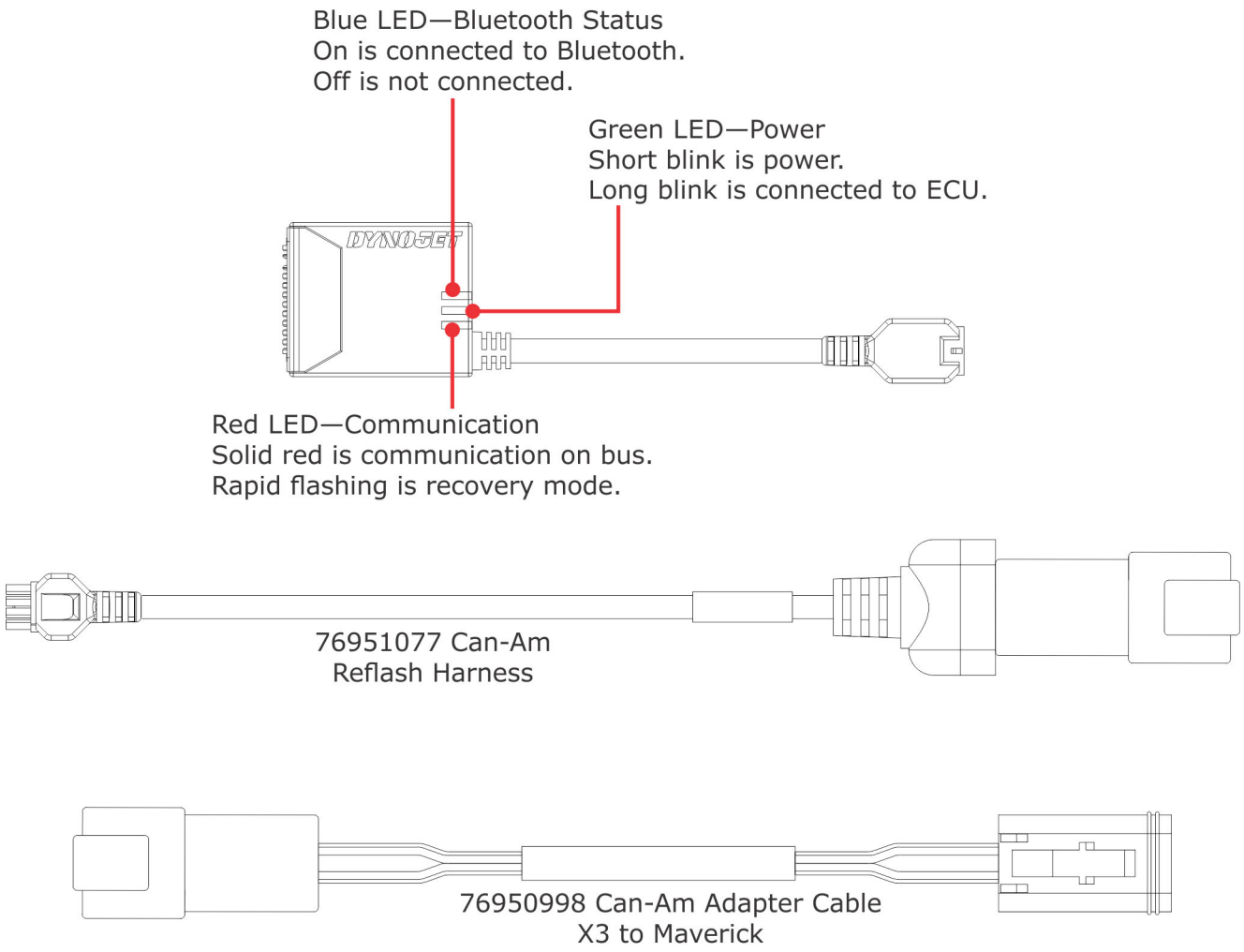


Figure 3-4: Power Vision 4 Overview—Can-Am

GETTING STARTED

GETTING STARTED—GENERAL

Use the following instructions to download the Dynojet PV4 app and connect the app to the vehicle.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

Note: For Gen 2 Harley-Davidson instructions, refer to "Getting Started—Gen 2 Harley-Davidson" on page 3-11.

1 Download and install the Dynojet Power Vision 4 app from the Apple App or Google Play Store.

Figure 3-5: Download the Power Vision 4 App

2 Verify Bluetooth and Location are enabled.

Location permission is required for the app to access the phone's Bluetooth hardware.

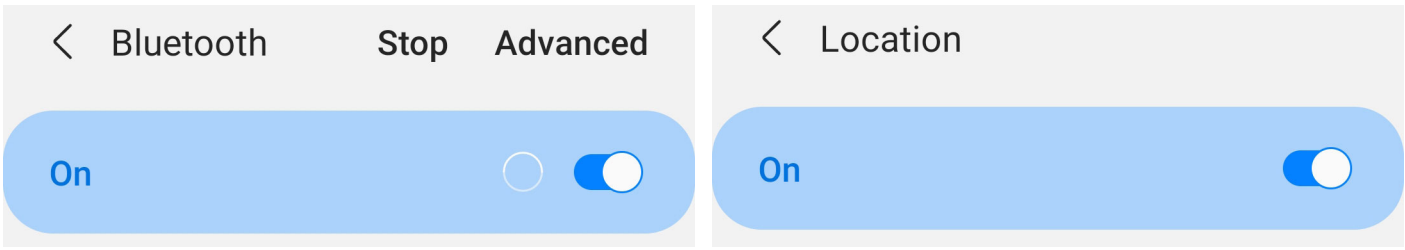


Figure 3-6: Enable Bluetooth and Location

3 Connect the Power Vision 4 to the vehicle diagnostic port.

Refer to "Locating the Diagnostic Connector—Harley-Davidson" on page 2-2.

Refer to "Locating the Diagnostic Connector—Polaris" on page 2-8.

Refer to "Locating the Diagnostic Connector—Indian" on page 2-13.

Refer to "Locating the Diagnostic Connector—Can-Am" on page 2-15.

4 Turn the key on.

5 Open the Dynojet Power Vision 4 app.

6 Log in using your Dynojet account login information.

Don't have a Dynojet account? Press **I don't have an account** and create one. Enter your information and a password and press **Register**.

Forgot your password? Press **Forgot your password?** and follow the on-screen prompts.

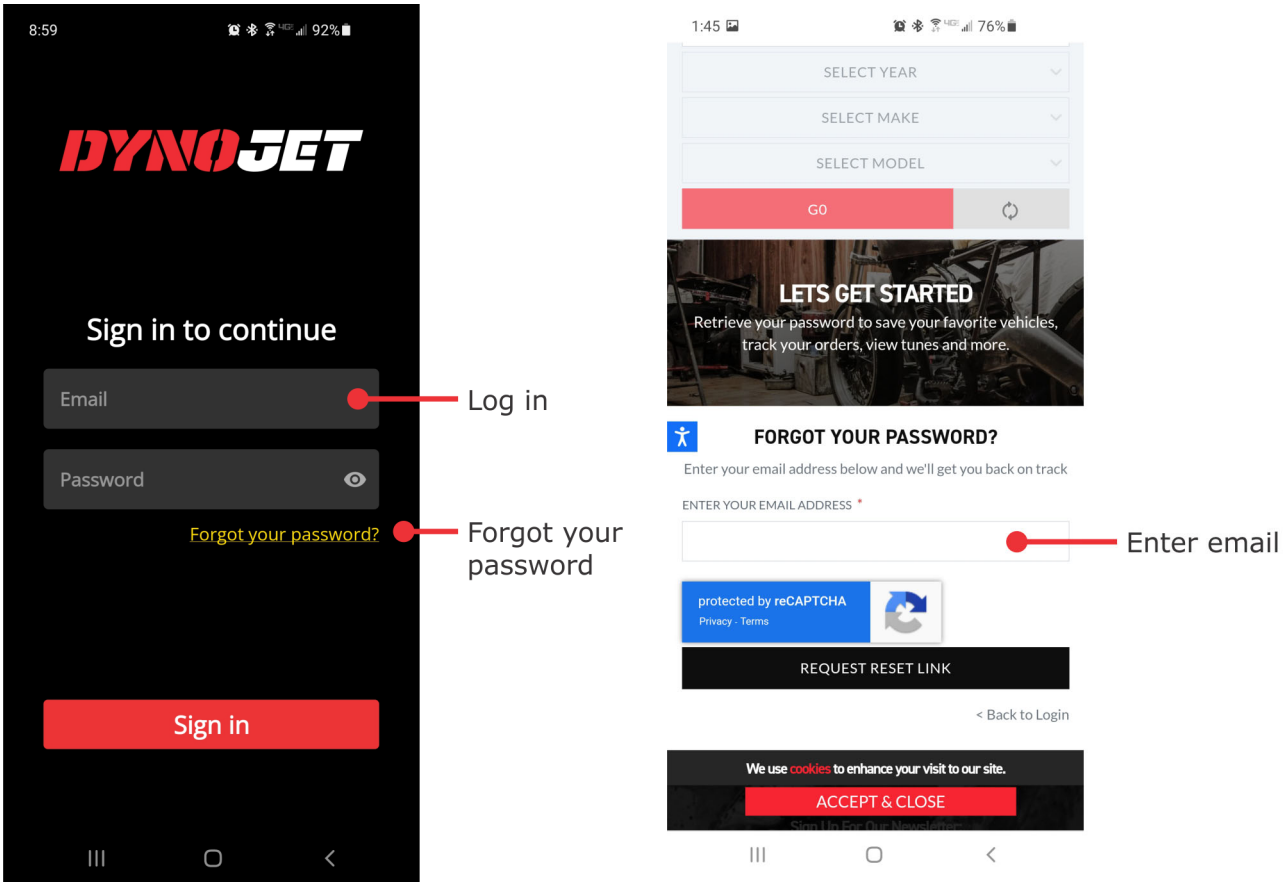


Figure 3-7: Logging In

7 Press **Scan**.
8 Press **Identify** to briefly illuminate the blue LED on your Power Vision 4.
9 Select the Dynojet PV4.

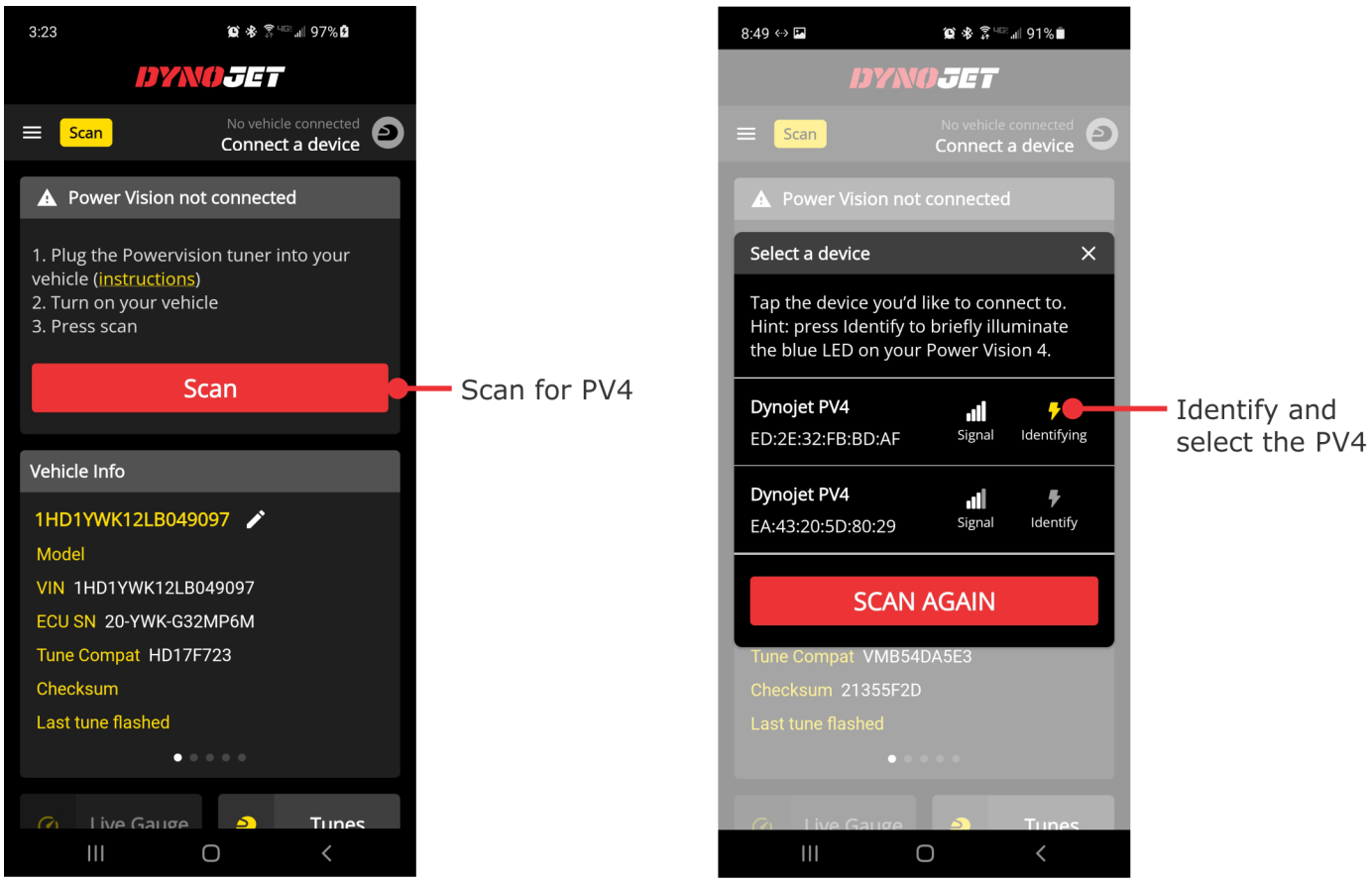


Figure 3-8: Scan For and Select PV4

10 Press the **Pencil** to edit the vehicle name. Press **Save** to save.

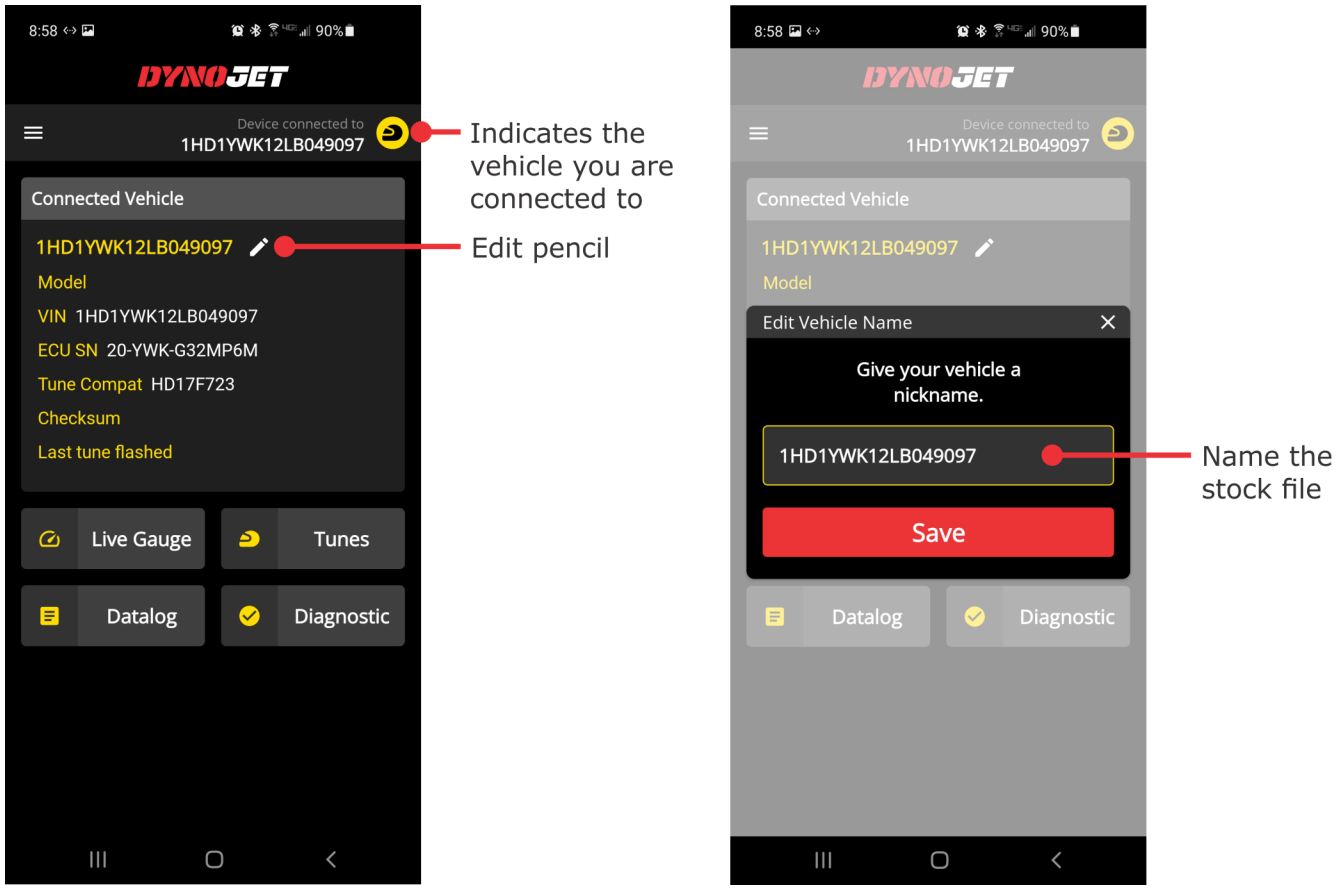


Figure 3-9: Edit Vehicle Name

11 Press **Tunes**.

Note: The first time you connect to a vehicle you will need to Read the ECU. Press **Yes** to continue.

Note: Read ECU not available on Can-Am models.

12 Type in a stock file name. The VIN is used by default.

This file will be uploaded to your Cloud account.

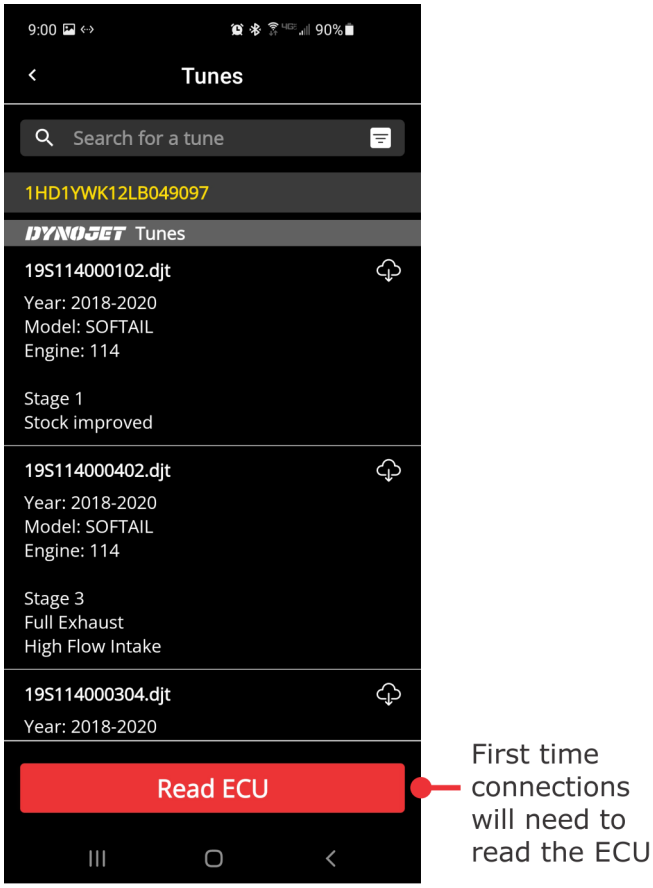


Figure 3-10: Read the ECU

GETTING STARTED—GEN 2 HARLEY-DAVIDSON

Use the following instructions to download the Dynojet PV4 app and connect the app to the vehicle.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Download and install the Dynojet Power Vision 4 app from the Apple App or Google Play Store.

Figure 3-11: Download the Power Vision 4 App

2 Verify Bluetooth and Location are enabled.

Location permission is required for the app to access the phone's Bluetooth hardware.

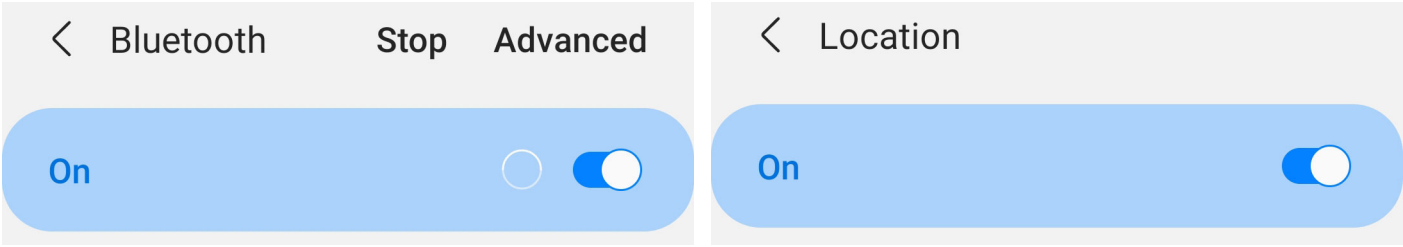


Figure 3-12: Enable Bluetooth and Location

3 Connect the Power Vision 4 to the vehicle diagnostic port.

Refer to "Locating the Diagnostic Connector—Harley-Davidson" on page 2-2.

4 Turn the ignition switch to the ON position and the run switch to the RUN position, but do not start the vehicle.

5 Open the Dynojet Power Vision 4 app.

6 Log in using your Dynojet account login information.

Don't have a Dynojet account? Press **I don't have an account** and create one. Enter your information and a password and press **Register**.

Forgot your password? Press **Forgot your password?** and follow the on-screen prompts.

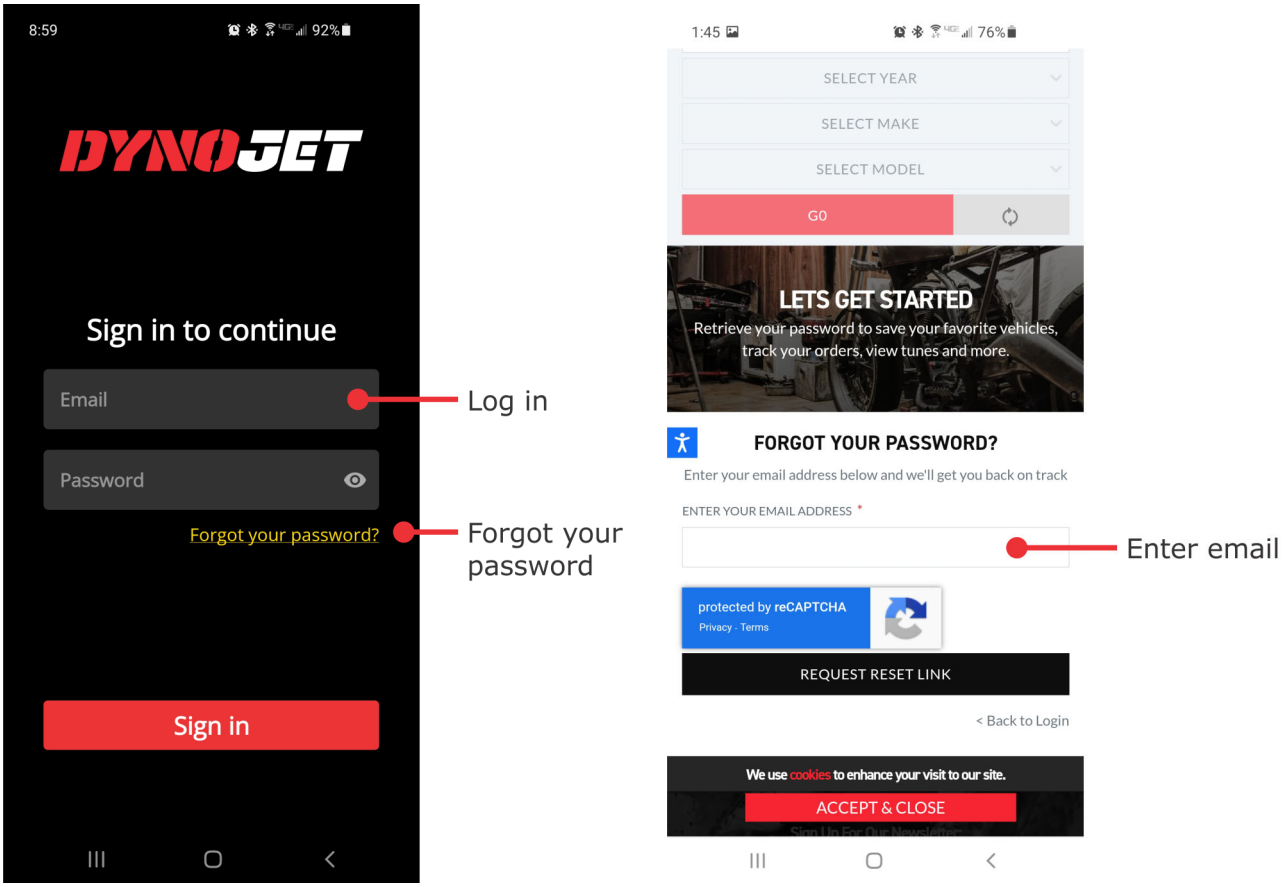


Figure 3-13: Logging In

7 Press **Scan**.
8 Press **Identify** to briefly illuminate the blue LED on your Power Vision 4.
9 Select the Dynojet PV4.

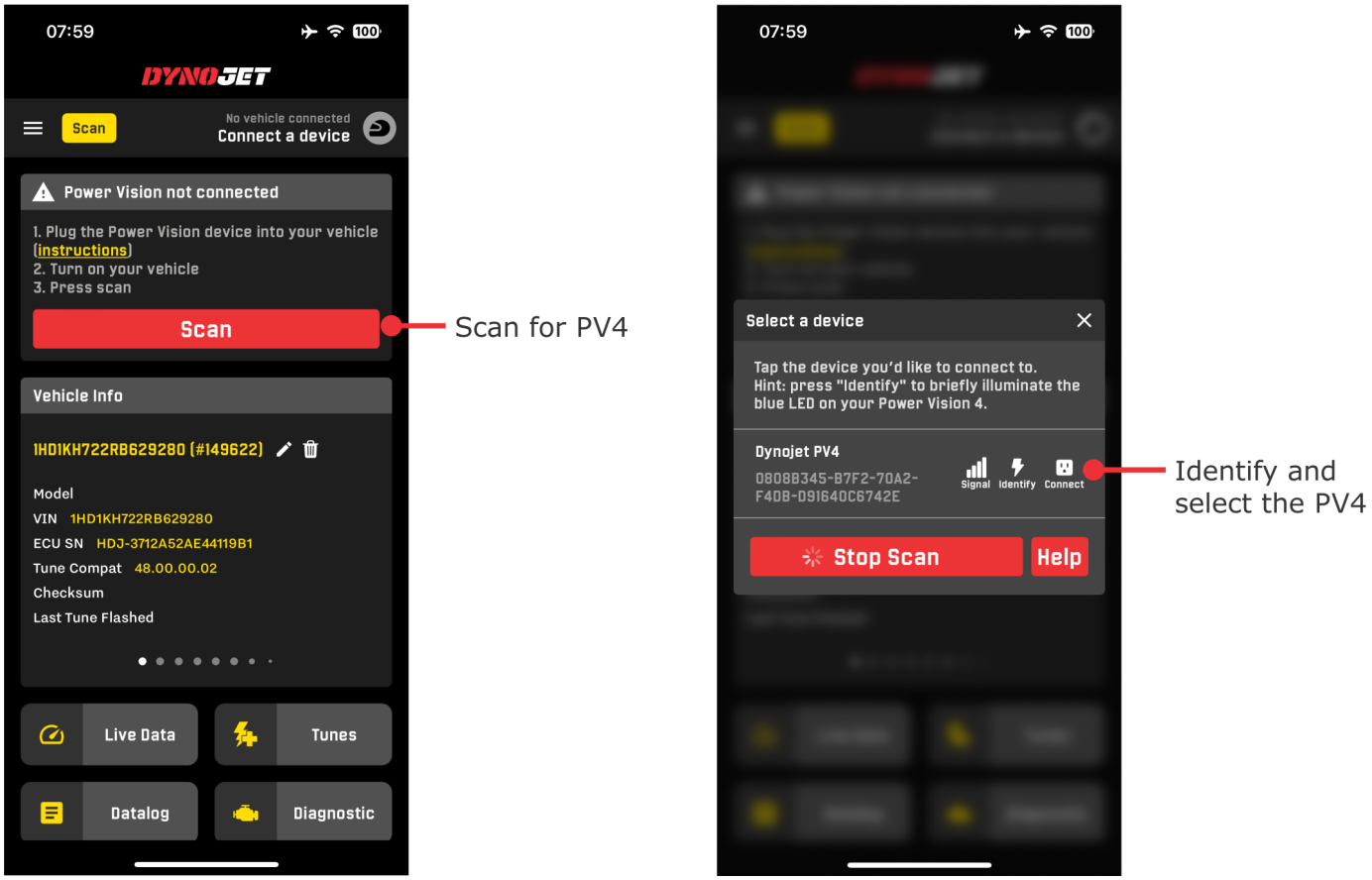


Figure 3-14: Scan For and Select PV4

Note: Not all Power Vision 4 devices support Gen 2 Harley-Davidson ECU's. Use the following instructions to upgrade your Power Vision 4.

9a You may be prompted to purchase a feature license for 10 Universal Tokens to enable flashing of this new ECU using your Power Vision 4.

9b Press **OK**.

9c Press **Purchase for 10.0 universal tokens** to enable this feature license.

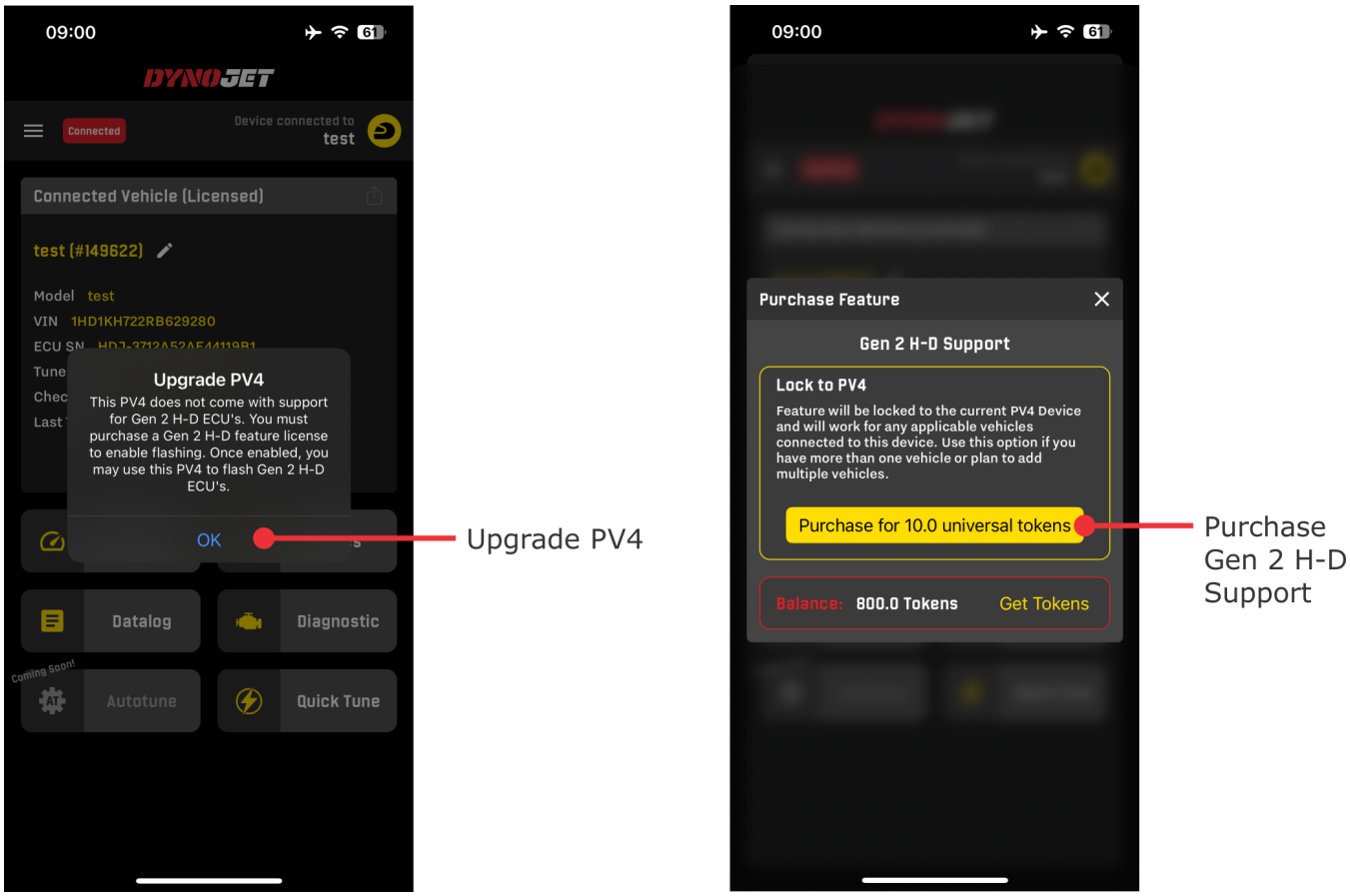


Figure 3-15: Upgrading the Power Vision 4

10 Press the **Pencil** to edit the vehicle name. Press **Save** to save.

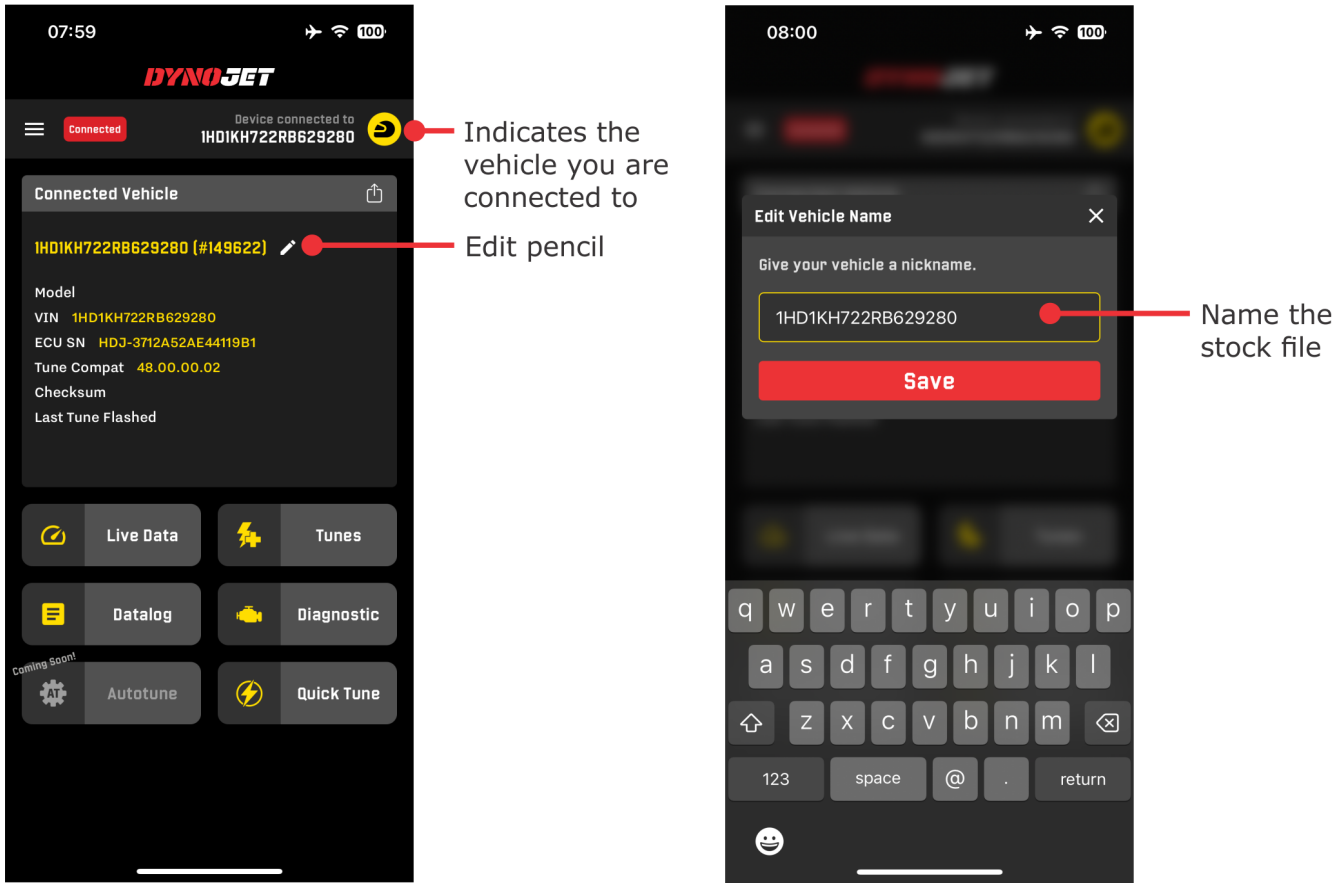


Figure 3-16: Edit Vehicle Name

11 Press **Tunes**.

Note: The first time you connect to a vehicle you will need to perform a one-time ECU Prep Flash.

Note: The battery must be fully charged or on a charger.

12 Press **Yes** to proceed with the installation.

13 Verify your battery is fully charged or connected to a battery charger.

14 Press **Begin Installation**.

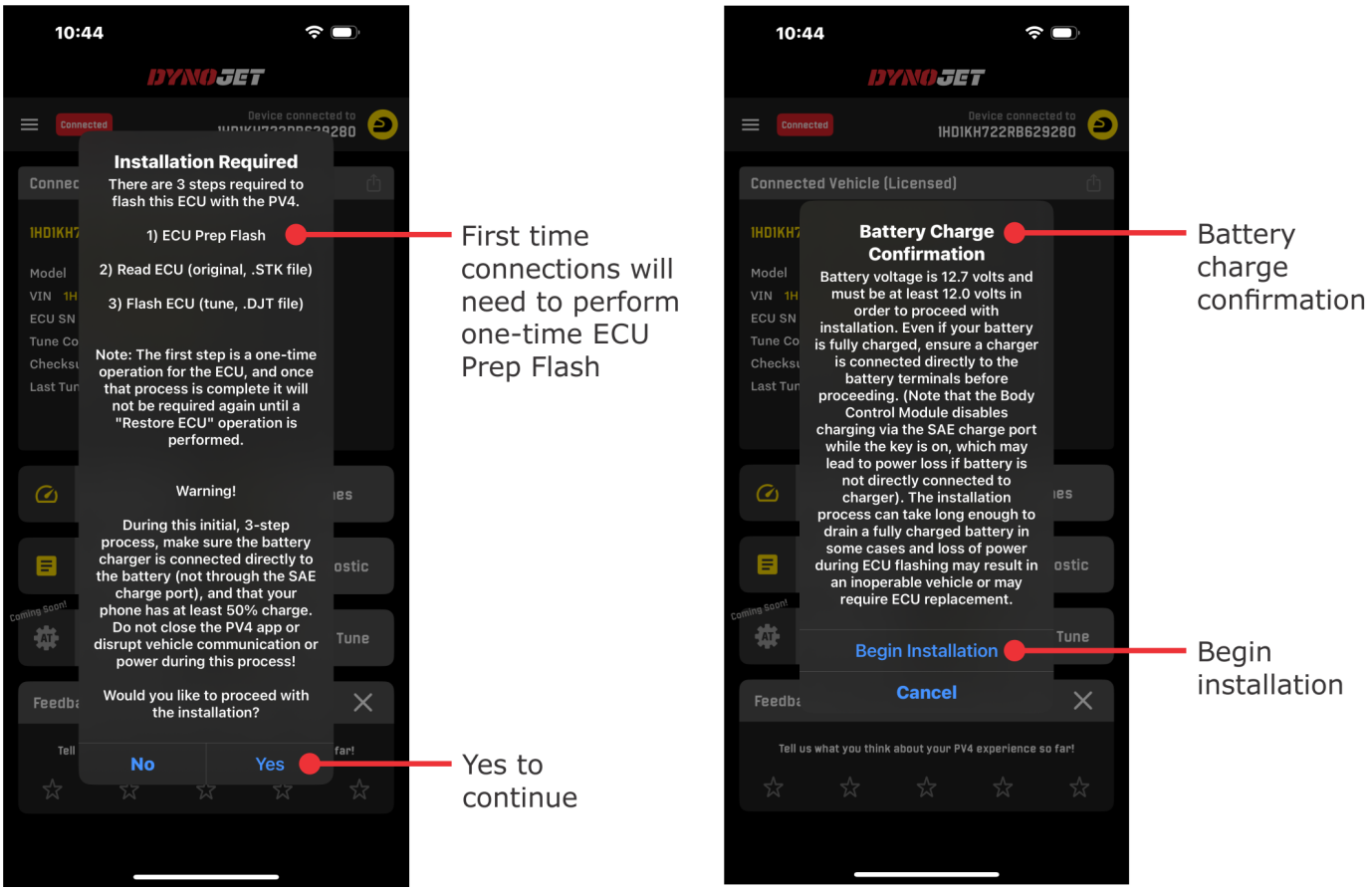


Figure 3-17: One-Time ECU Prep Flash

15 Once the installation is complete, turn the key off or place the run switch in the off position. Once the vehicle is powered off, please wait 30 seconds to reset the ECU.

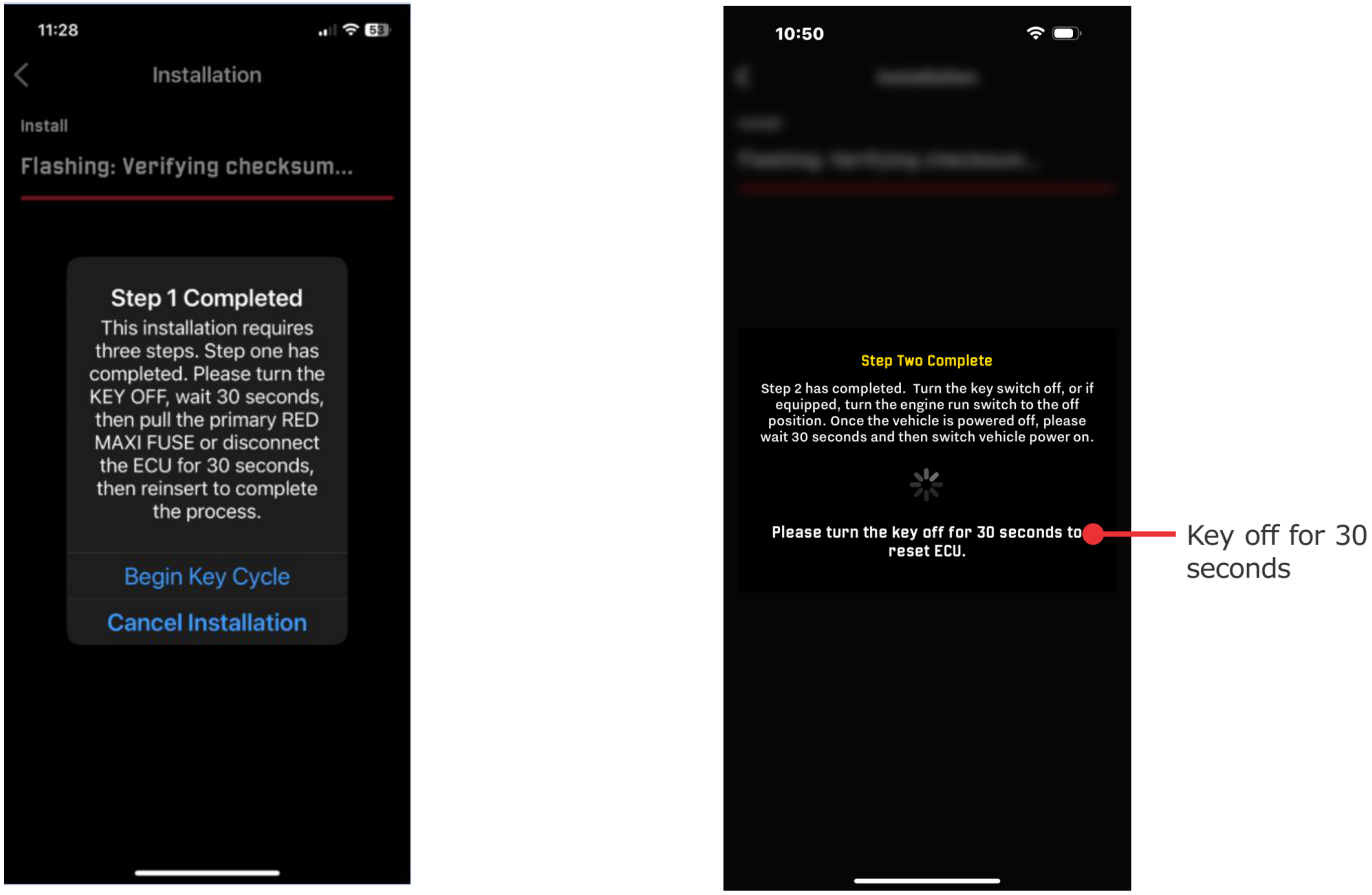


Figure 3-18: Key Off for 30 Seconds

Installation is complete. You may now flash a tune!

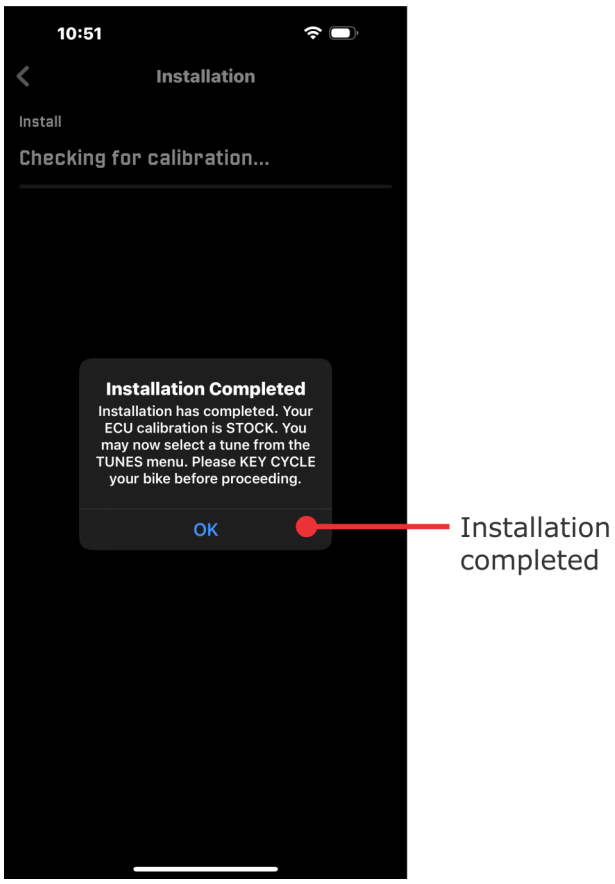


Figure 3-19: Installation Complete

FLASHING YOUR VEHICLE

Your Power Vision 4 comes with one available Flash License. Before you can flash additional vehicles, you will need to purchase additional flash licenses from Dynojet.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Press **Tunes**.

2 Select a tune file (.djt) from the list of available Dynojet Tunes or press **Filter** to filter the list of tunes.

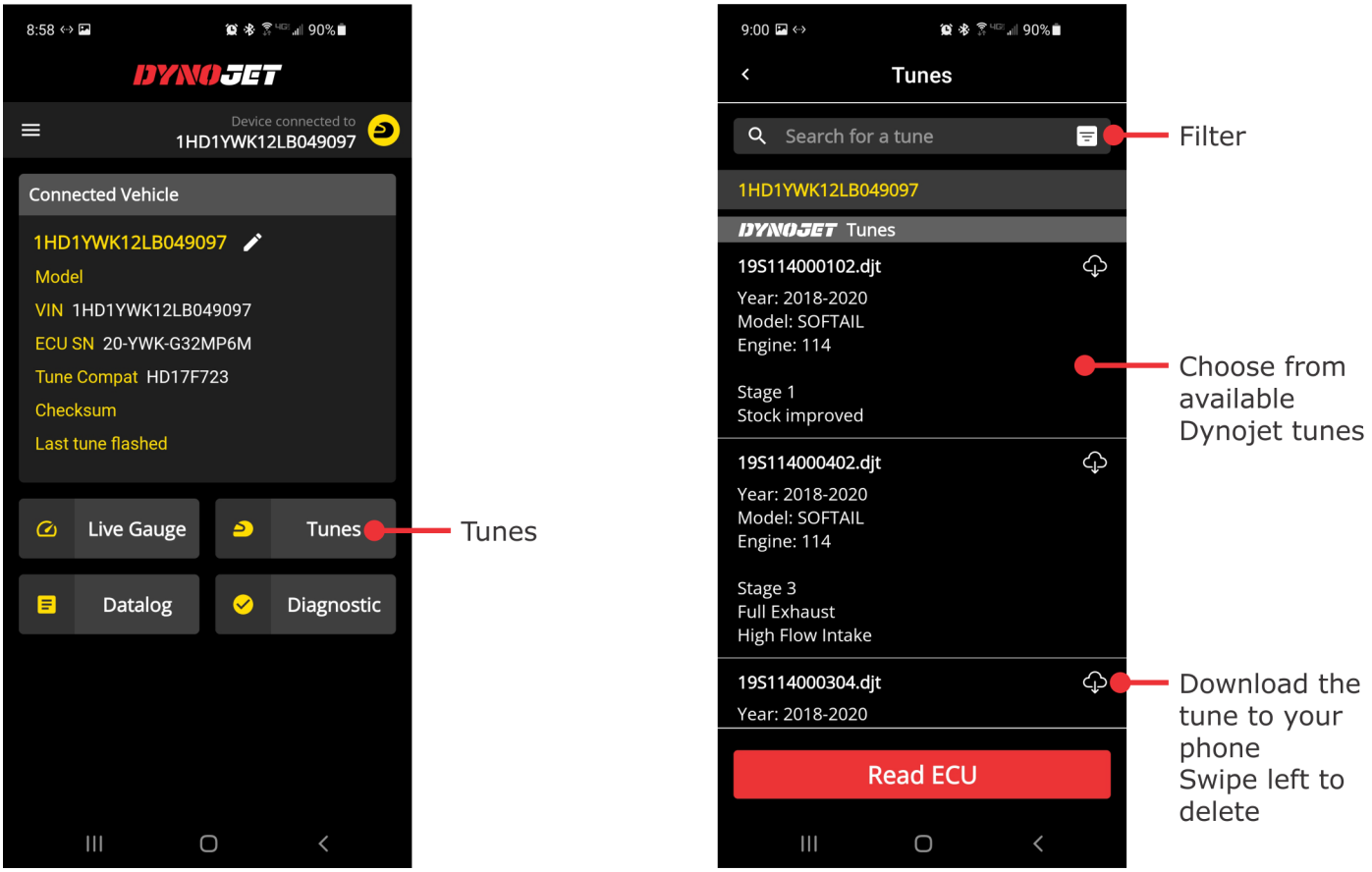


Figure 3-20: Viewing Tunes

3 Tap a Filter field and select from the available options to narrow the results.

4 Once you have selected your filters, slide the filter screen to the left. Select a tune file from the filtered list.

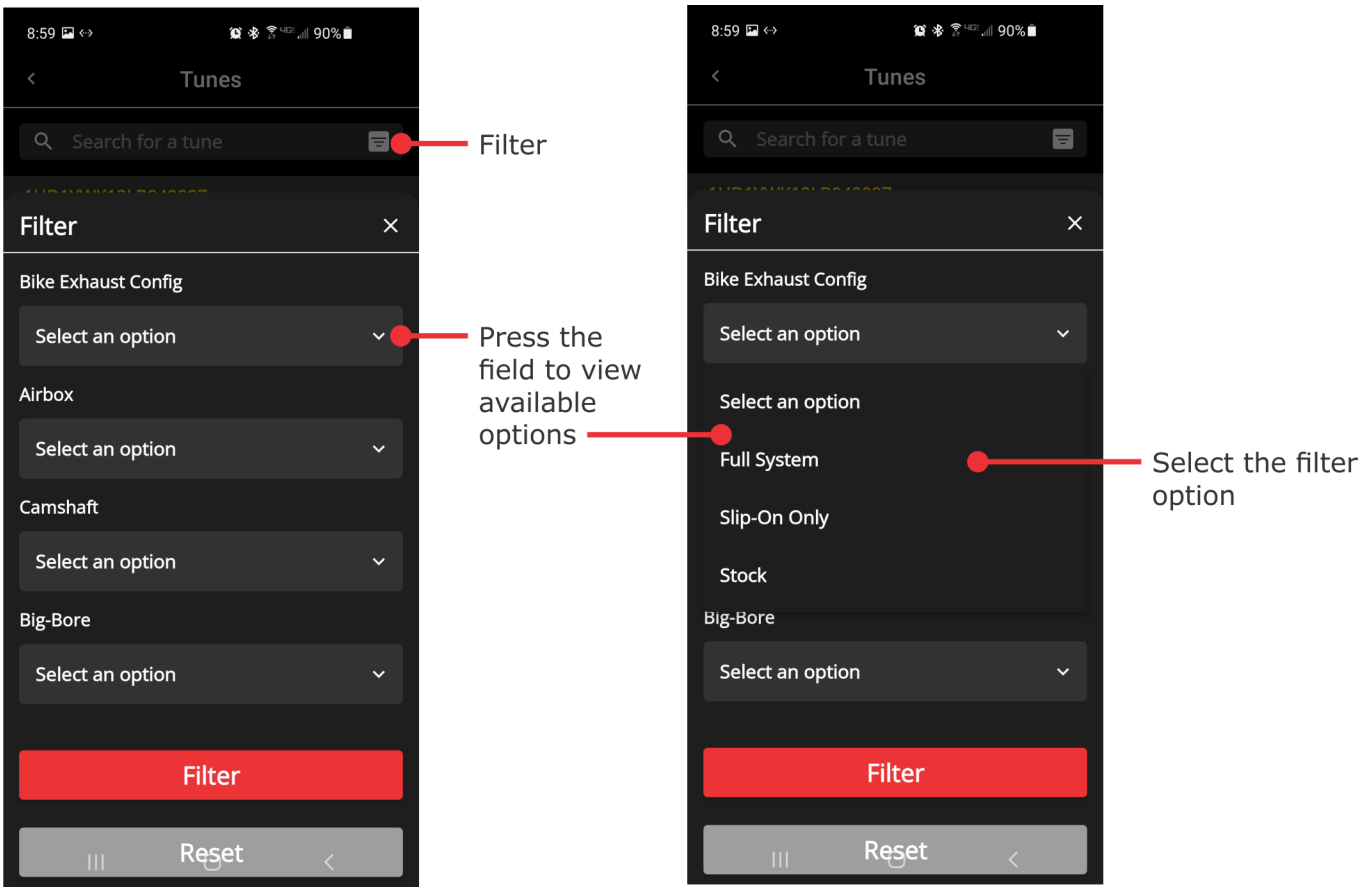


Figure 3-21: Filtering Tunes

5 Press **Flash Tune** to use the included Flash License to lock the PV4 to this vehicle and flash the tune. Do no turn off the key switch or disconnect the Power Vision 4 during the flash.

Note: Battery must be fully charged or connected to a charger.

6 Once the flash is completed, turn the key off for at least ten seconds to reset the ECU.

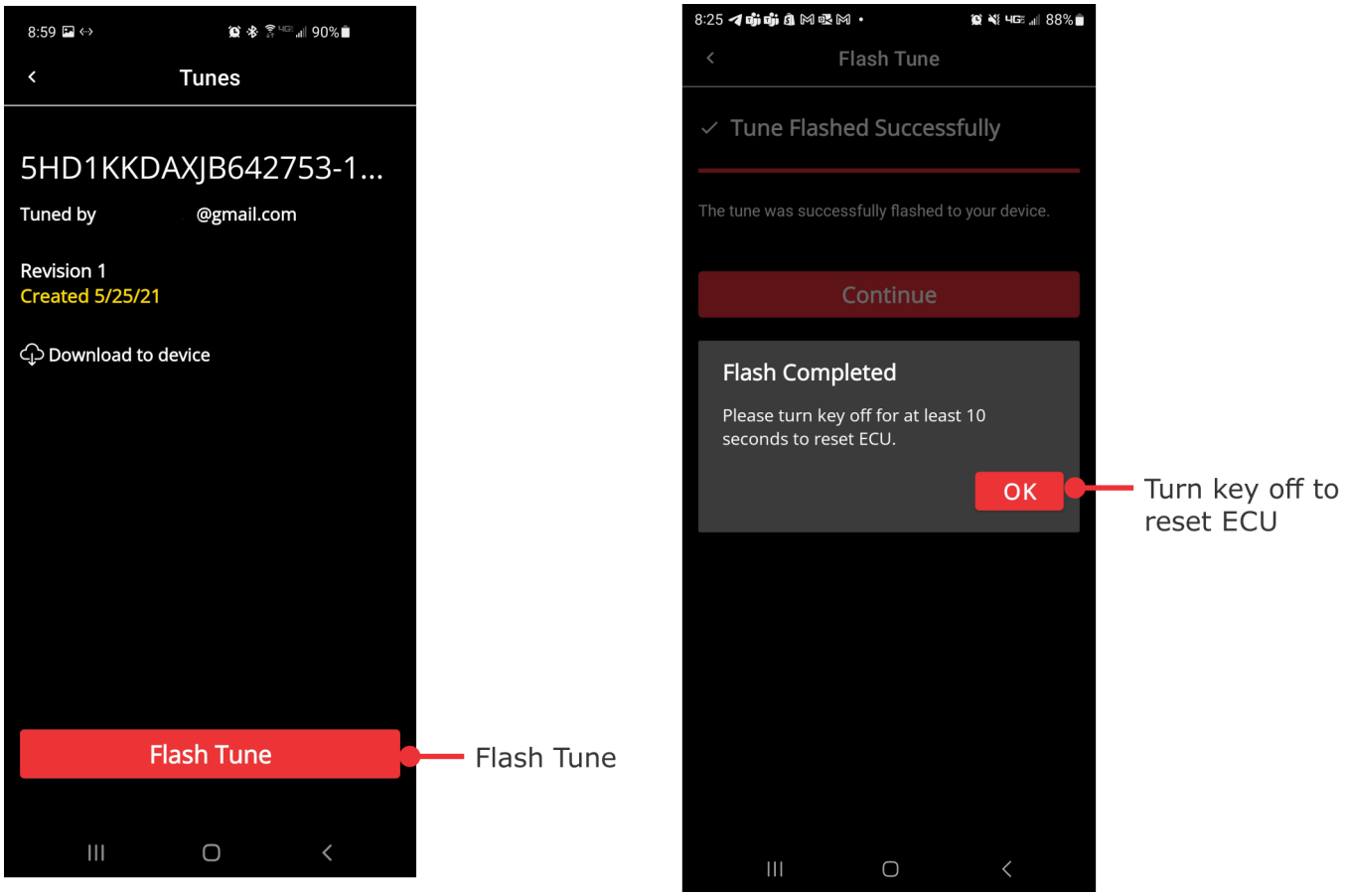


Figure 3-22: Flash Tune

EDITING TUNES WITH POWER CORE

Use Dynojet's Power Core software to edit the ECU tables to improve your riding experience. When you are finished editing, save the tune to the Cloud and use the Dynojet PV4 App to flash your tune to the vehicle.

1 Open the Dynojet Power Core software.
2 Open C3 Tuning.
3 From the Home ribbon, click **Browse Cloud**. Log in using your Dynojet account login information.

Note: If you do not already have an account, you will need to create an account to use the Browse Cloud feature. Click **Create Account** and follow the on-screen instructions.

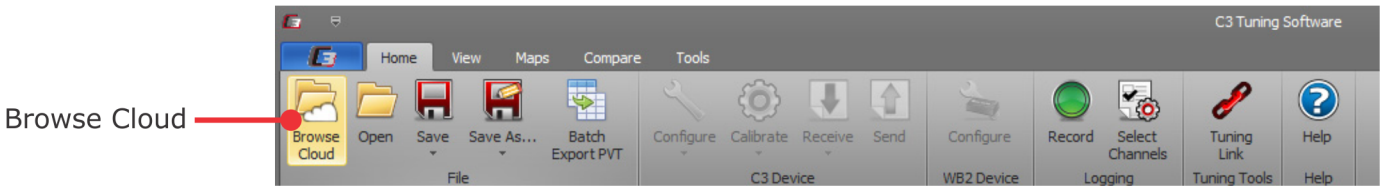


Figure 3-23: Browse Cloud

4 Using the drop-down arrow, select your vehicle.
5 Select the stock file to edit and click **Open**.

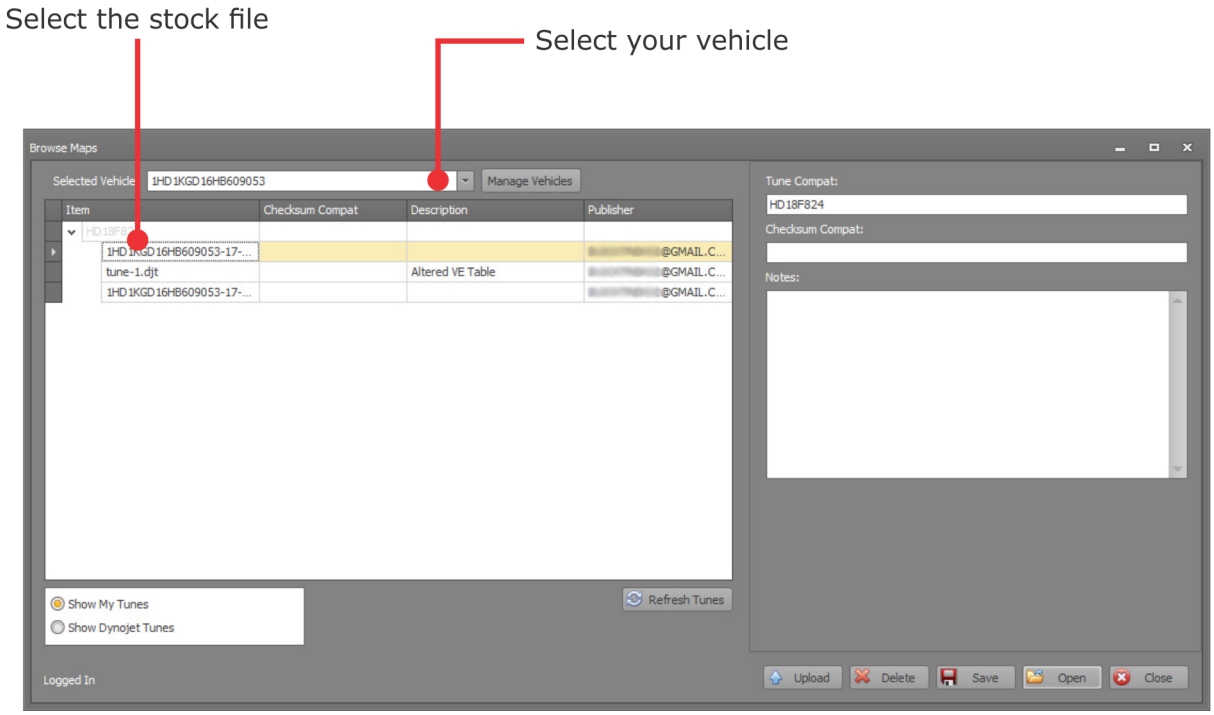


Figure 3-24: Select Stock File

6 Make your desired changes to the table(s).
7 Click **Save As>Save to Cloud as**.
8 Enter a name for your tune and click **Save**.
9 Enter any important tune notes and click **Save**.

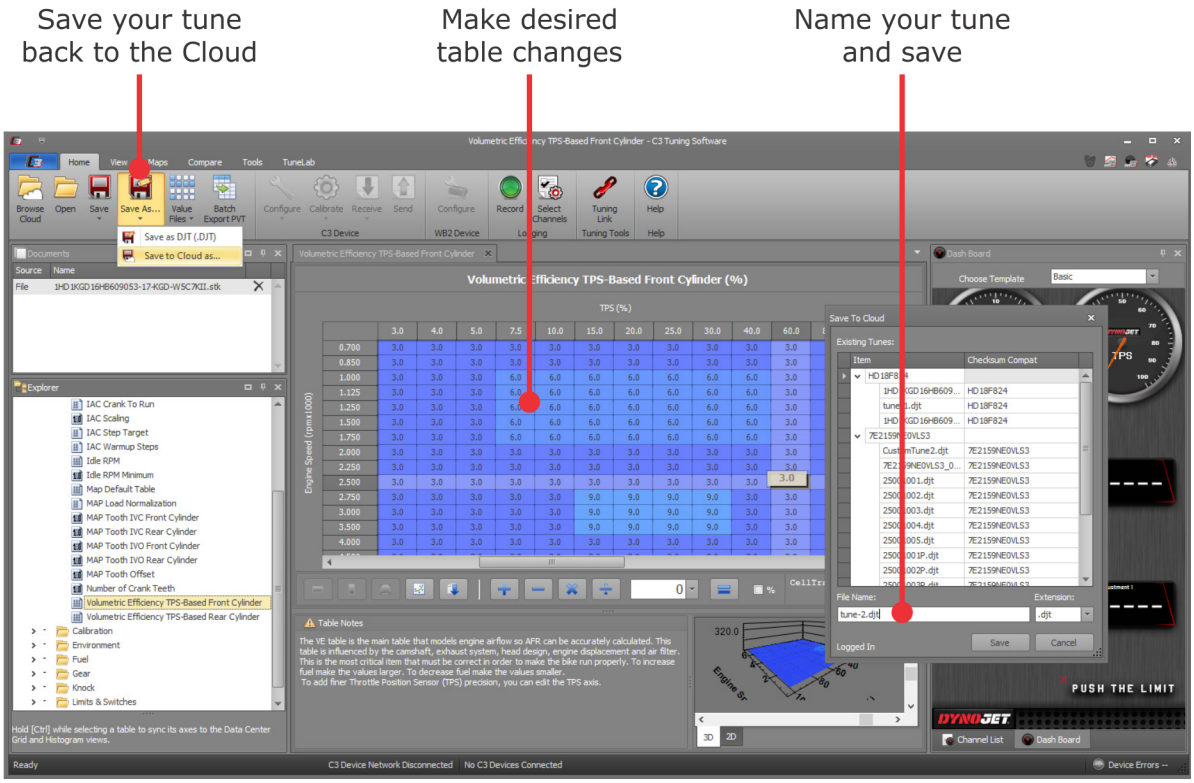


Figure 3-25: Edit and Save Tune

10 Return to your phone and refresh the tunes list in the app.
11 Select your modified tune and press **Flash Tune**.

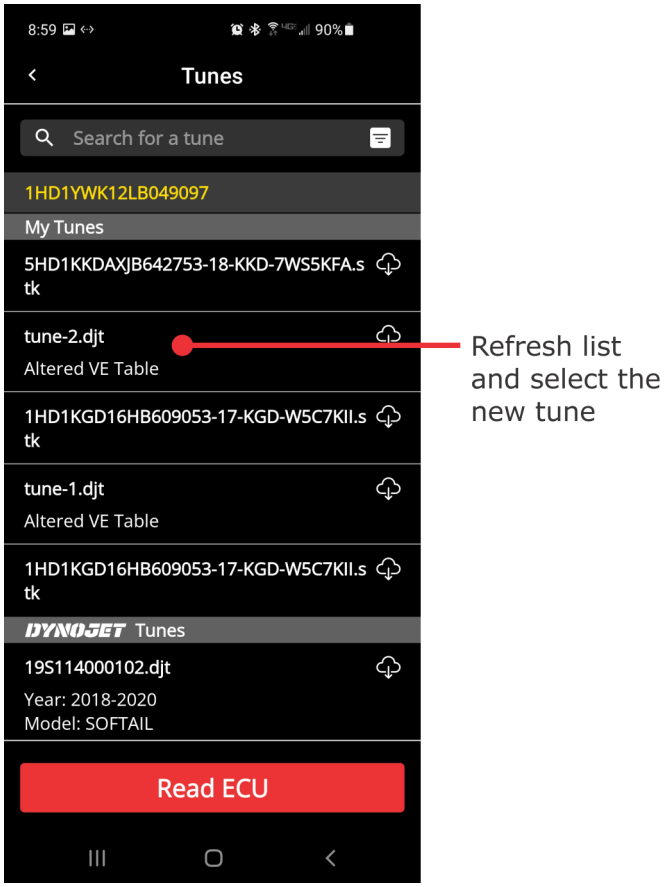


Figure 3-26: Refresh Tunes List

LICENSE MANAGER

View available tokens, redeem and purchase tokens, and download a license to store on your phone for use off line. Your Power Vision 4 comes with one available Flash License. Flash Licenses are transferable. For more detailed information about Dynojet Universal Tokens, refer to https://www.dynojet.com/dynojet-universal-tokens/.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Press **Menu** and press **License Manager**.
2 View your available tokens or press **Purchase Tokens** to purchase tokens.
3 Press **Download** to download and store your tune license on your phone.
4 Press **Create License** to purchase a license for a vehicle.

You will need available tokens to create a license.

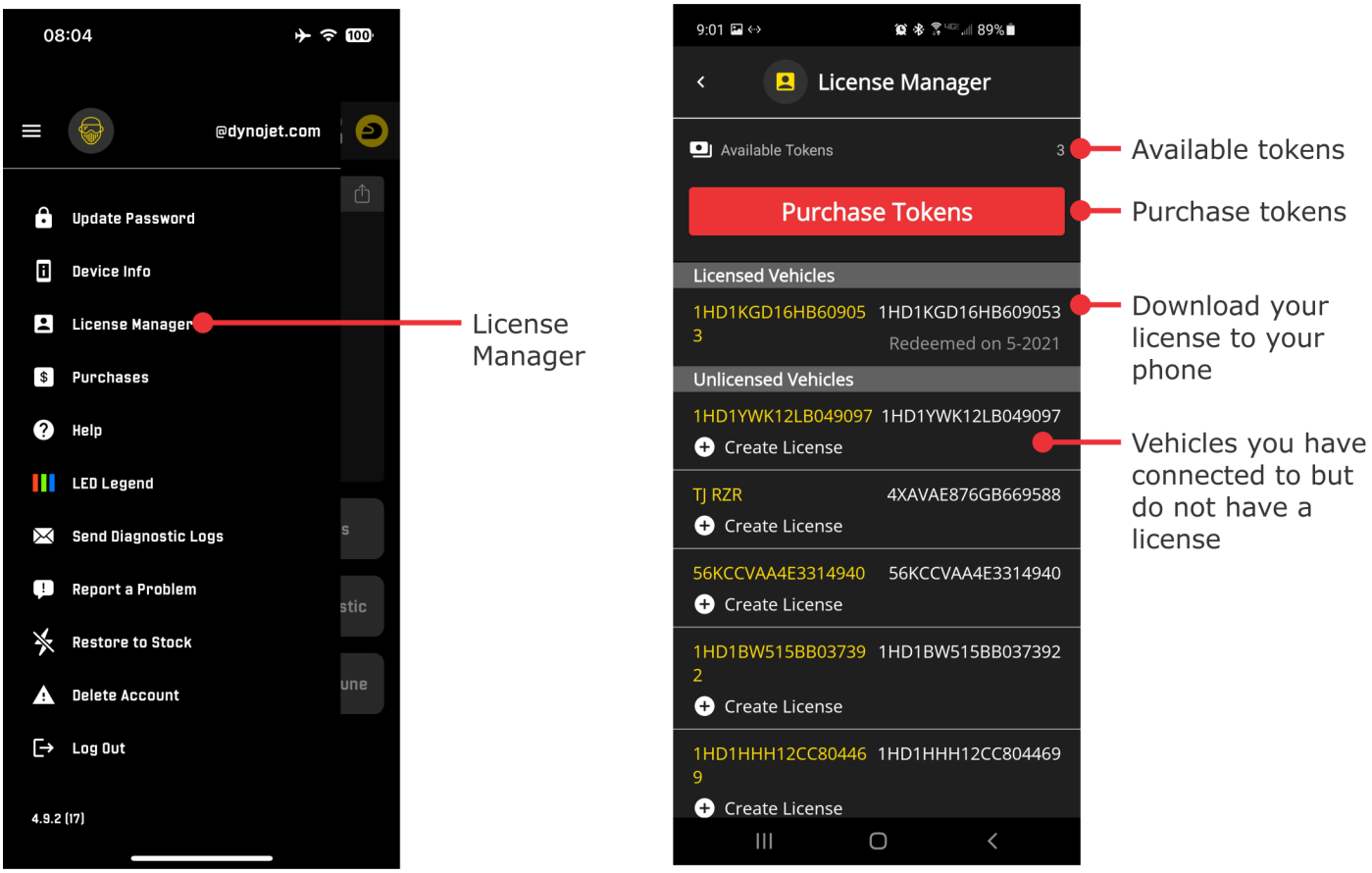


Figure 3-27: Purchase Tokens

5 Press **Redeem Universal Tokens** to redeem your tokens for a license.

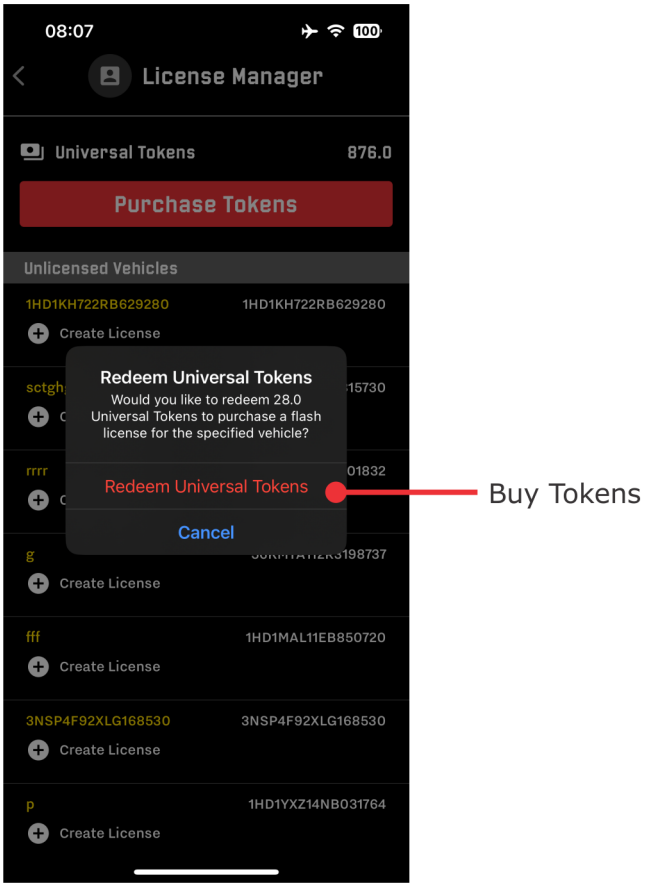


Figure 3-28: Redeem Universal Tokens

6 If you receive an error when flashing a tune, you may need to purchase tokens before you can flash your tune. Press **Menu** and press **License Manager**
7 Press **Purchase Tokens** to purchase tokens.
8 Use the **+** button to increase how many tokens then press **Buy Tokens**.
9 Verify and press **Purchase** to complete your purchase. Once your purchase is complete, your tokens will be shown under Available Tokens.

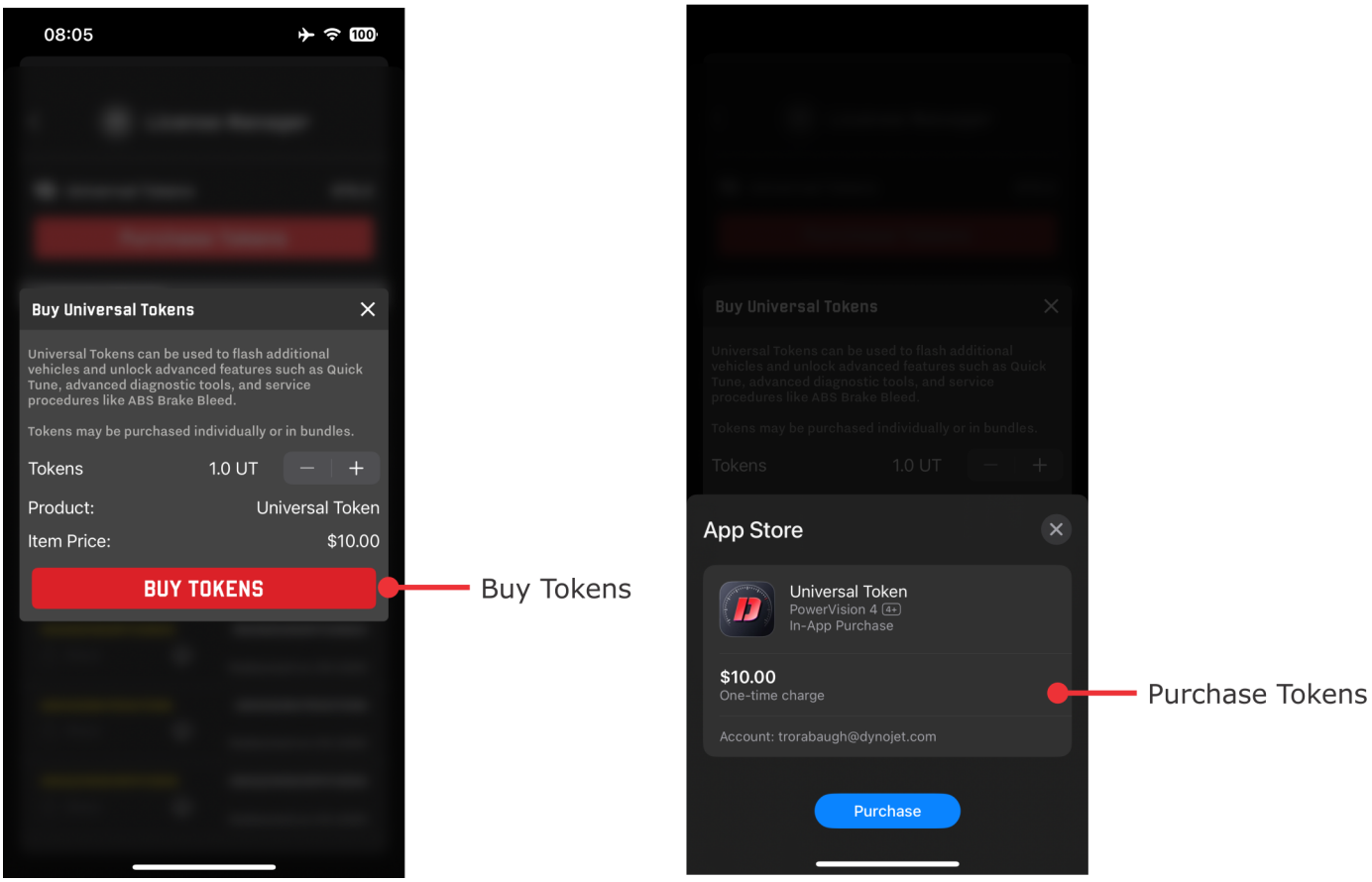


Figure 3-29: Buy Universal Tokens

VIEWING LIVE GAUGES

Your Power Vision 4 allows you to configure gauges for real time monitoring. The Power Vision must be connected to the vehicle to view live gauges.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Connect the Power Vision to the vehicle diagnostic port.
2 Start the vehicle.
3 Press **Live Gauge**.
4 Press the **Record** button to record and save a log file. Refer to "Recording the Gauges" on page 3-37.
5 Swipe the screen left or right to create and view additional gauge screens.
6 Press **+** to add a gauge.

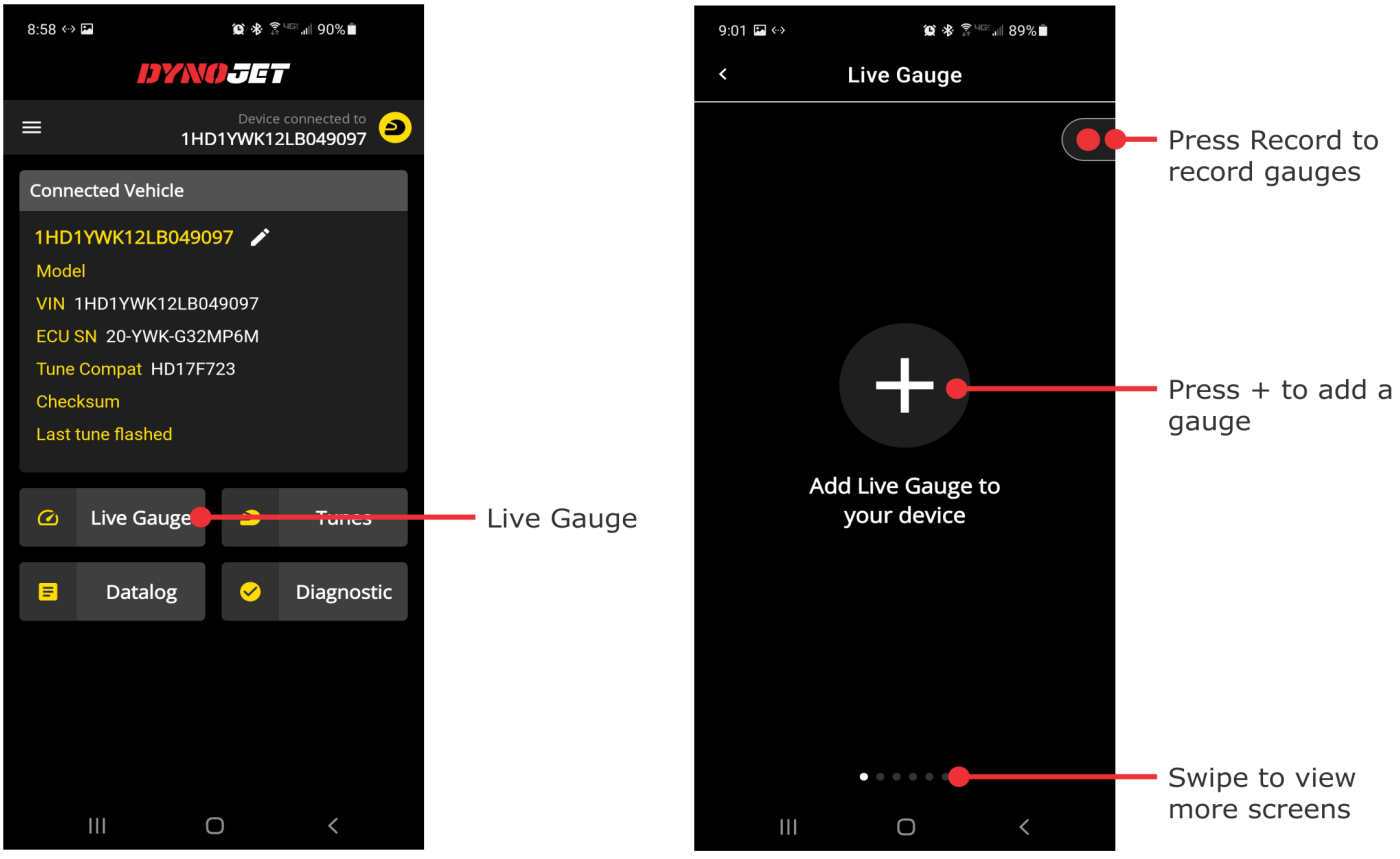


Figure 3-30: Adding a Live Gauge

ADDING A CIRCULAR GAUGE

Gauge options will vary depending on which Data value is chosen.

1. Press the **Data** drop-down arrow to view a list of the available options for that field.
2. Press the **Gauge Type** drop-down arrow and select circular.
3. Press the **Min Value** and **Max Value** and use the number pad to enter your values.
4. Press the **Number of ticks** drop-down arrow to view a list of the available options for that field.
5. Press the **Multiplier** drop-down arrow to view a list of the available options for that field.
6. Press the **Precision** drop-down arrow to view a list of the available options for that field.
7. Use the slider buttons to choose a color range.
8. Once you have completed your selections, press **Add Gauge**.

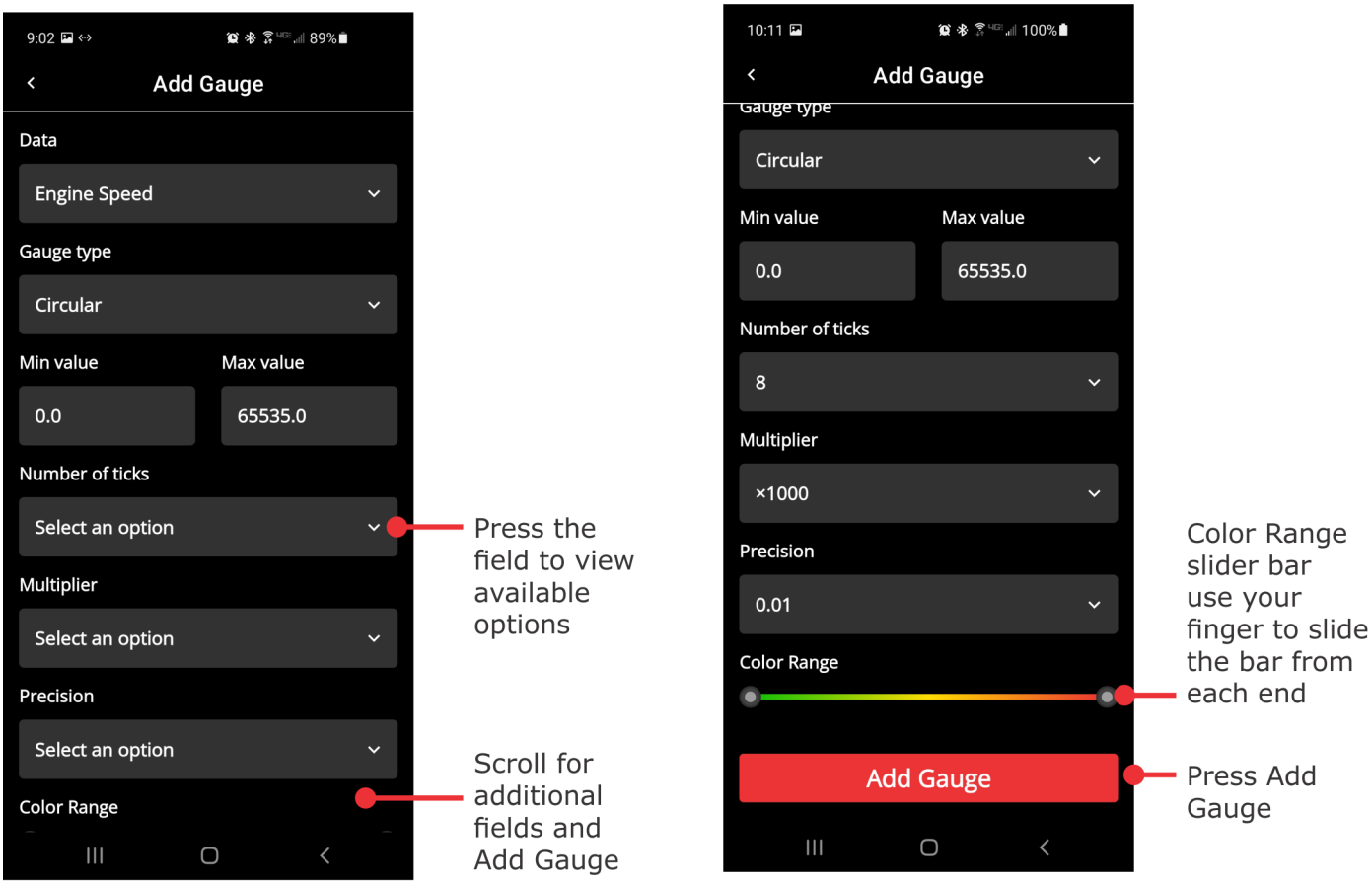


Figure 3-31: Add a Circular Gauge

9 Press **Edit** to edit or remove the gauge.

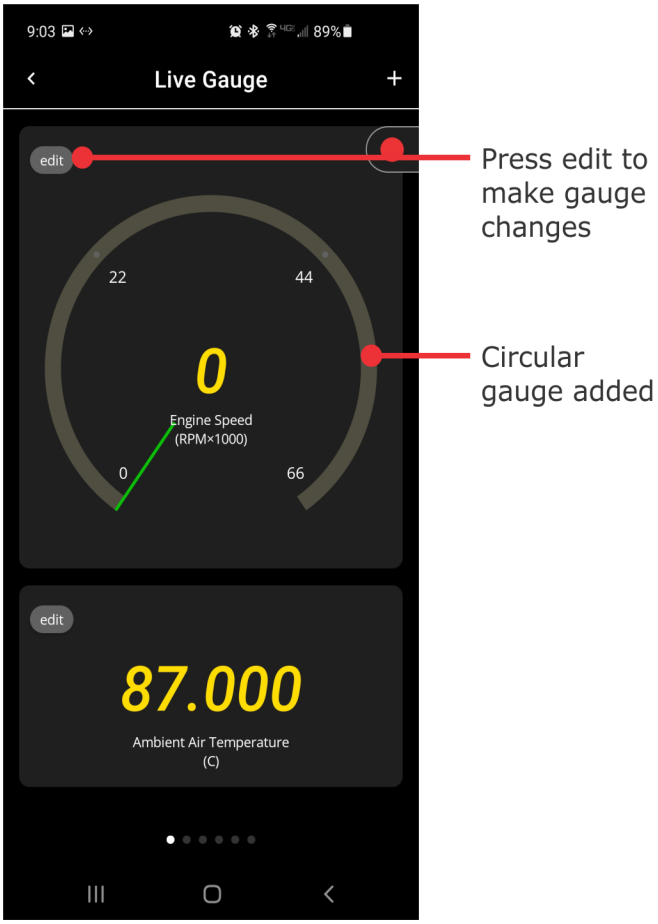


Figure 3-32: Edit the Circular Gauge

ADDING A DIGITAL GAUGE

Gauge options will vary depending on which Data value is chosen.

1 Press the **Data** drop-down arrow to view a list of the available options for that field.
2 Press the **Gauge Type** drop-down arrow and select digital.
3 Press the **Min Value** and **Max Value** and use the number pad to enter your values.
4 Press the **Multiplier** drop-down arrow to view a list of the available options for that field.
5 Press the **Precision** drop-down arrow to view a list of the available options for that field.
6 Once you have completed your selections, press **Add Gauge**.

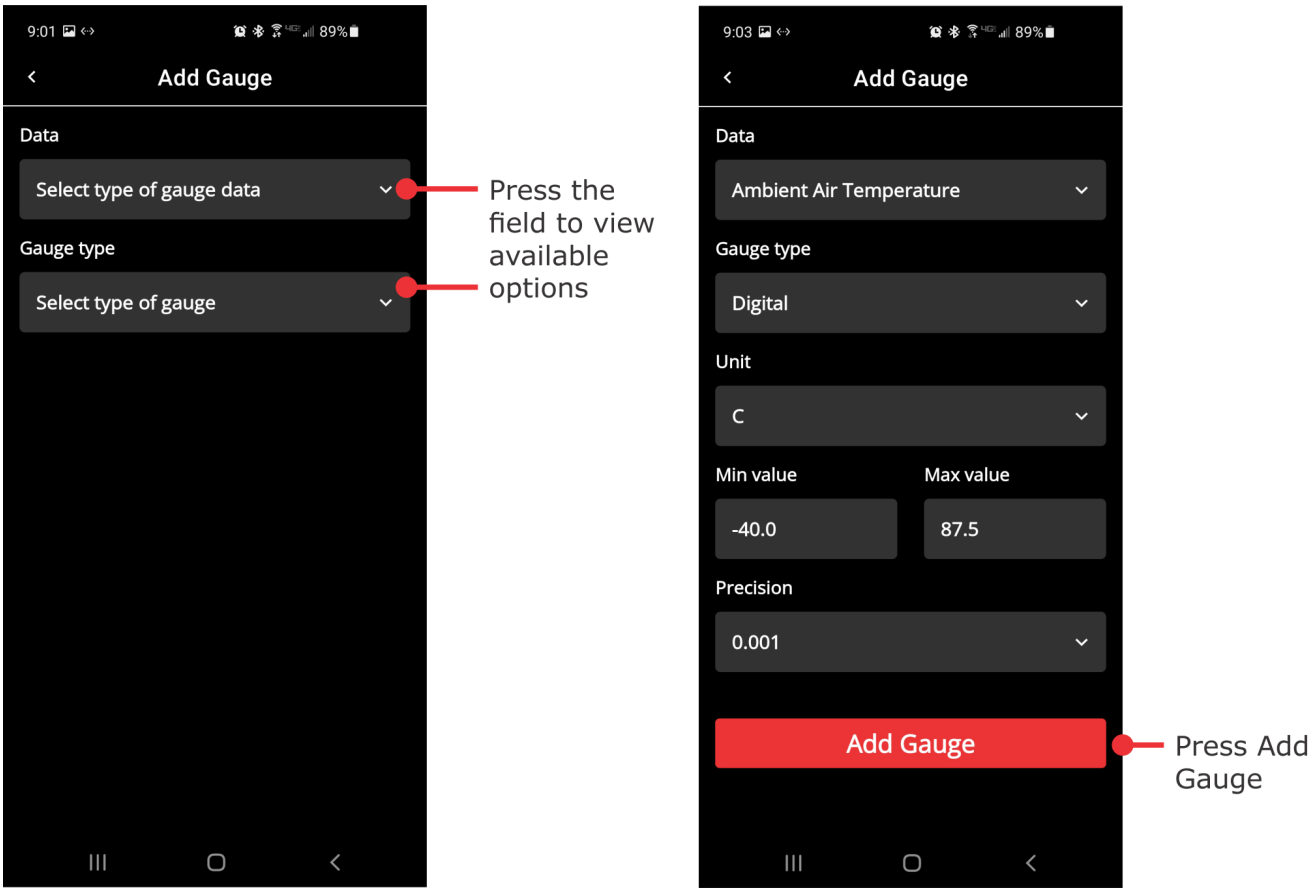


Figure 3-33: Add a Digital Gauge

7 Press **Edit** to edit or remove the gauge.

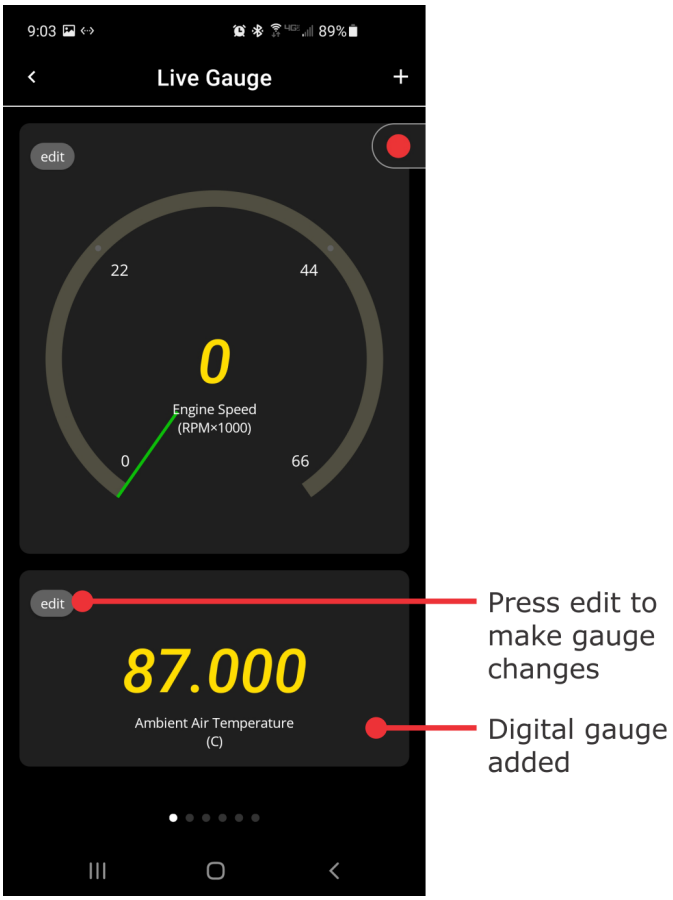


Figure 3-34: Edit the Digital Gauge

ADDING A GRAPH GAUGE

Gauge options will vary depending on which Data value is chosen.

1 Press the **Data** drop-down arrow to view a list of the available options for that field.
2 Press the **Gauge Type** drop-down arrow and select graph.
3 Press the **Min Value** and **Max Value** and use the number pad to enter your values.
4 Press the **Number of ticks** drop-down arrow to view a list of the available options for that field.
5 Press the **Multiplier** drop-down arrow to view a list of the available options for that field.
6 Use the slider buttons to choose a color range.
7 Once you have completed your selections, press **Add Gauge.**

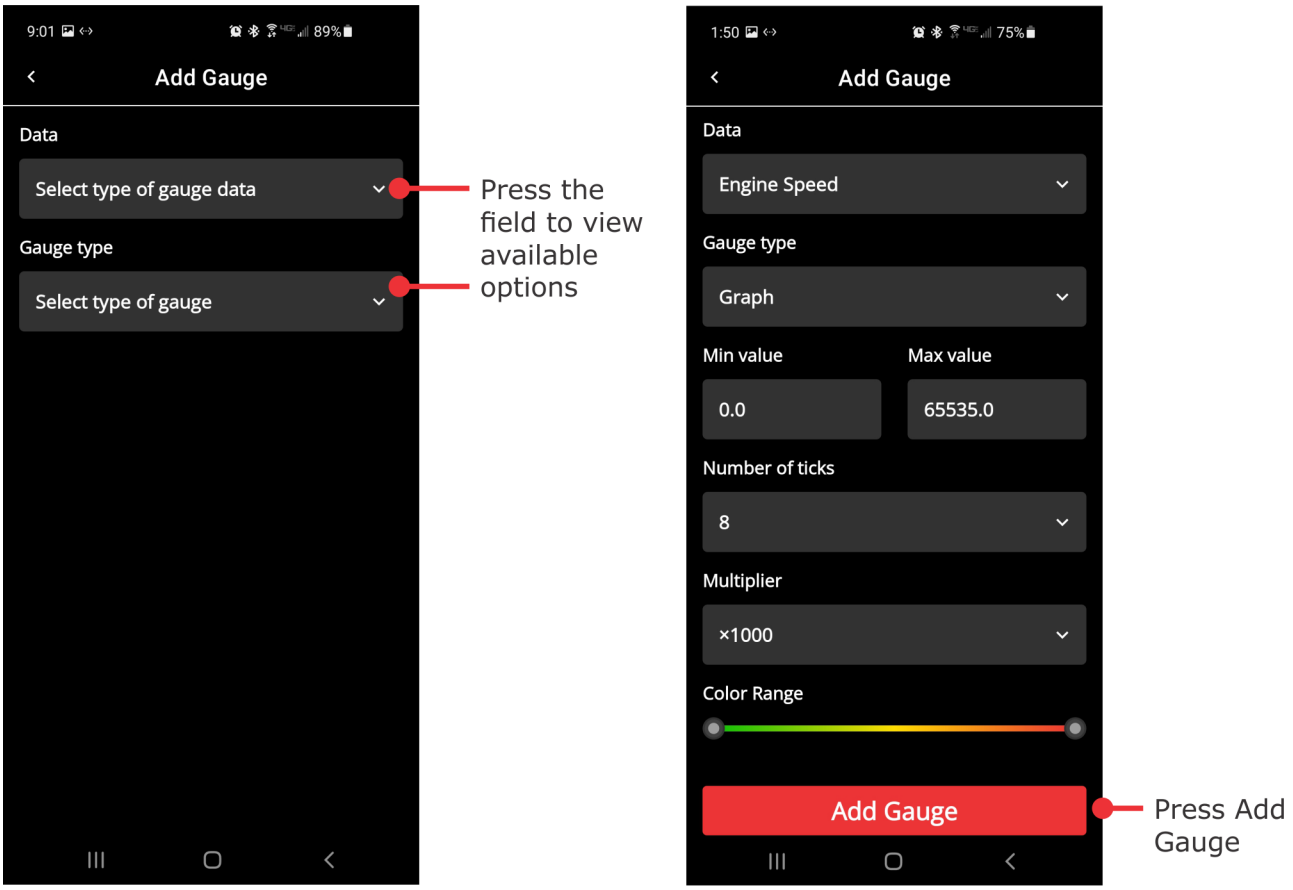


Figure 3-35: Add a Graph Gauge

8 Press **Edit** to edit or remove the gauge.

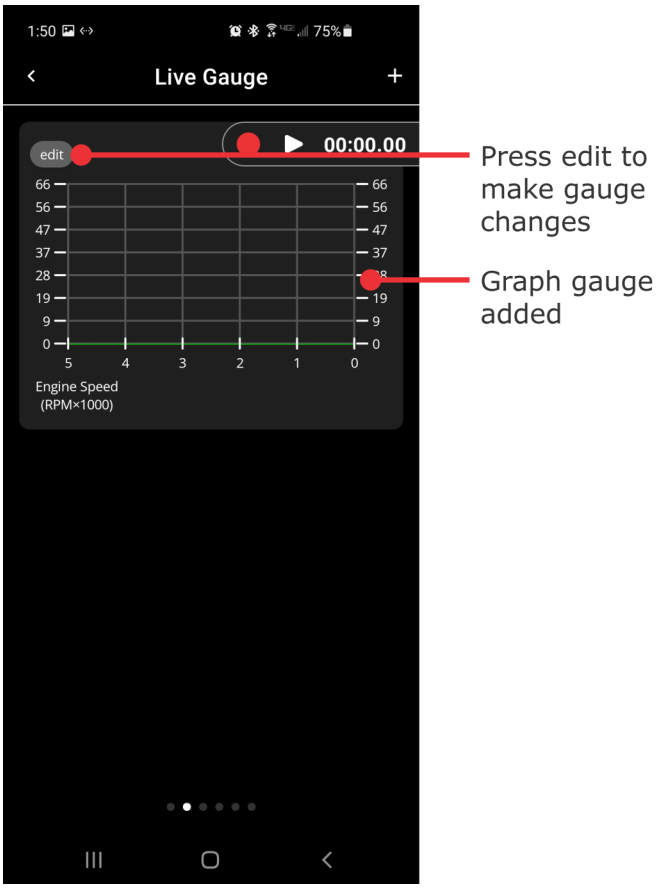


Figure 3-36: Edit the Graph Gauge

ADDING A VERTICAL GAUGE

Gauge options will vary depending on which Data value is chosen.

1 Press the **Data** drop-down arrow to view a list of the available options for that field.
2 Press the **Gauge Type** drop-down arrow and select vertical.
3 Press the **Min Value** and **Max Value** and use the number pad to enter your values.
4 Press the **Number of ticks** drop-down arrow to view a list of the available options for that field.
5 Press the **Multiplier** drop-down arrow to view a list of the available options for that field.
6 Press the **Precision** drop-down arrow to view a list of the available options for that field.
7 Use the slider buttons to choose a color range.
8 Once you have completed your selections, press **Add Gauge.**

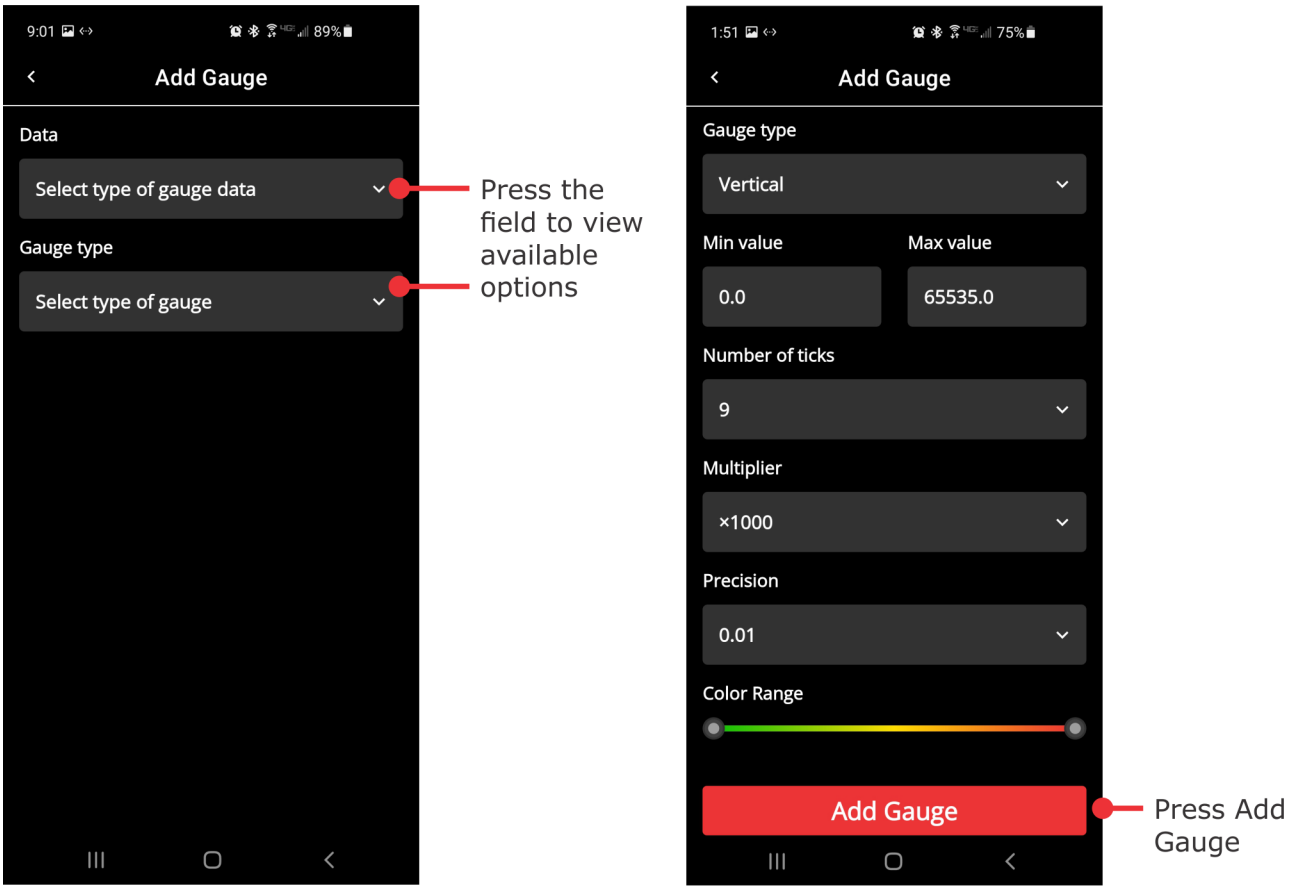


Figure 3-37: Add a Vertical Gauge

9 Press **Edit** to edit or remove the gauge.

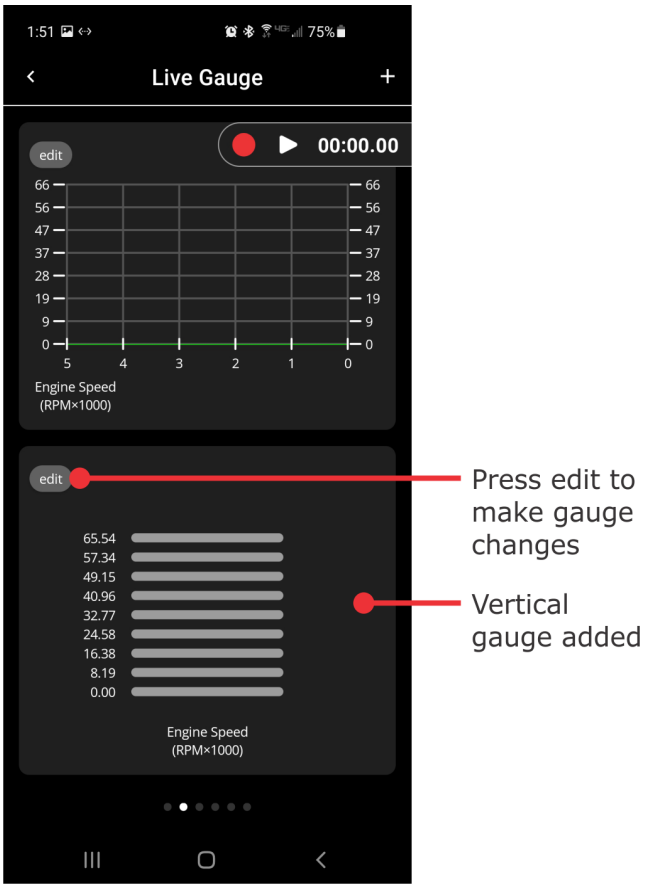


Figure 3-38: Edit the Vertical Gauge

RECORDING THE GAUGES

1 Press **Play** to begin recording.
2 Press **Stop** to stop recording and save as a logfile.

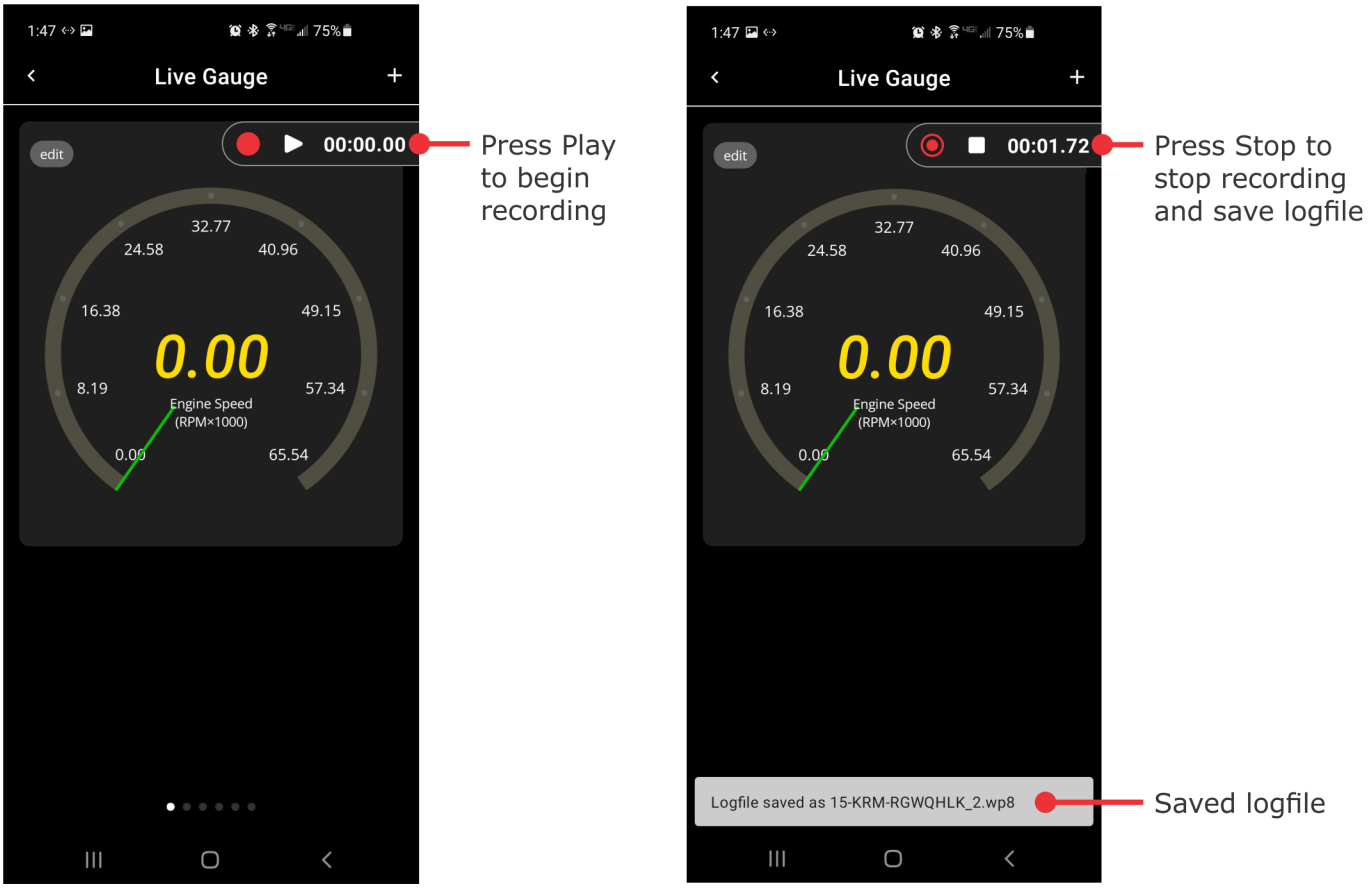


Figure 3-39: Record the Gauges

Flyout Screen Descriptions

Flyout screens and options will vary depending on which Data value is selected. Some examples of the different flyout screens are shown below.

Data and Gauge Type Drop-down screens:

Press the drop-down arrow to view a list of options. Use your finger to scroll through and select an option.

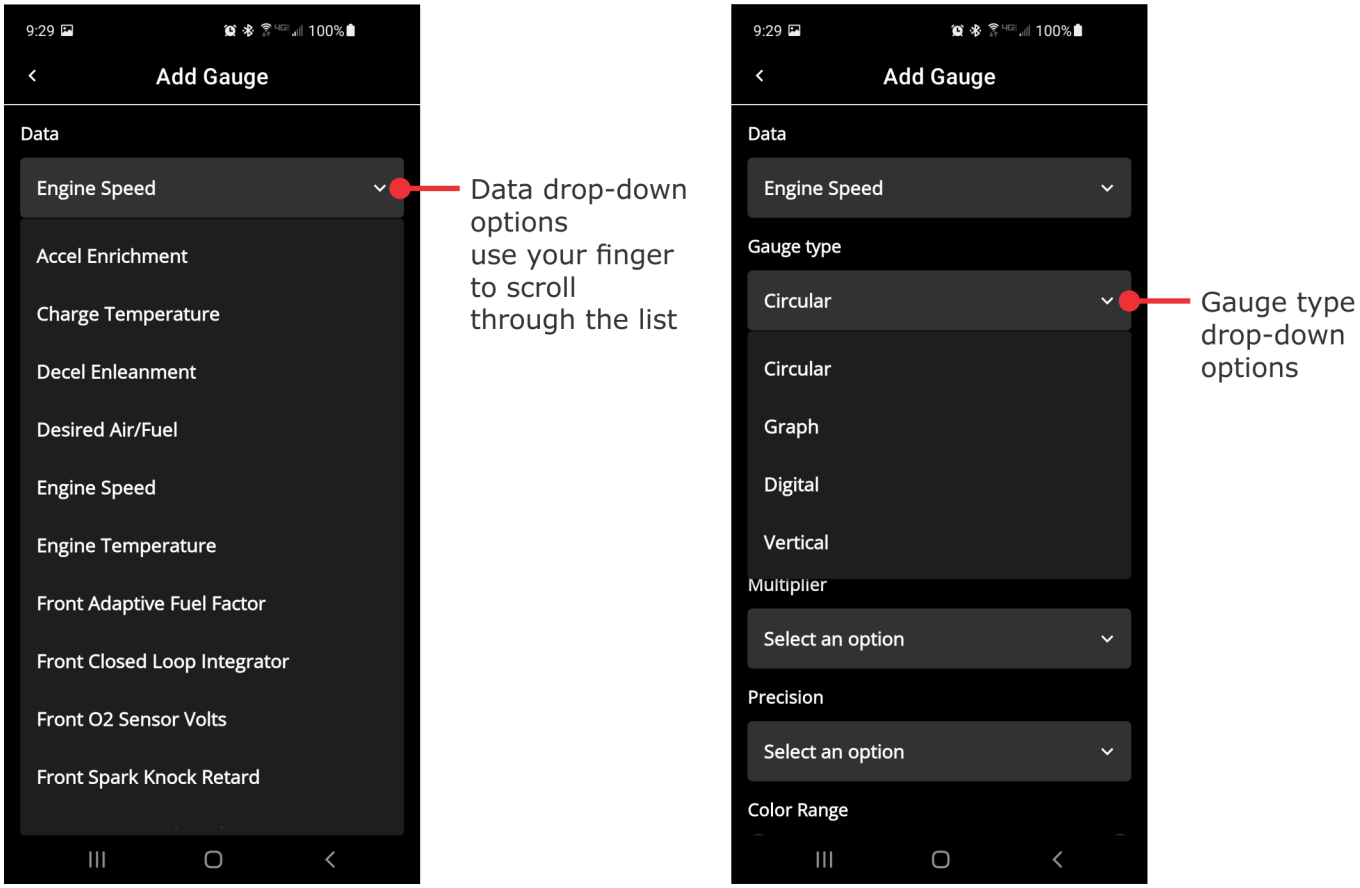


Figure 3-40: Drop-Down Options

MIN VALUE AND MAX VALUE KEYBOARD SCREENS:

Press the field to bring up a number keyboard. Use your finger to enter your values in the field.

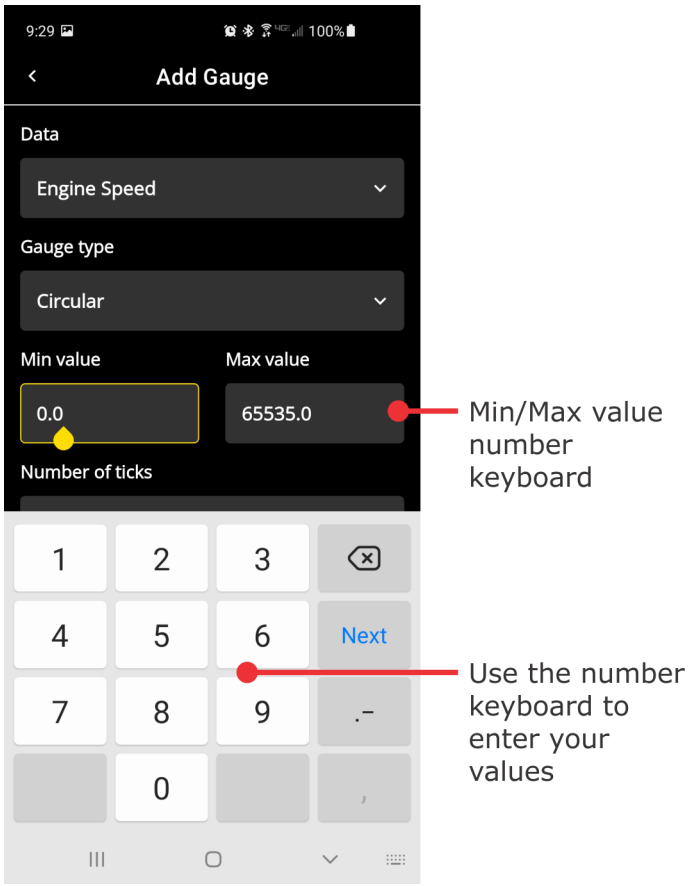


Figure 3-41: Using the Keyboard

NUMBER OF TICKS AND MULTIPLIER DROP-DOWN SCREENS:

Press the drop-down arrow to view a list of options. Use your finger to scroll through and select an option

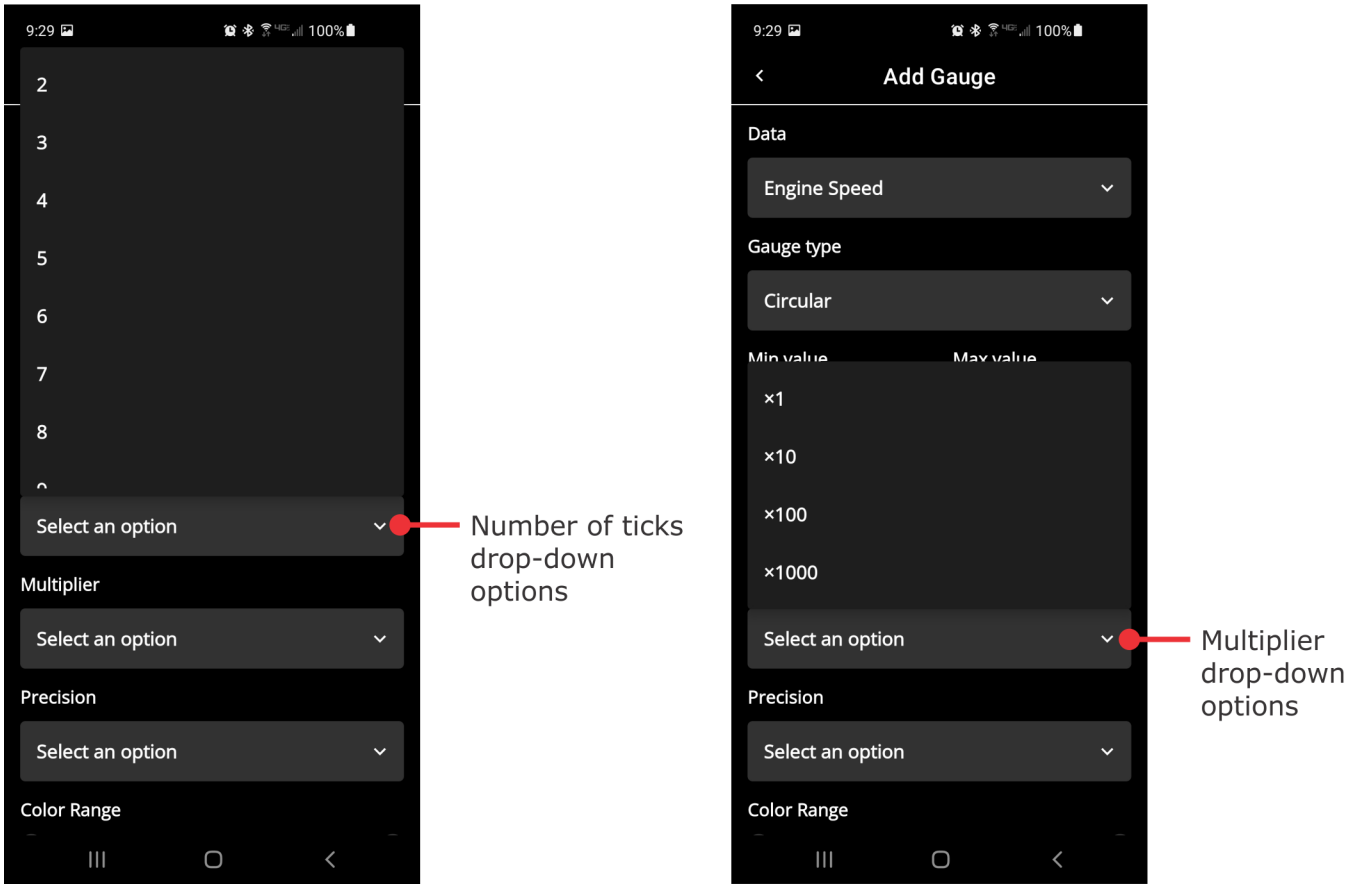


Figure 3-42: Ticks and Multipliers

PRECISION DROP-DOWN AND COLOR RANGE SLIDER BAR SCREENS:

Press the drop-down arrow to view a list of options. Use your finger to scroll through and select an option.

Use your finger to slide the bars left or right to change the color range for the selected gauge.

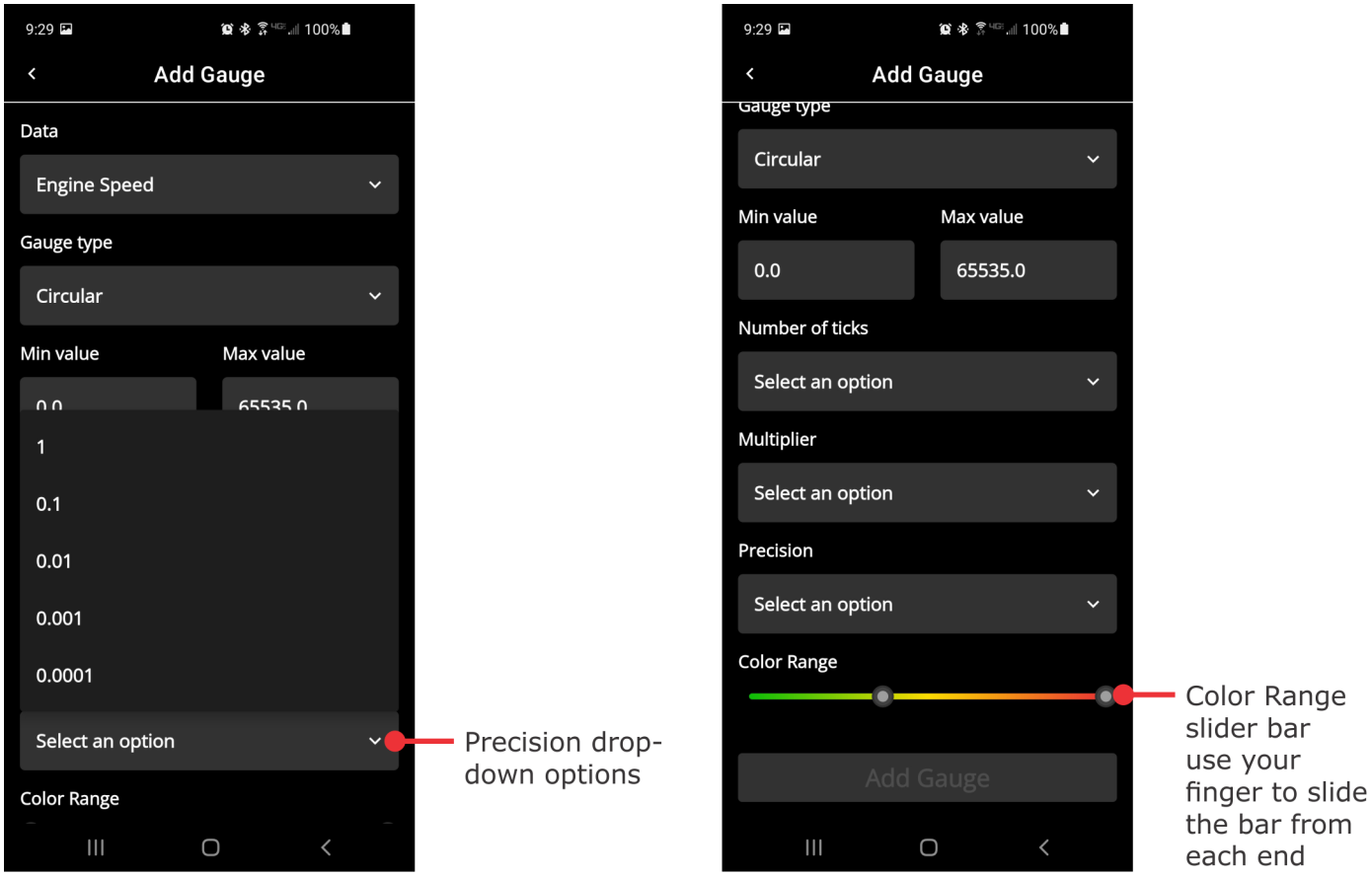


Figure 3-43: Precision and Color Range

LOGGING DATA

The Power Vision 4 allows you to log data and view the log using Power Core software.

1 Press **Start** to begin logging data.
2 Press **Lap** to save the current log and start a new log.
3 Press **Stop** to stop logging.
4 Press **Share** to share your log.

   Email the log to your computer and open the log using Power Core. Refer to the Power Core Help.

5 Swipe to delete the log.

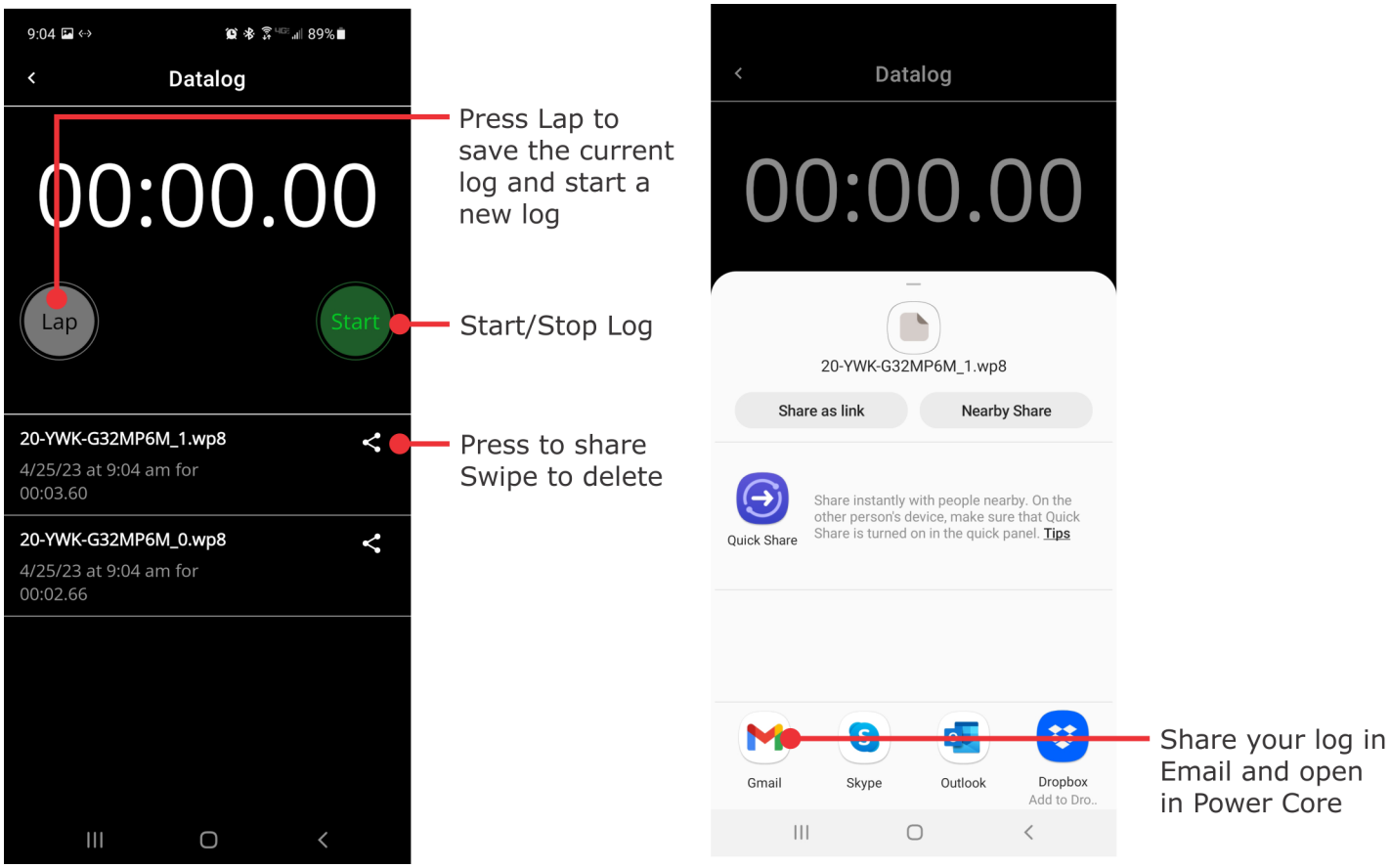


Figure 3-44: Logging Data

VIEWING THE DIAGNOSTIC CODES

Diagnostics allows you to read and clear diagnostic trouble codes (DTCs).

1 Press **Read DTCs** to read codes.
2 Press **Clear DTCs** to clear the codes.
3 Press the **Share** button to share your log.

If requested by Dynojet Technical Support, Email the file to PV4tech@dynojet.com.
Harley-Davidson customers, Email the file to PV4hdtech@dynojet.com.

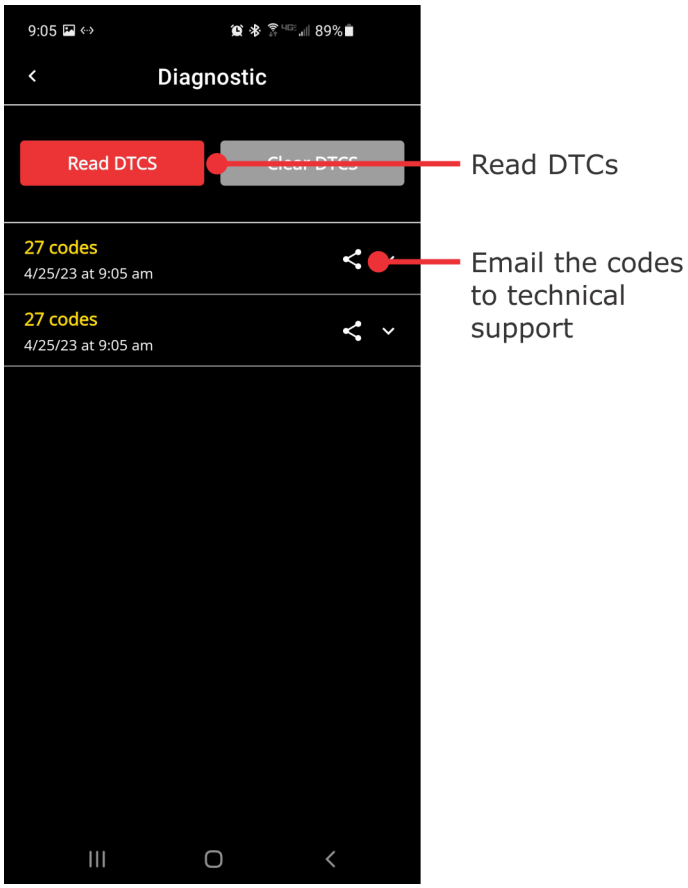


Figure 3-45: Diagnostic Codes

USING AUTOTUNE—HARLEY-DAVIDSON ONLY

Note: Autotune is only available with Harley-Davidson motorcycles.

Autotune is a process that tunes the VE Tables in an ECU using AFR (air/fuel ratio) readings from O2 sensors. The VE Tables control how much fuel is injected into the engine. The more accurate the VE Tables in the ECU are, the better the engine will run. Autotune uses a process that compares the known target AFR values in the Target AFR Tables of the ECU to the actual AFR values read from the O2 sensors in the exhaust pipe and tunes the VE Tables in the ECU to more accurately reflect the real VE of the engine.

Autotune Basic utilizes the stock narrowband O2 sensors to read the AFR data from the engine's exhaust. Additionally, because Autotune Basic uses narrowband O2 sensors to read AFR data, the Target AFR Tables of the ECU are set to a single constant value very close to stoich AFR (14.7) when Autotune is enabled so that the narrowband O2 sensors are able to read AFR across the whole range of the VE Tables. The Target AFR Tables are then restored to the original values when Autotune Basic is disabled. It is especially important to remember to disable Autotune when finished using Autotune Basic for this reason.

Autotune Basic can be used on most bikes that have factory narrowband O2 sensors installed.

Autotune Pro uses accessory wideband O2 sensors to read the AFR data from the engine's exhaust. Autotune Pro however requires the installation of an appropriate Dynojet wideband O2 kit (P/N AT-110, AT-300, etc.).

Target Tune feeds wideband data directly into the ECU for full-time wideband closed-loop operation. Optionally, you may also calculate and finalize a correction based on learned fuel trims, but this step is not required. Enabling Target Tune without first installing a Target Tune module may result in serious engine damage.

ENABLING AUTOTUNE

Note: Autotune will only be available when connected to a Harley-Davidson motorcycle.

1 Connect the Power Vision 4 to the vehicle. Refer to "Getting Started" on page 3-6 for more information.

2 Verify the vehicle is on and the engine is off.

3 Press **Autotune**.

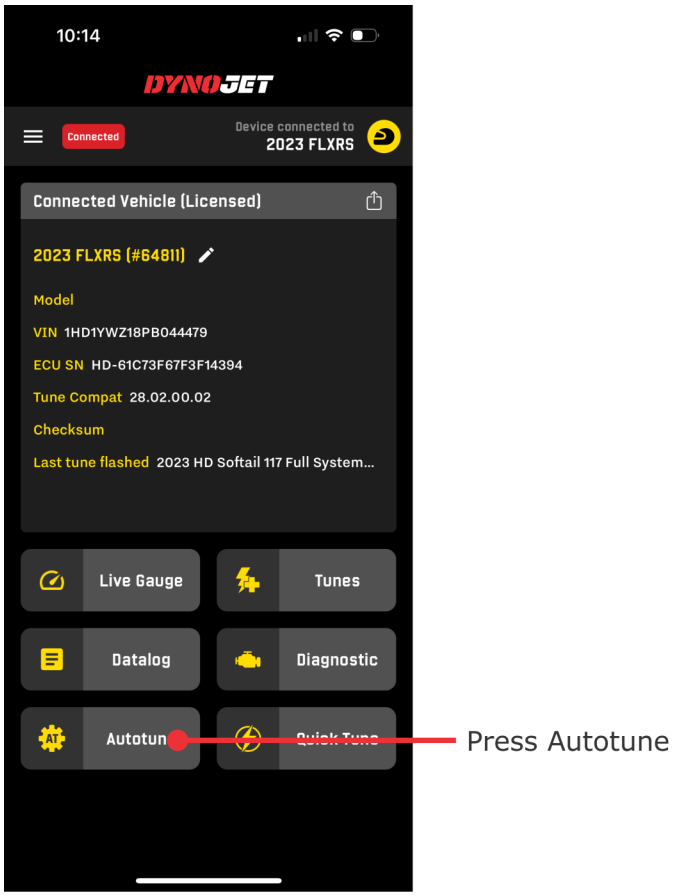


Figure 3-46: Autotune

4 Autotune will prompt you to select a base file. Press **Select Tune**. All Autotune corrections will be built on top of this base file. You may select the original stock file read from the ECU or another tune file.

5 Review your tune information and press **Select for Autotune**.

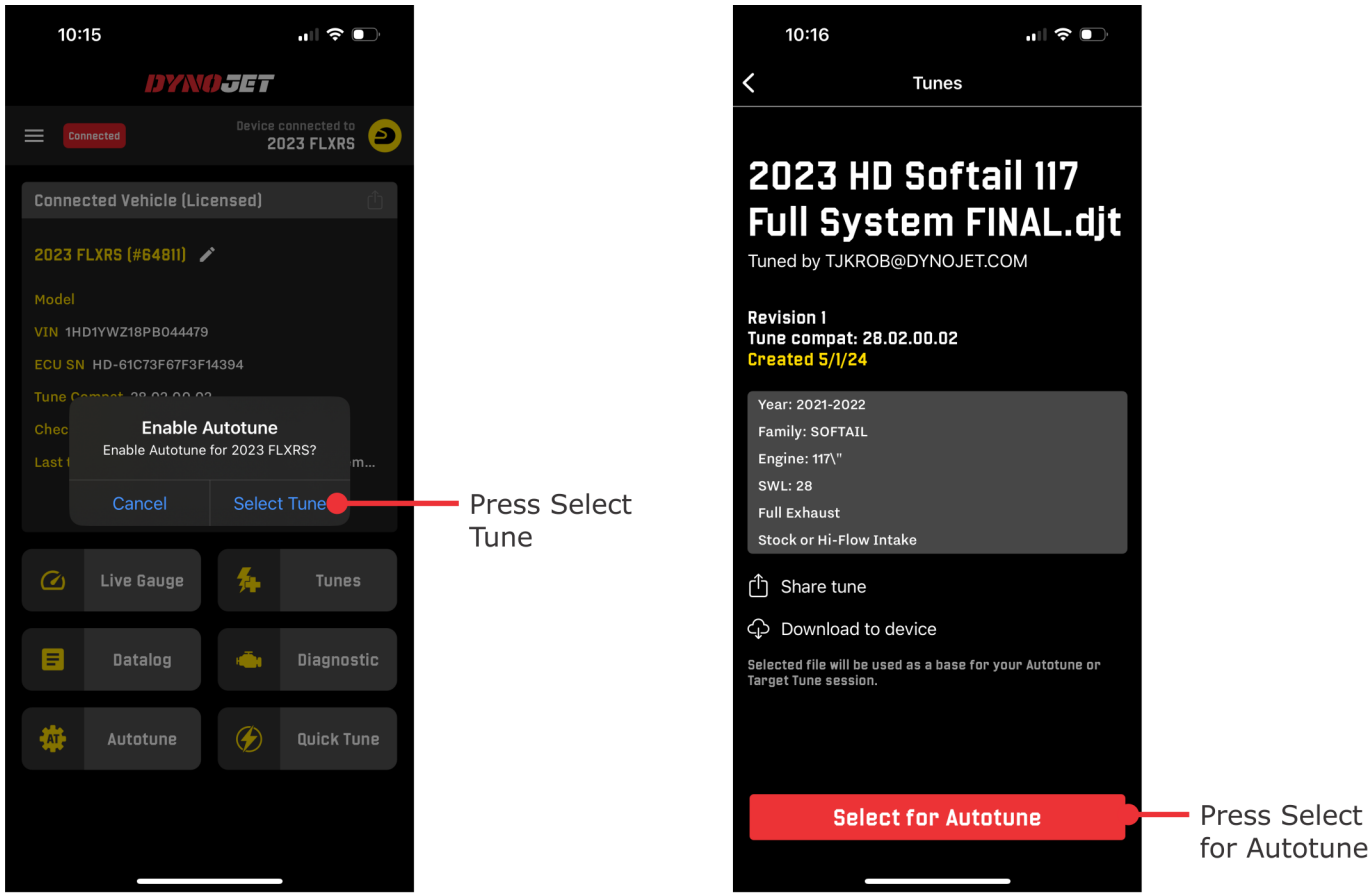


Figure 3-47: Select a Tune

6 Select Autotune Basic, Autotune Pro, or Target Tune from the tabs at the top of the screen.

Autotune Basic utilizes your factory narrowband sensors to automatically adapt the current tune to match any aftermarket components installed on your vehicle.

AT Pro will generate a more accurate tune and do so more quickly while generating substantially less heat within your engine.

Autotune Pro utilizes a Dynojet Wideband 2 module (Dynojet P/N AT-300) to automatically adjust the selected tune to match the Target AFR table.

Target Tune feeds wideband data directly into the ECU for full-time wideband closed-loop operation. Optionally, you may also calculate and finalize a correction based on learned fuel trims, but this step is not required.

Note: Enabling Target Tune without first installing a Target Tune module may result in serious engine damage.

7 Using your finger, select the **Min Hit Count**.

8 Make any additional adjustments as necessary.

9 Press **Enable**.

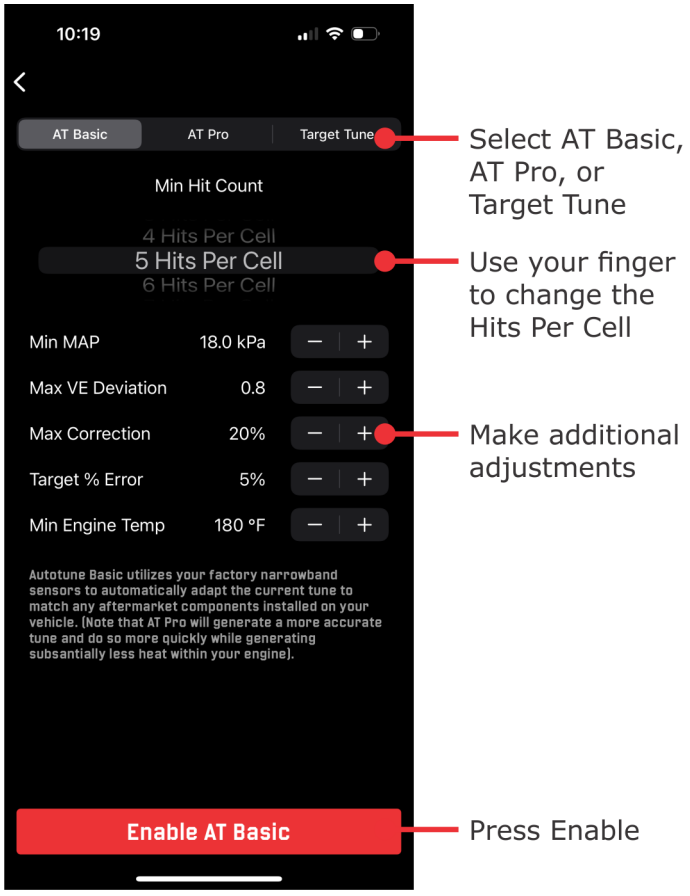


Figure 3-48: Enable Autotune

Your Autotune session is ready to start. The currently selected file has embedded values that allow the Autotune process to collect data from your ECU during normal riding and must be flashed to complete Autotune setup.

Note: While in collection mode, the vehicle may have reduced performance during rapid throttle transitions and that Autotune should be disabled when the correction process is completed.

10 Press **Flash the Autotune File**.

Note: Battery must be fully charged or connected to a charger.

Note: Do not tamper with the vehicle during this process and follow all prompts accordingly. Failure to follow this guidance may damage the ECU and render the vehicle inoperable.

11 Press **Yes** to flash the file to your vehicle.

12 Once the flash is completed, turn the key off for at least ten seconds to reset the ECU.

13 Press **OK**.

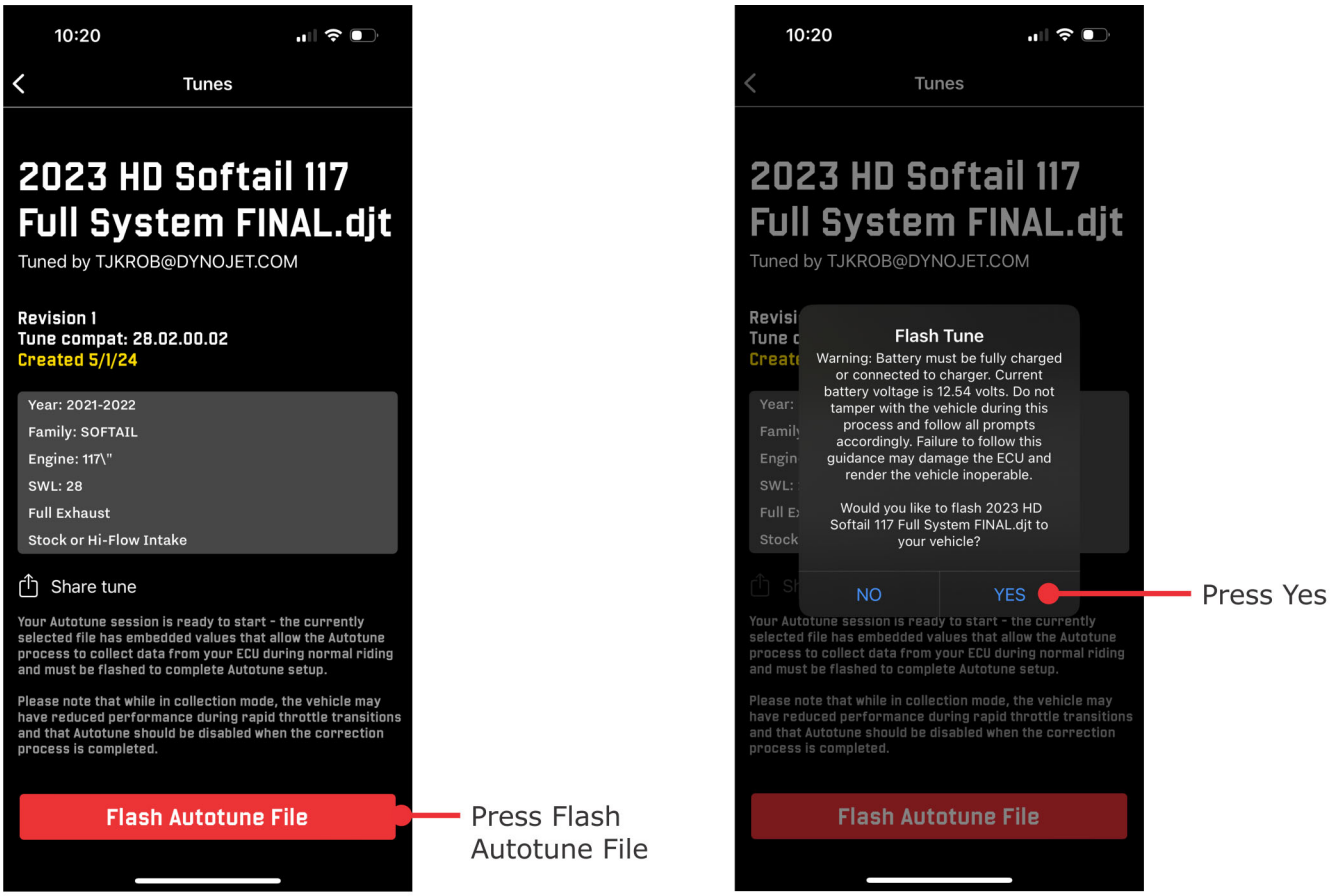


Figure 3-49: Flash Autotune File

LOGGING SAMPLE DATA

1 Start the vehicle. Autotune will automatically log sample data when the engine is running.

Spend time driving under all conditions that you would like Autotune to correct. Spend time in as many combinations of RPM and throttle positions as possible. The more varied the driving conditions, the more areas of the map Autotune will be able to correct.

2 Select **Table View** to view your data in table format.

3 Press **Live Gauge** to view your data as live gauges. Refer to "Viewing Live Gauges" on page 3-28 for information on adding gauges.

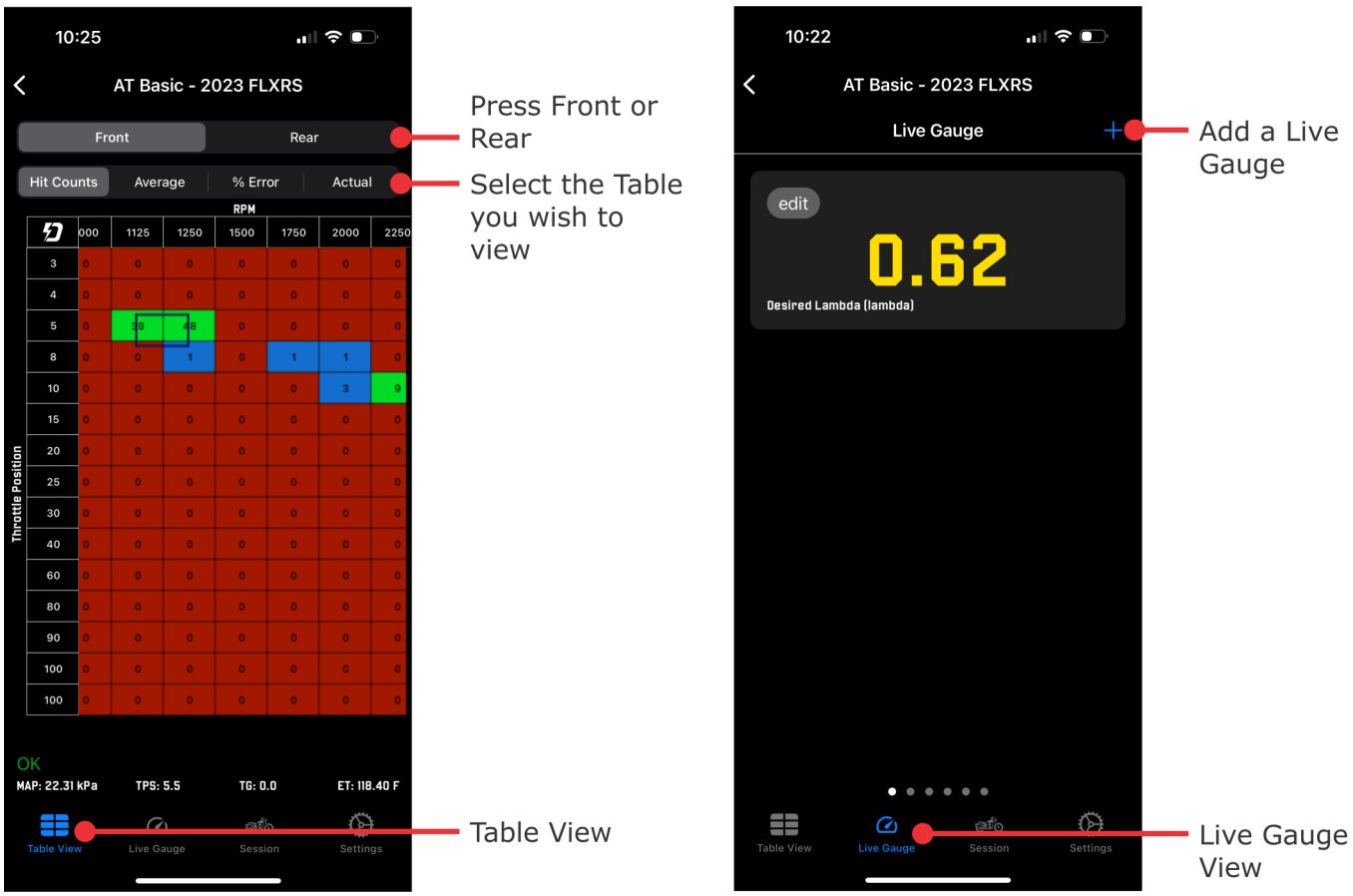


Figure 3-50: Logging Sample Data

APPLYING VALUES

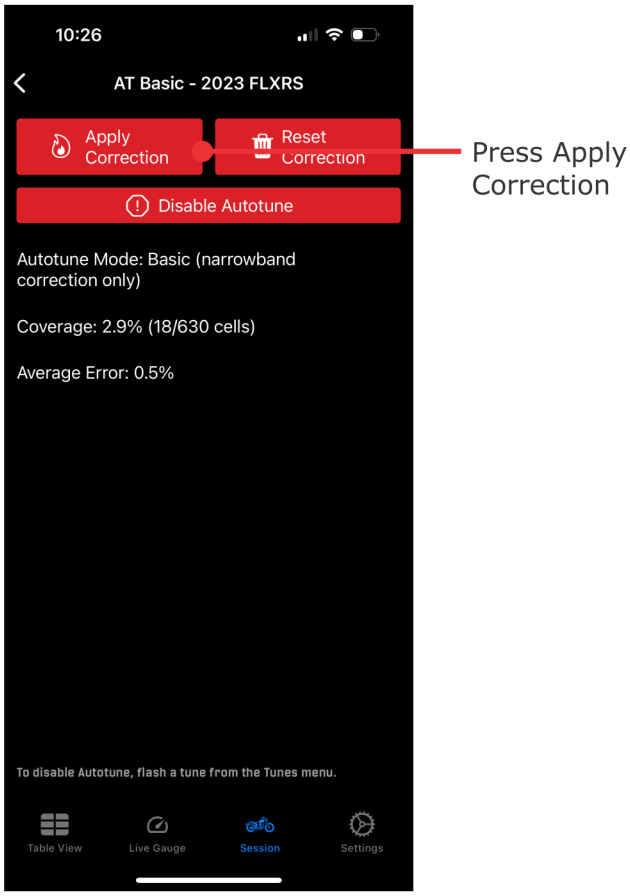
1 Verify the vehicle is on and the engine is off. Press **Apply Correction**.

2 Using your finger, you may adjust the Table Filtering slider bars. This step is optional.

- Detection Sensitivity determines how much a cell's value must differ from neighboring cells before it is classified as bad.
- Smoothing Strength decides how much of the original cell value to use (100% will ONLY use the values of surrounding cells).

3 Review the correction and press **Accept Correction**.

- Autotune will process the internal log files recorded during the logging session(s) and determine the corrections needed to tune the VE Tables.
- Autotune will apply the corrections and save a new correction file to the PV4 system.
- Autotune will turn on the Autotune enable settings inside the correction file (this allows the ECU to remain in an Autotune state) and automatically flash this tune file to the ECU.



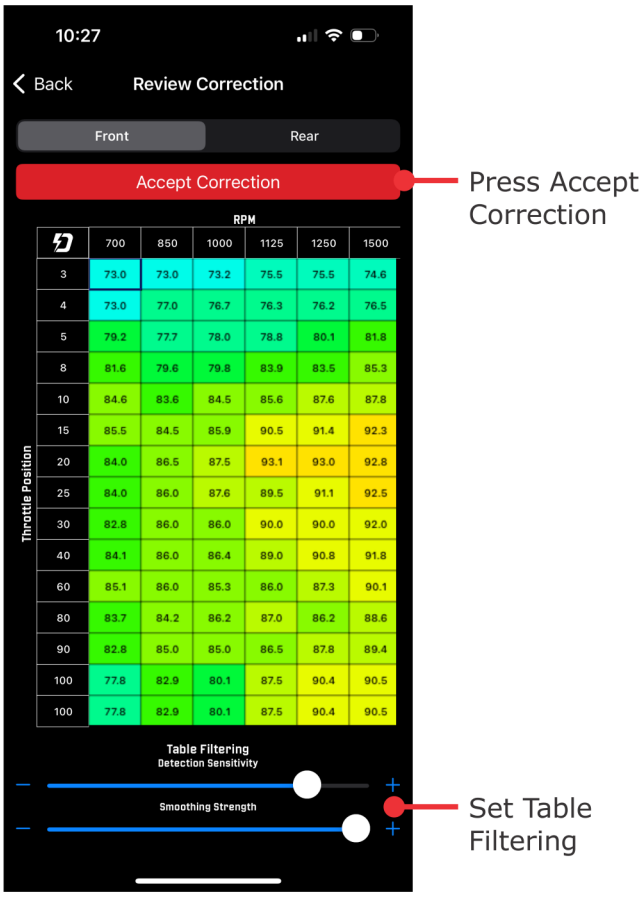


Figure 3-51: Apply Correction

4 Enter a name for your tune and press **Save**.
5 Press **Begin Next Autotune Session** or **Flash Tune and Exit AT**.
6 Once the tune has flashed successfully, turn the key off for at least ten seconds to reset the ECU.
7 For best results, Dynojet recommends repeating the Logging Sample Data and Applying Values sections two to three times

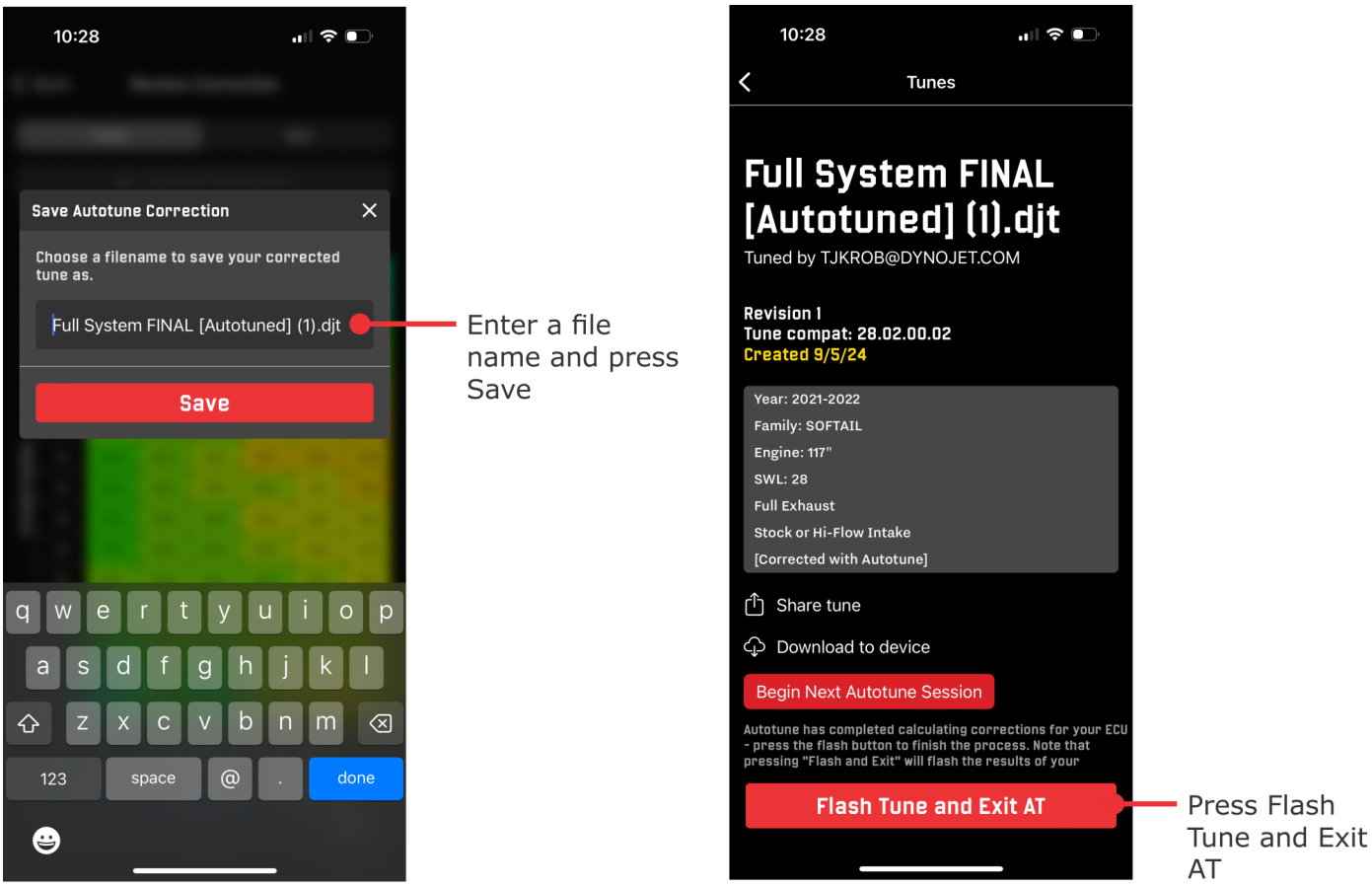


Figure 3-52: Flash Tune

DISABLING AUTOTUNE

1 Verify you are finished logging and applying values and the desired results have been achieved.

2 Verify the vehicle is on and the engine is off. Press **Sessions** and press **Disable Autotune**.

Autotune will automatically flash the last saved correction file to the ECU. This correction file contains the latest Autotune corrections but without the Autotune enable settings turned on, therefore disabling Autotune.

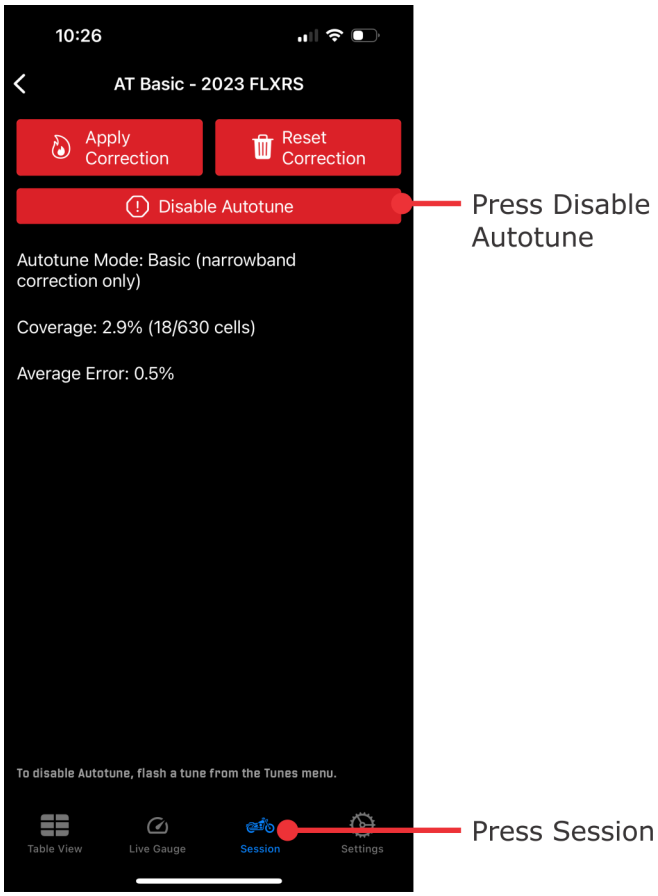


Figure 3-53: Disabling Autotune

RESETTING THE CORRECTION

Reset Correction can only be used during the Autotune logging session.

Press **Sessions** and press **Reset Correction** to reset the Autotune correction values. This will delete all existing Autotune log files allowing you to start over fresh.

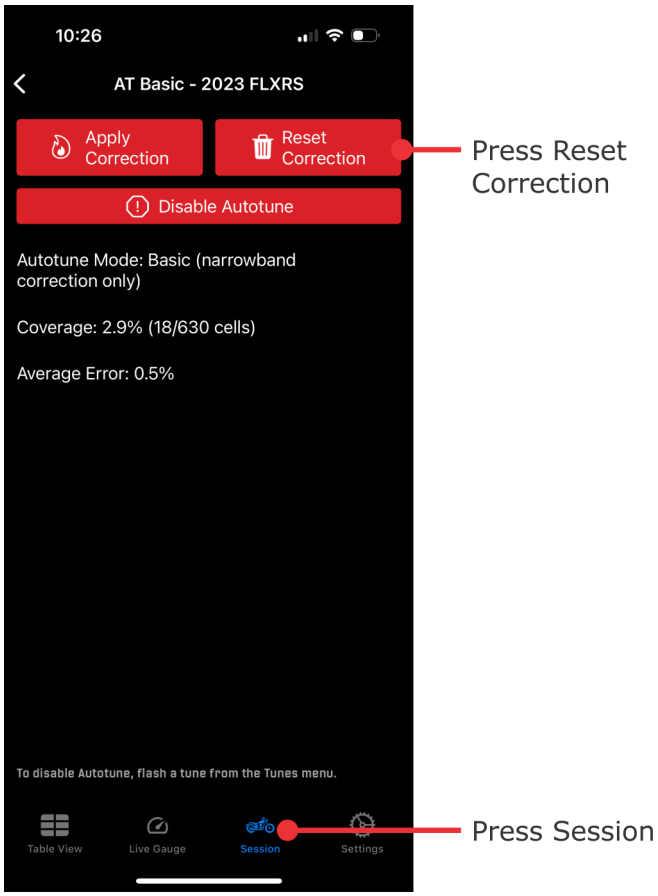


Figure 3-54: Resetting the Correction

QUICK TUNE–HARLEY-DAVIDSON ONLY

Note: Quick Tune is only available with Harley-Davidson motorcycles.

Quick Tune allows you to change EEPROM settings stored in your ECU. These settings are stored separately from the flash area where the calibration (tune) is written. Some items may be disabled, but can be unlocked by completing a Dynojet certification course. Flashing an STK file restores Quick Tune items to the value initially read from the ECU.

USING QUICK TUNE

Note: Quick Tune is only available with Harley-Davidson motorcycles.

1 Press **Quick Tune**.

2 Make your desired changes.

Some items may be disabled, but can be unlocked by completing a Dynojet certification course.

3 Press **Write EEPROM Values** to write the new settings back to the ECU.

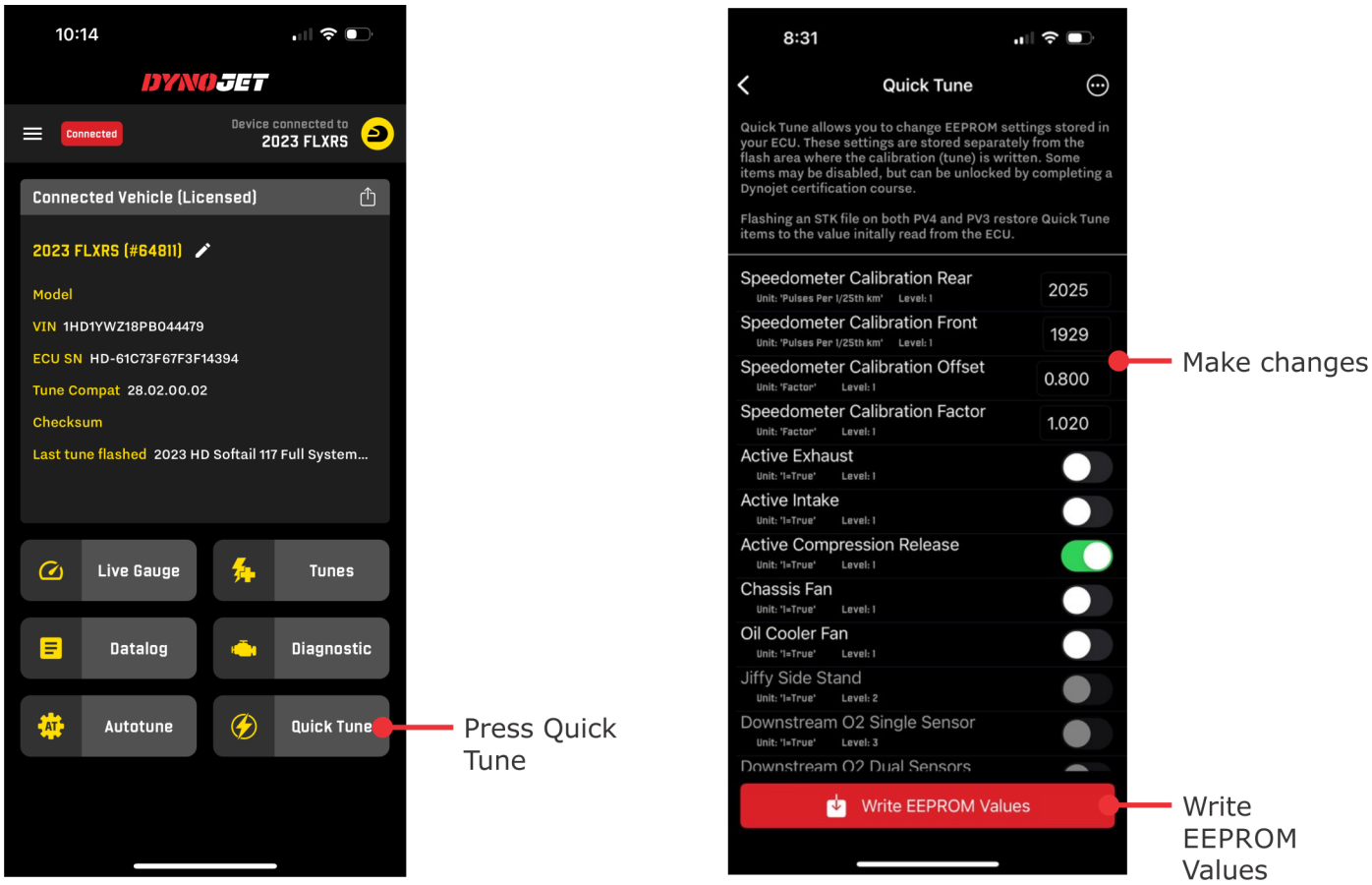


Figure 3-55: Write EEPROM Values

EXPORTING ITEMS

1 Press the **Menu** button.
2 Press **Export Items**.
3 Email your items to your computer or your desired location.

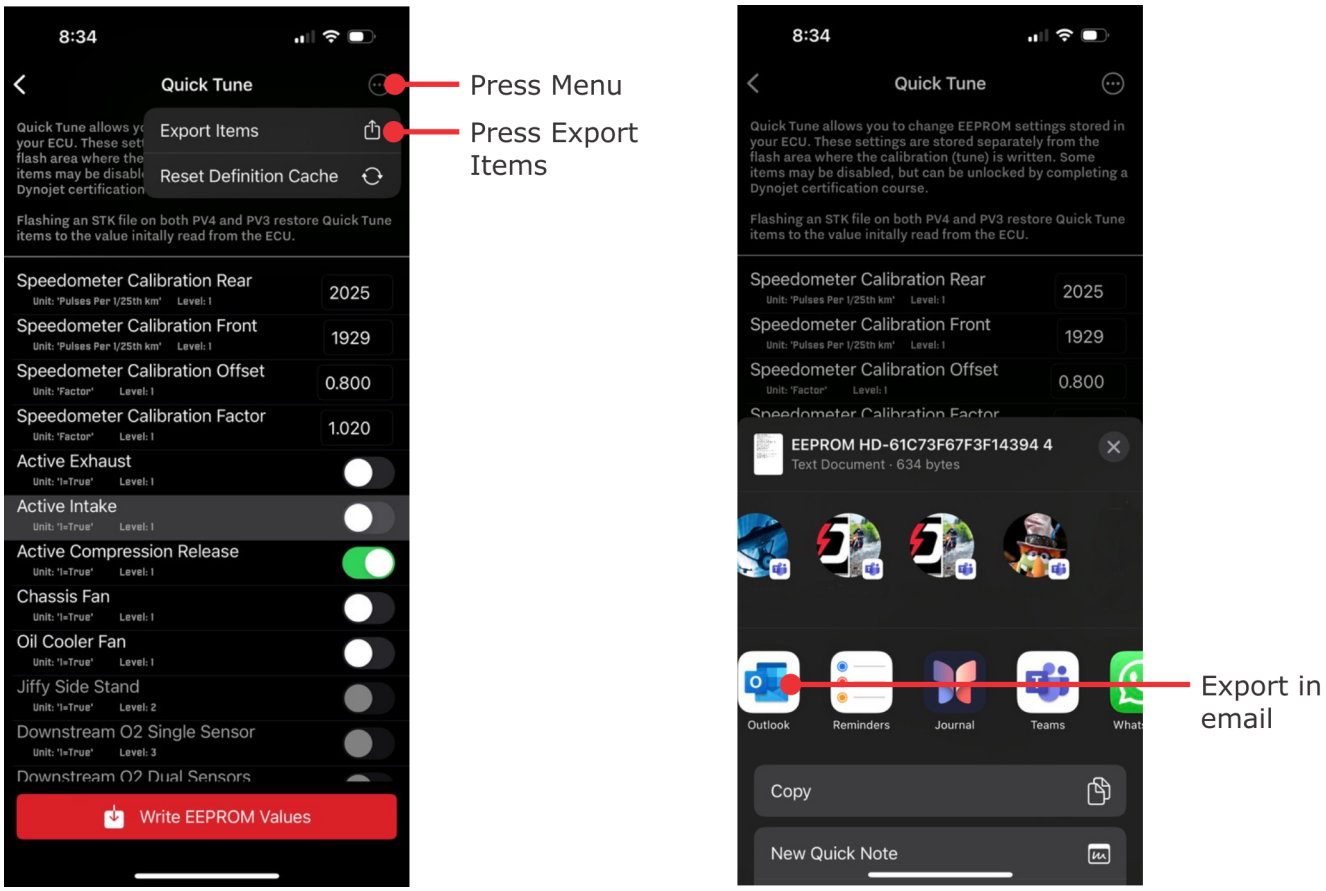


Figure 3-56: Exporting Items

Resetting the Definition Cache

This will reset the definitions to what is currently on the ECU.

1 Press the **Menu** button.
2 Press **Reset Definition Cache**.

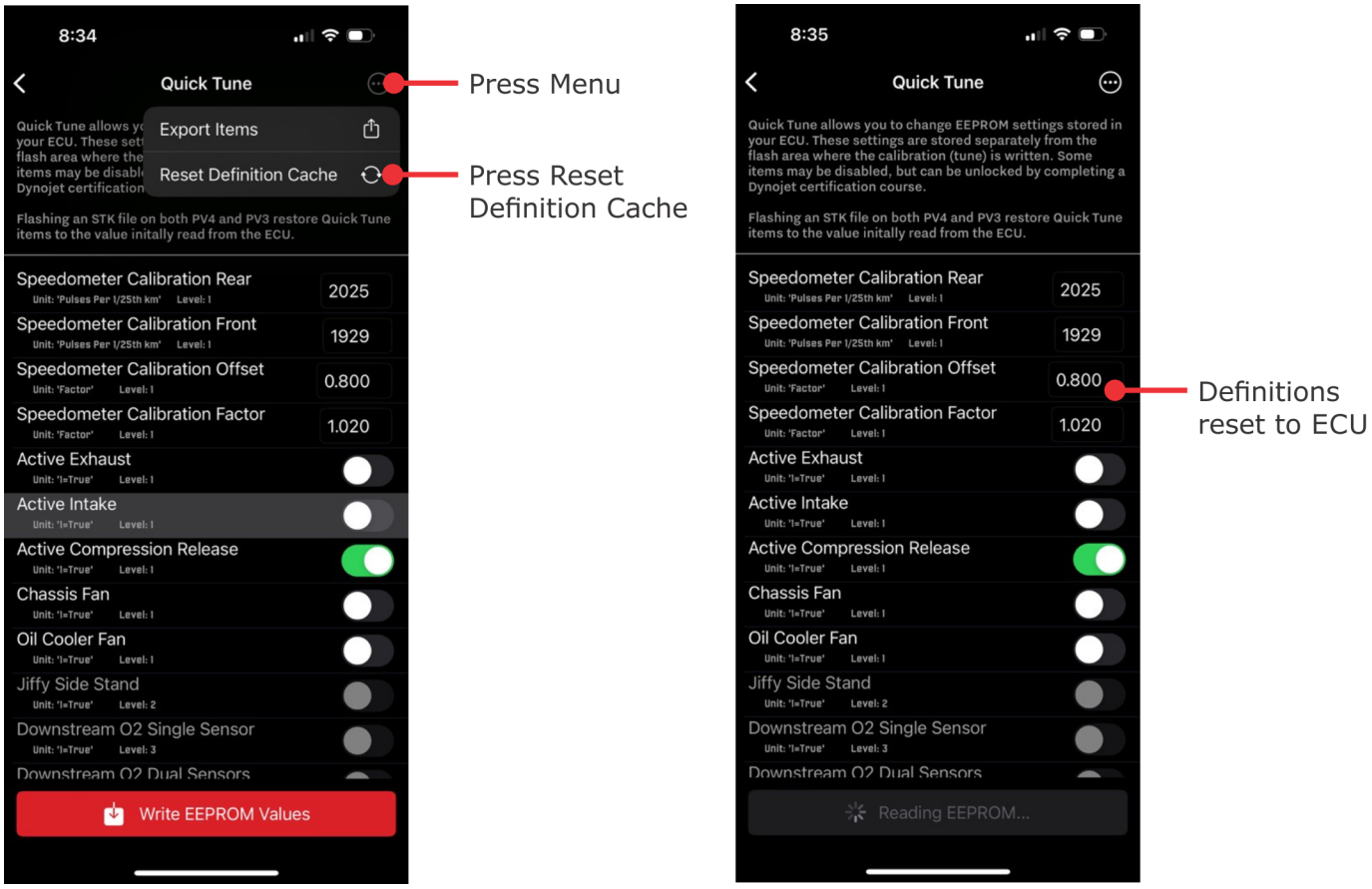


Figure 3-57: Reset the Definition Cache

VIEWING THE DEVICE INFORMATION

Device Information allows you to view the Power Vision 4 firmware version, serial number and address, and copy the device information if needed, while Firmware type allows you to toggle between different brands of firmware.

Note: Changing the Firmware type will require you to use the correct diagnostic cable.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

Note: For more information on the Disconnect button, refer to "Disconnecting from the Power Vision 4" on page 3-59.

1 Press **Menu** and press **Device Info**.

2 Press **Copy device information** to save a copy of the device information to your clipboard.

3 Press **Firmware Update** to view a list of the available options.

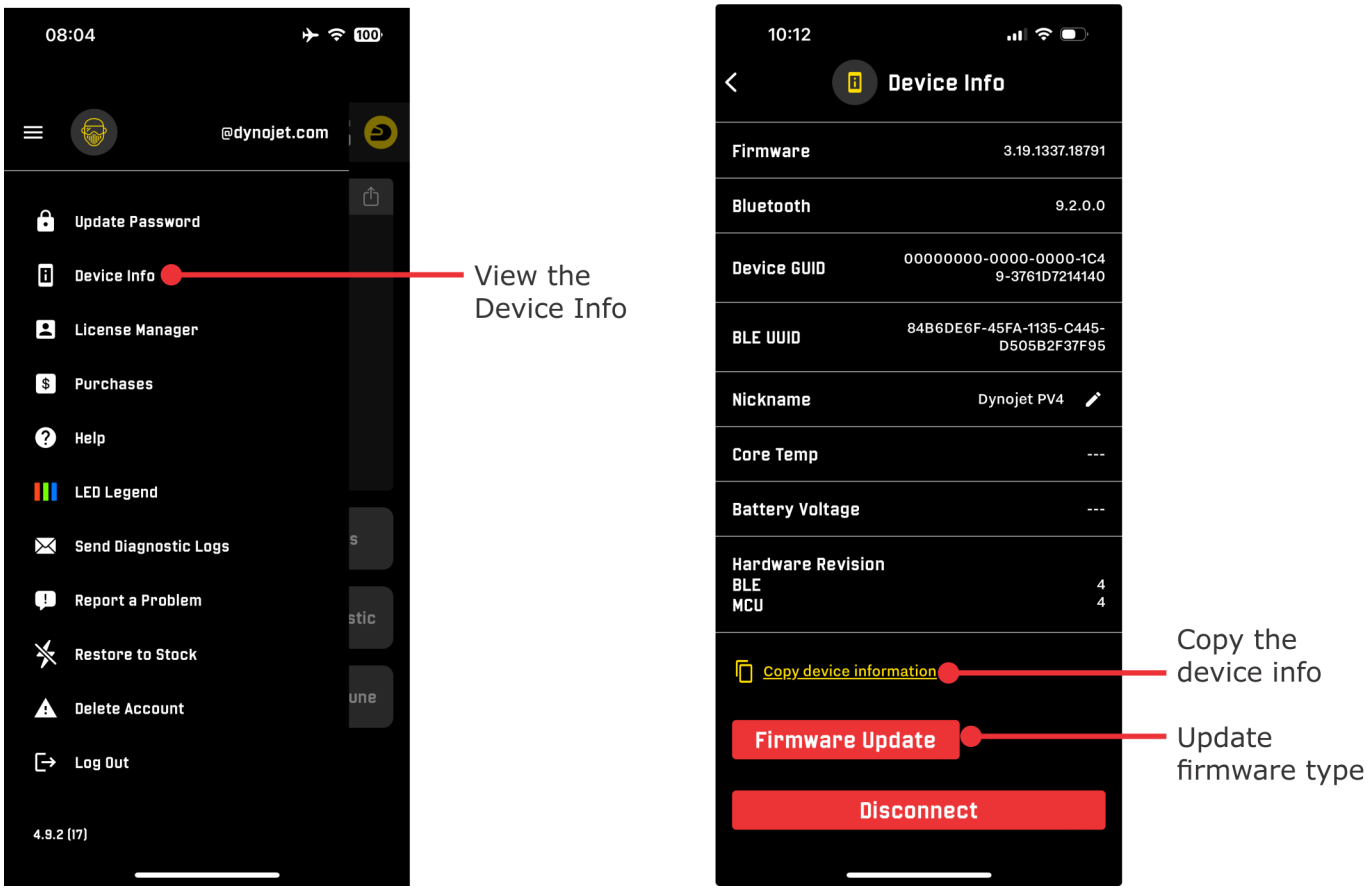


Figure 3-58: Firmware Update

4 Scroll through the brands and select a brand.
5 Press **Update** to confirm the firmware update.

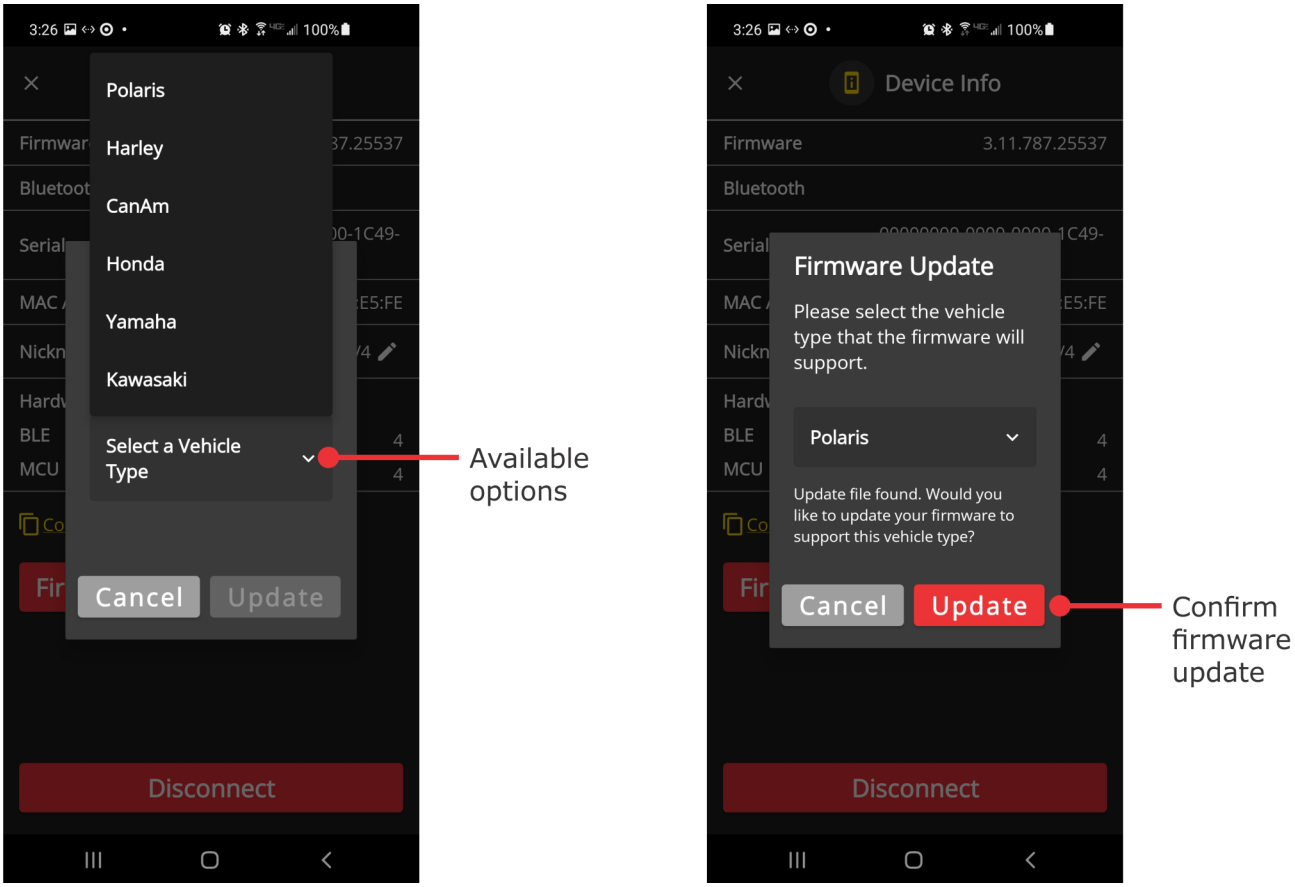


Figure 3-59: Firmware Update

DISCONNECTING FROM THE POWER VISION 4

Manually disconnect from your Power Vision 4 and connect to another device.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Press **Menu** and press **Device Info.**

2 Press **Disconnect**.

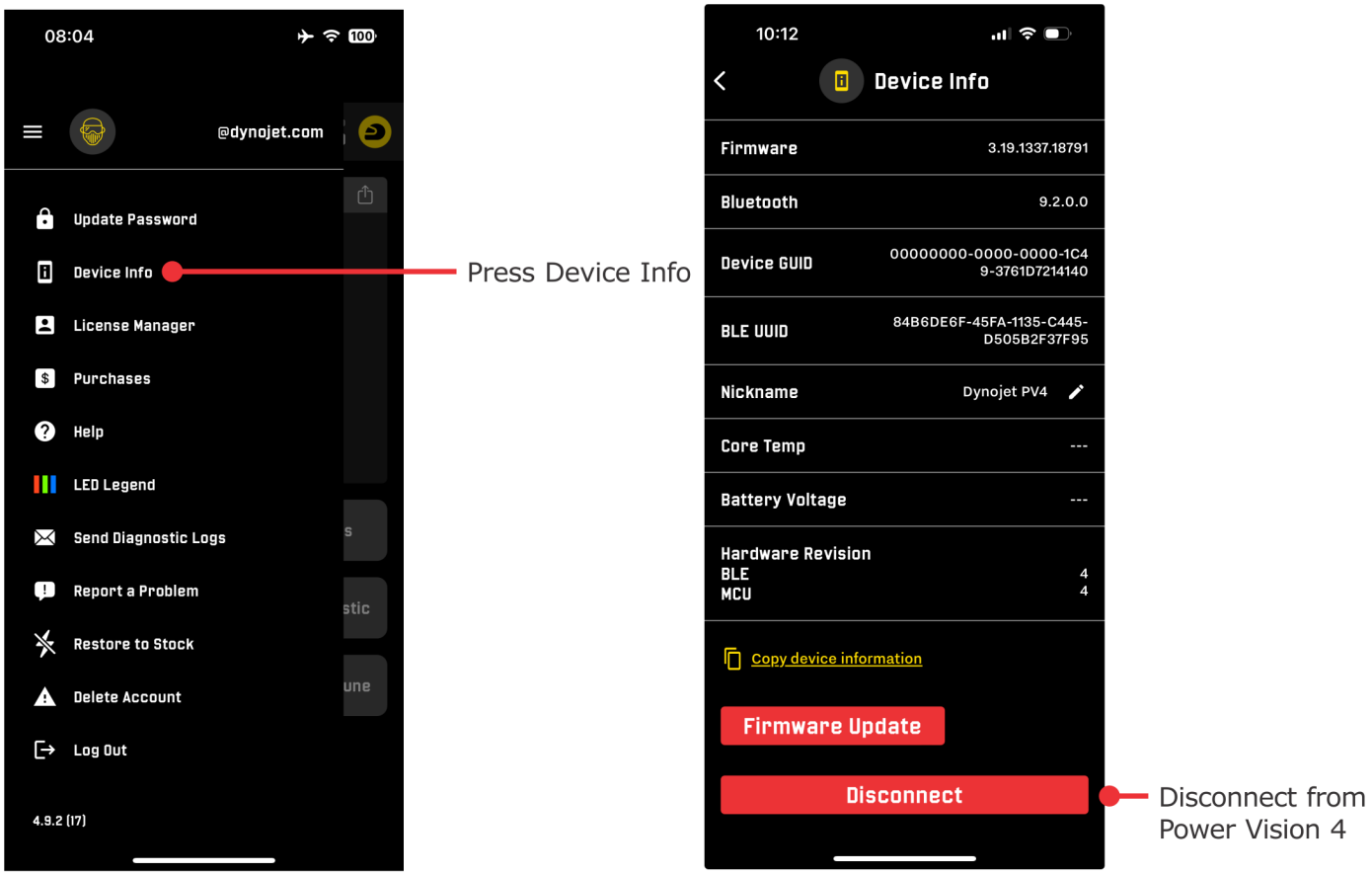


Figure 3-60: Disconnect the PV4

RESTORE TO STOCK

Use Restore to Stock to uninstall the Power Vision 4 from the connected ECU. In most situations, this option is to required and is provided for special scenarios only.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Press **Menu** and press **Restore to Stock.**

2 Press **Uninstall**.

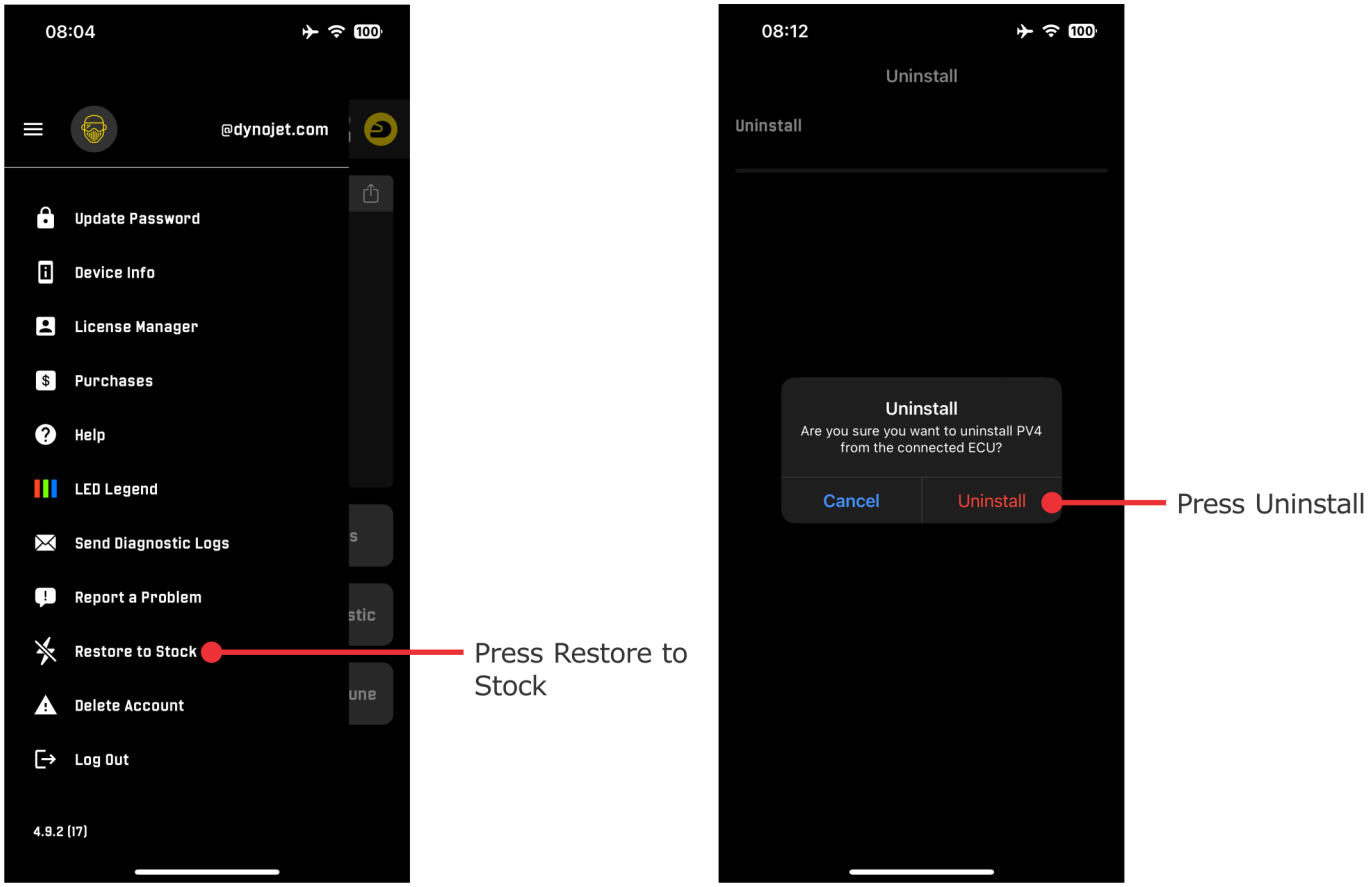


Figure 3-61: Restore to Stock

REPORT A PROBLEM

Have questions or need additional support? Email Dynojet's state of the art technical support team.

Note: Application features and displays may vary and depend on the vehicle and firmware selected. The images shown in the PV4 help are a representation only and are not intended to be a perfect match; however, the instructions are the same unless otherwise noted.

1 Press **Menu** and press **Report a Problem.**

2 Select which email app to use.

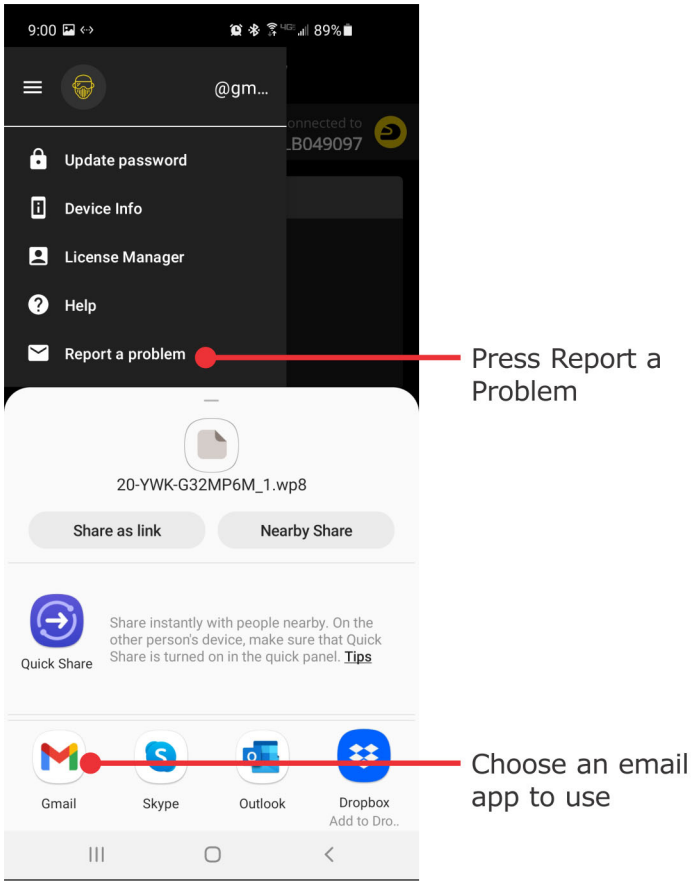


Figure 3-62: Report a Problem

APPENDIX A

POWER CORE SOFTWARE DOWNLOAD

Power Core software provides Dynojet dyno owners with the most advanced tuning software platform available in the industry! Control the dyno, analyze data, adjust/create calibrations and maps for Dynojet products and utilize real-time on-board data from vehicles running on the dyno. Whether you are new to the benefits of dyno testing and tuning or an experienced performance leader, Power Core software will give you the professional results you require.

Find more information in the Power Core Help or the Power Core User Guide P/N 98100020.

INSTALLING THE POWER CORE SOFTWARE

Download the Power Core software and take tuning to a whole new level.

Note: Downloading the Power Core software is optional and not necessary to use the Power Vision 4.

1 Go to www.dynojet.com/PowerVision.

2 From the top navigation menu, select **Support>Downloads**.

3 Click **Power Vision 3**.

4 Click the **Software** drop-down and download the Dynojet Power Core software.

5 Double-click the saved file to begin installation. Follow the on-screen prompts.

Refer to the Power Core Help in the software for more information.

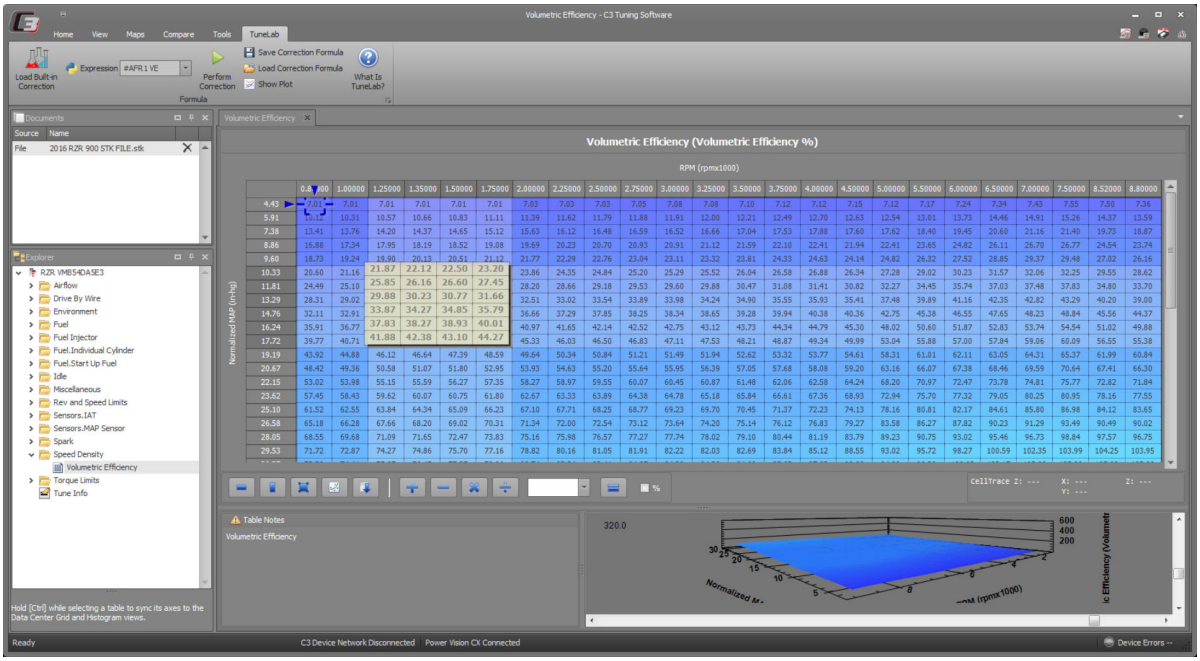


Figure A-1: Power Core Software

APPENDIX B

TROUBLESHOOTING

A few of the more common troubleshooting topics are listed in this chapter. For more topics, visit the Dynojet Knowledge Base.

For additional help, please contact Dynojet Technical Support.

This appendix is divided into the following categories:

- Unable to Connect to Bluetooth, on page B-2
- Phone Will Not Connect to Power Vision 4, on page B-2
- Error Flashing, on page B-2
- How to Recover the ECU, on page B-2
- How to Update the Firmware, on page B-3
- No Tune License Available, on page B-3
- How to Purchase a Tune License, on page B-3
- How to Manage Existing Tune Licenses, on page B-4
- Unable to Log Into My Dynojet Account, on page B-4

UNABLE TO CONNECT TO BLUETOOTH

If you are having trouble connecting to Bluetooth, check some of these Bluetooth basics.

1 Verify you are within range.
2 Turn Bluetooth off and then on again.
3 Restart your phone. Restart the Power Vision 4.
4 Have your phone "forget" the Power Vision 4.
5 Clear the Bluetooth cache.

PHONE WILL NOT CONNECT TO POWER VISION 4

Location permission is required for the app to access the phone's Bluetooth hardware.

1 Locate the Dynojet PV4 app on your phone.
2 Long-press the app until the context menu appears.
3 Press **App Info** or the ⓘ icon.
4 Press **Permissions**.
5 Verify Location and Storage are both allowed.

ERROR FLASHING

1 Verify the Power Vision 4 app is up to date through either the Apple App Store or Google Play Store.
2 Connect the Power Vision 4 to your bike.
3 Verify the firmware is up to date. Refer to "How to Update the Firmware" on page B-3.
4 Install the firmware update(s) if available.
5 Contact Dynojet Technical Support if the error persists.

HOW TO RECOVER THE ECU

You may need to recover the ECU if the flash was interrupted leaving the ECU is in an incomplete flash state.

1 The Power Vision 4 will ask you to recover the ECU. Press **OK**.

This will automatically flash the stock (.stk) file.

2 If the Power Vision 4 does not automatically flash the stock (.stk) file, you will need to manually choose a stock (.stk) file from your tunes. Ideally, this would be the stock (.stk) file you saved from your initial read.
3 Once the ECU is recovered, you will need to re-flash your tune. Refer to "Flashing Your Vehicle" on page 3-19.

HOW TO UPDATE THE FIRMWARE

1 Every time you connect the Power Vision 4, the device automatically checks for new firmware. If newer firmware is found, the Power Vision 4 will ask you to update the device.
2 Press **Yes** to update the device.
3 You must update the device to continue. If you choose not to update the device, the app will close.

NO FLASH LICENSE AVAILABLE

Your new Power Vision 4 comes with one available flash license. Before you can flash additional vehicles, you will need to purchase additional flash licenses.

1 Log in to your Dynojet.com account.
2 Click **My Account>Tune Licenses**.

View your purchased licenses. This will also show you which VIN the license file is associated with.

3 Purchase additional flash licenses to use with additional vehicles.

HOW TO PURCHASE A FLASH LICENSE

Through a Flash License, you can unlock the true potential of your Dynojet Power Vision 4. Your Power Vision 4 will be able to tune the original vehicle, as well as additional vehicles. Your Power Vision 4 device will remember each license that is applied, essentially multiplying the power of your device.

Each license is vehicle specific, not device specific, allowing you to load a Flash License onto any number of Power Vision units you have.

1 Visit https://www.dynojet.com/digital-products/.
2 Log in to your Dynojet account.
3 Select the Tune License for the vehicle you wish to purchase and add to your cart.
4 Check out following the prompts.

HOW TO MANAGE EXISTING FLASH LICENSES

Manage your existing Flash Licenses on the Dynojet website.

1 Log in to your Dynojet.com account using the same account you used to log into the Power Vision 4 app.
2 Click **My Account** in the top, right corner of the browser.
3 Click **Tune Licenses**.

UNABLE TO LOG INTO MY DYNOJET ACCOUNT

1 Verify there is not a typo in your email or password.
2 Verify you have entered your full email address. For example, “username@dynojet.com” instead of “username@dynojet.”
3 Passwords are case sensitive. Verify caps lock is turned off.
4 Refresh your browser and try signing in again.
5 Did you forget your password? Click **Forgot my password** and follow the prompts.
6 Still having trouble, contact Dynojet Technical Support.

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